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Providentia Training Session 2

26/02/2026

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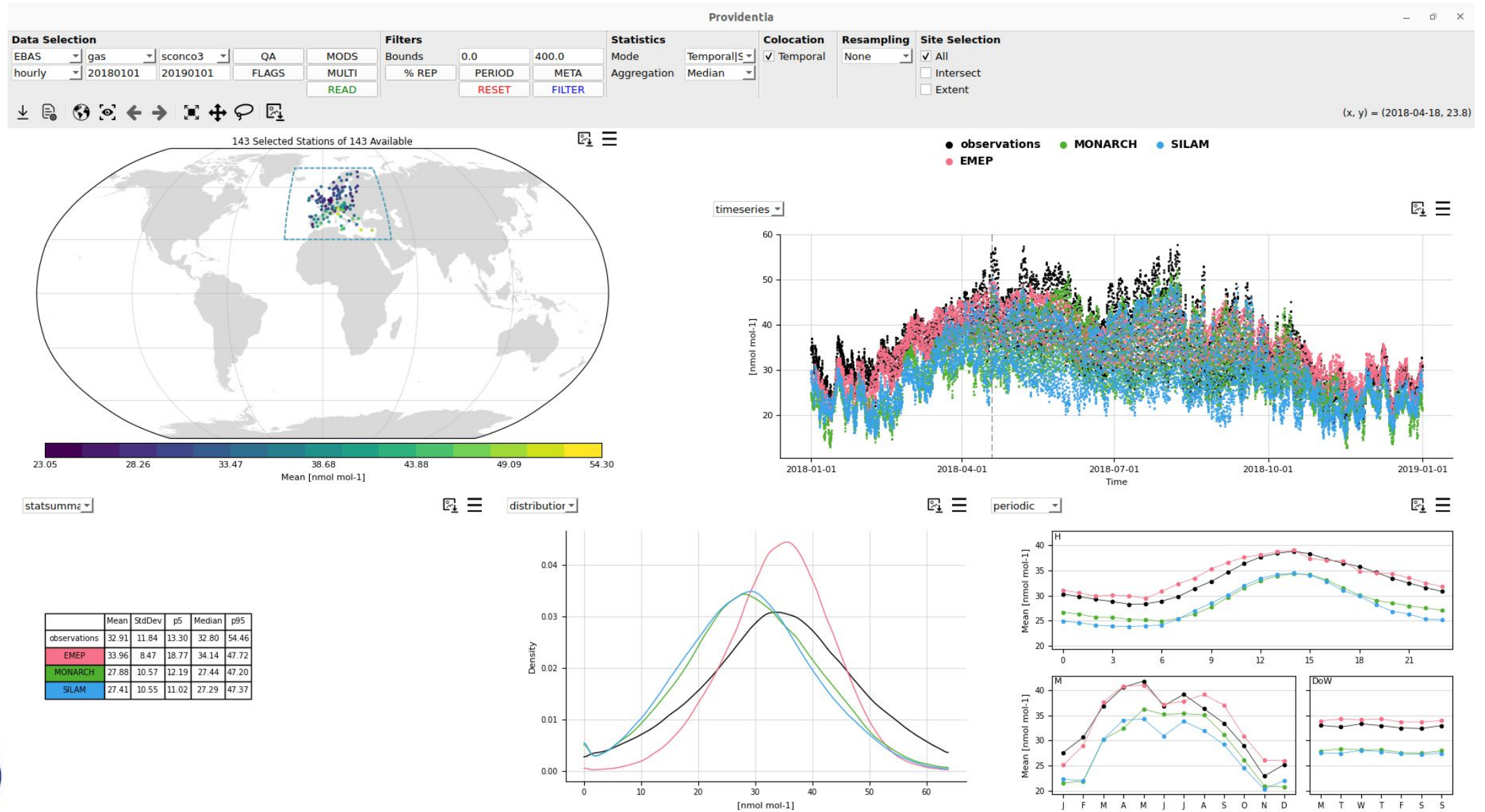
Dashboard



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Dashboard



Main menu

Data Selection					Filters			Statistics		Colocation	Resampling	Site Selection
EBAS	gas	sconco3	QA	MODS	Bounds	0.0	400.0	Mode	Temporal S	☒ Temporal	None	☒ All
hourly	20180101	20190101	FLAGS	MULTI	% REP	PERIOD	META	Aggregation	Median			☐ Intersect
READ					RESET FILTER							☐ Extent
												

Data selection menu

Network	Matrix	Species	Quality assurance (GHOST)	Models
EBAS ▾	gas ▾	sconco3 ▾	QA	MODS

Temporal resolution	Start date	End date	Data flags (Provider)	Multispecies filtering
hourly ▾	20180101	20190101	FLAGS	MULTI

Filters menu

	Data lower bound	Data upper bound
Bounds	0.0	400.0

Representativity filters	Time period filters	Metadata filters
% REP	PERIOD	META

Statistics menu

Statistics

Mode

Aggregation

Statistic mode

Statistic aggregation

Colocation menu



Colocate models vs. observations, removing temporal gaps

Resampling menu



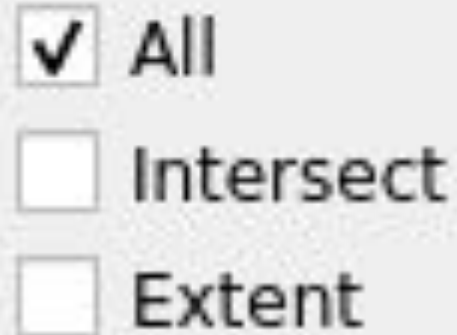
Temporal resolution to resample your data to (always lower than the selected resolution)

Stations selection menu

Select all stations

Select intersecting stations
within all model domains

Select stations on current map view



A screenshot of a software menu for selecting stations. It contains three items, each with a checkbox on the left and a label on the right. The first item, 'All', has a checked checkbox. The second item, 'Intersect', has an unchecked checkbox. The third item, 'Extent', has an unchecked checkbox.

☒	All
☐	Intersect
☐	Extent

Icons menu

Upload
configuration files

Set original
view

Forward

Pan

Save
canvas



Download data and
configuration files

Set world
view

Back

Zoom to
rectangle

Lasso

Interactive features



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Navigation tips

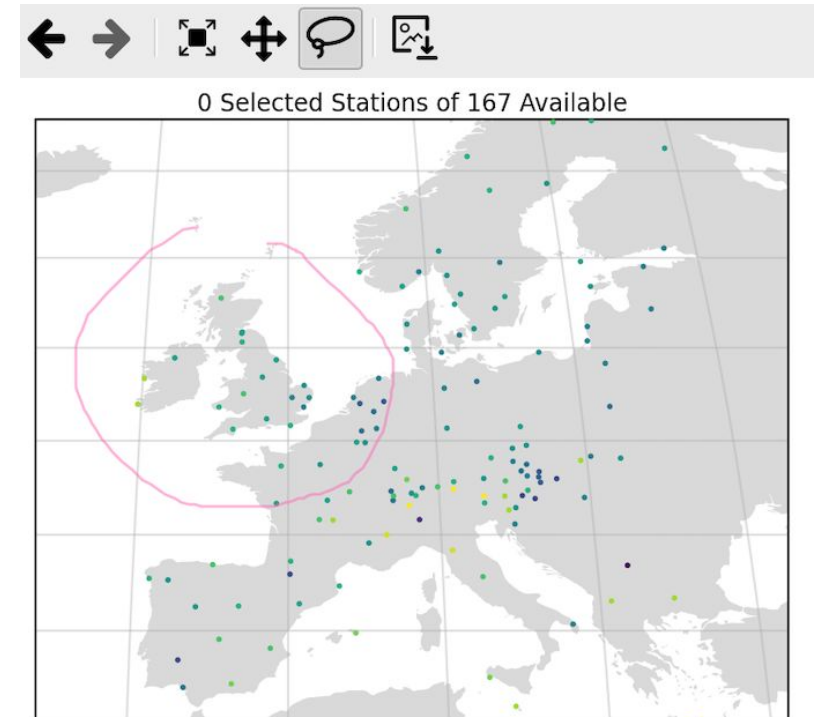
Individual stations can be selected by left clicking on them on the map.

Multiple stations can be selected by activating the lasso, and drawing it round all desired stations.

To unselect stations simply click on an empty space on the map.

When multiple stations are selected, to add or remove any individual station right click on the given station.

To quickly zoom on the map, the mouse or trackpad scroll wheel can be used.



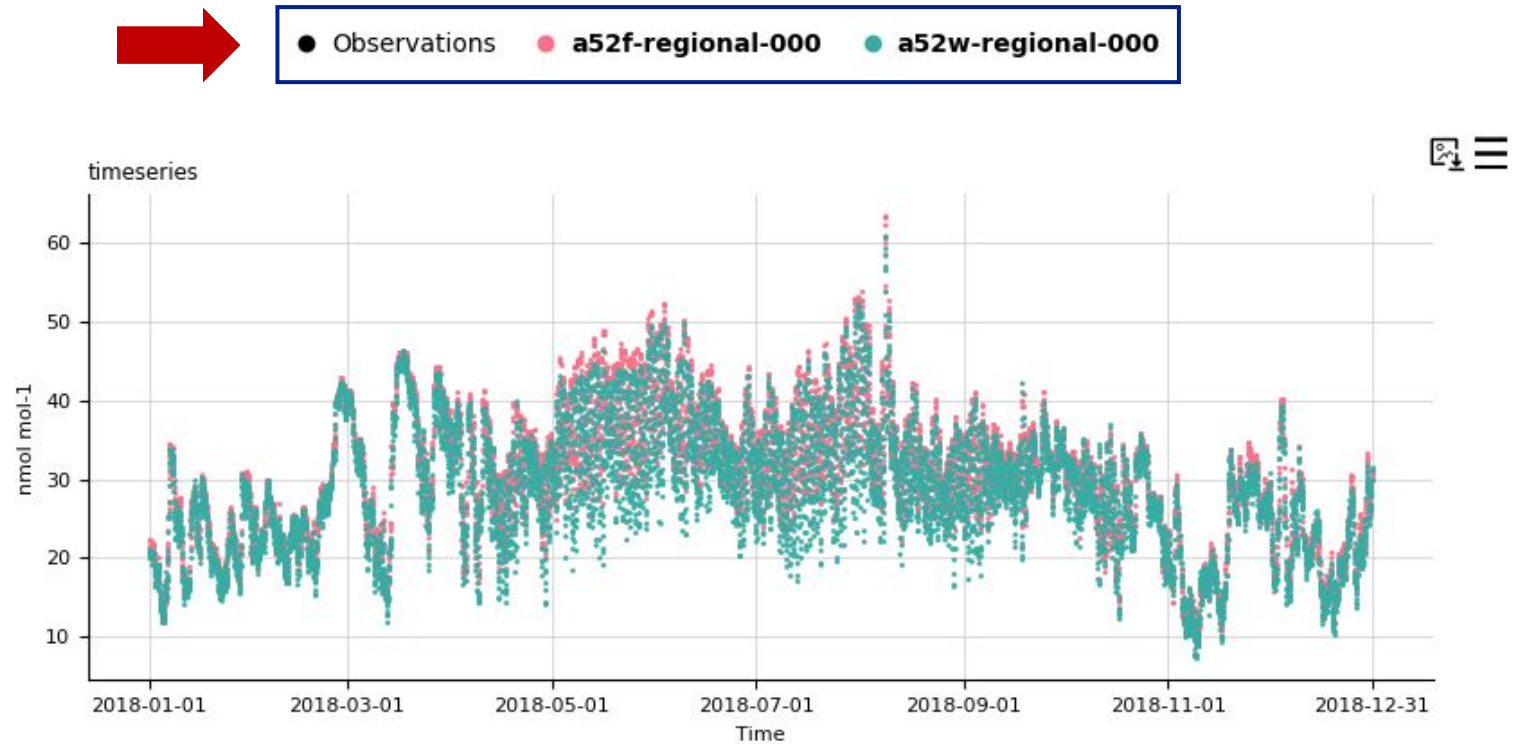
Example of lasso selection

Legend picking

Clicking on the legend labels will remove or add data to each of the plots

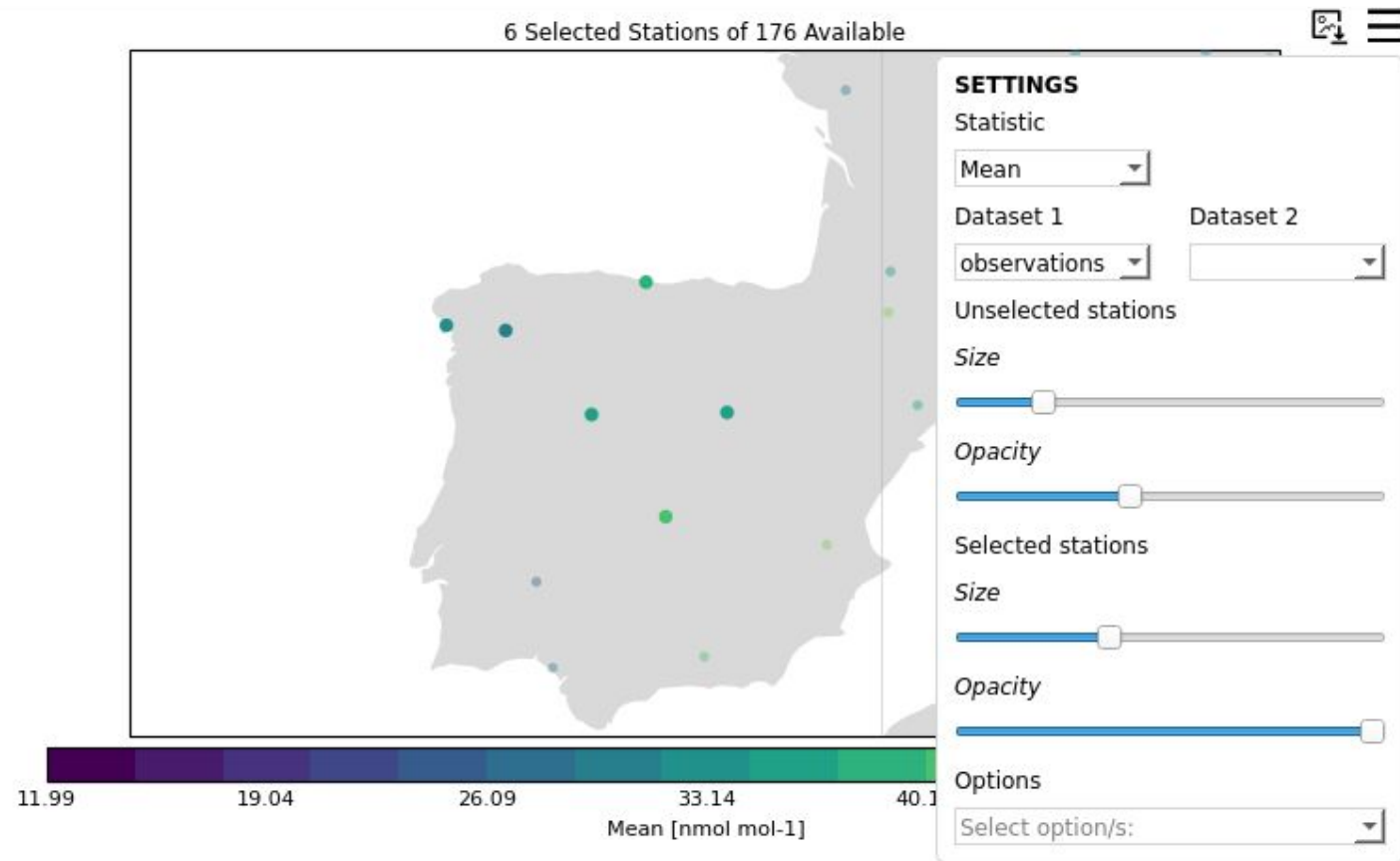
Bold = Visible

Roman = Invisible



Customise the plot style

The style of the plots can be edited by clicking on the **burger menus** and changing the settings.



Choosing the statistics

The statistics in the statsummary can be updated from the burger menu.

statsummary

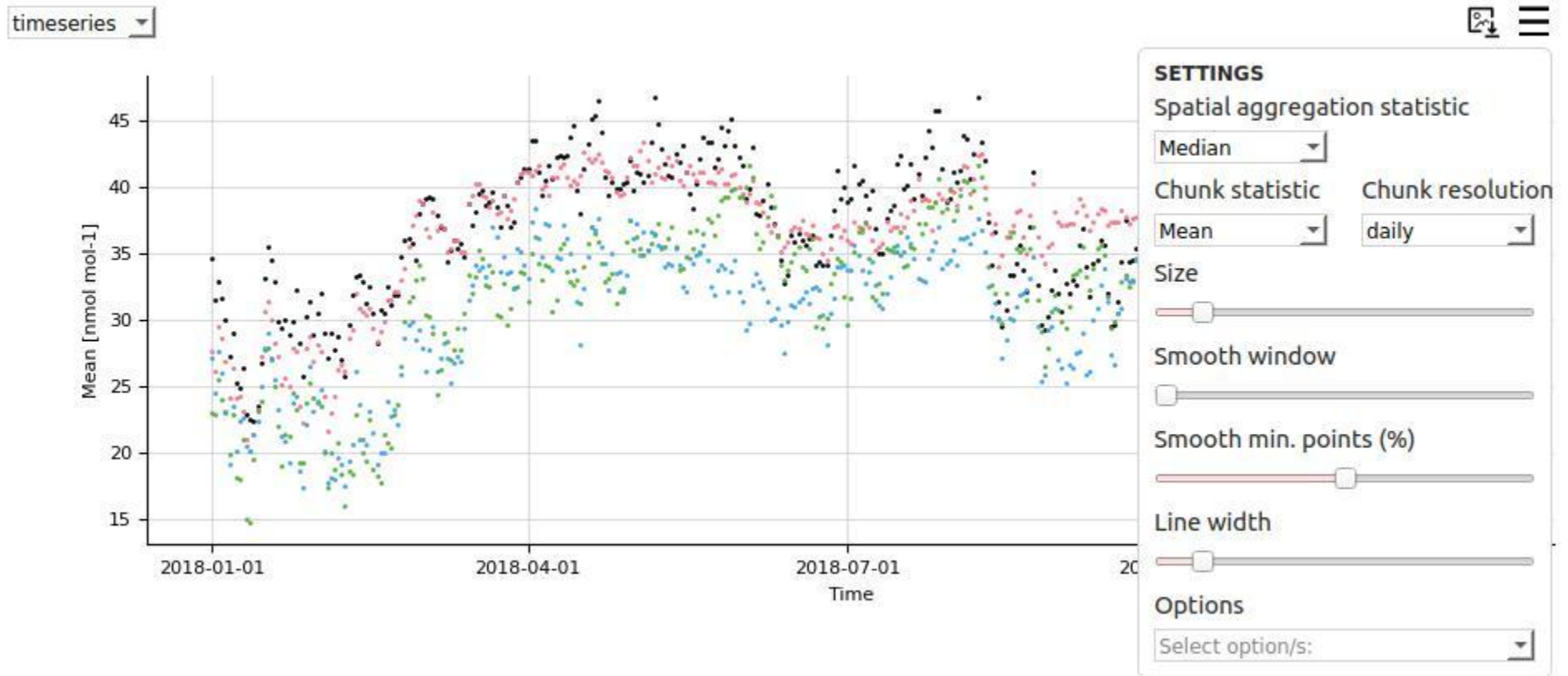


SETTINGS
Periodic cycle: None
Statistic: , Median, p95
Periodic mode: Independent
Periodic aggregation: Mean
Options: Select option/s:

	Mean	StdDev	p5				
Observations	34.01	13.98	10.96				
cams61_camsra_ph2-eu-000	30.21	13.27	7.40				
cams61_chimere_ph2-eu-000	36.24	9.37	20.07	36.76	50.57	100.98	79.17
cams61_dehm_ph2-eu-000	42.40	10.27	24.92	42.74	58.55	117.41	79.17

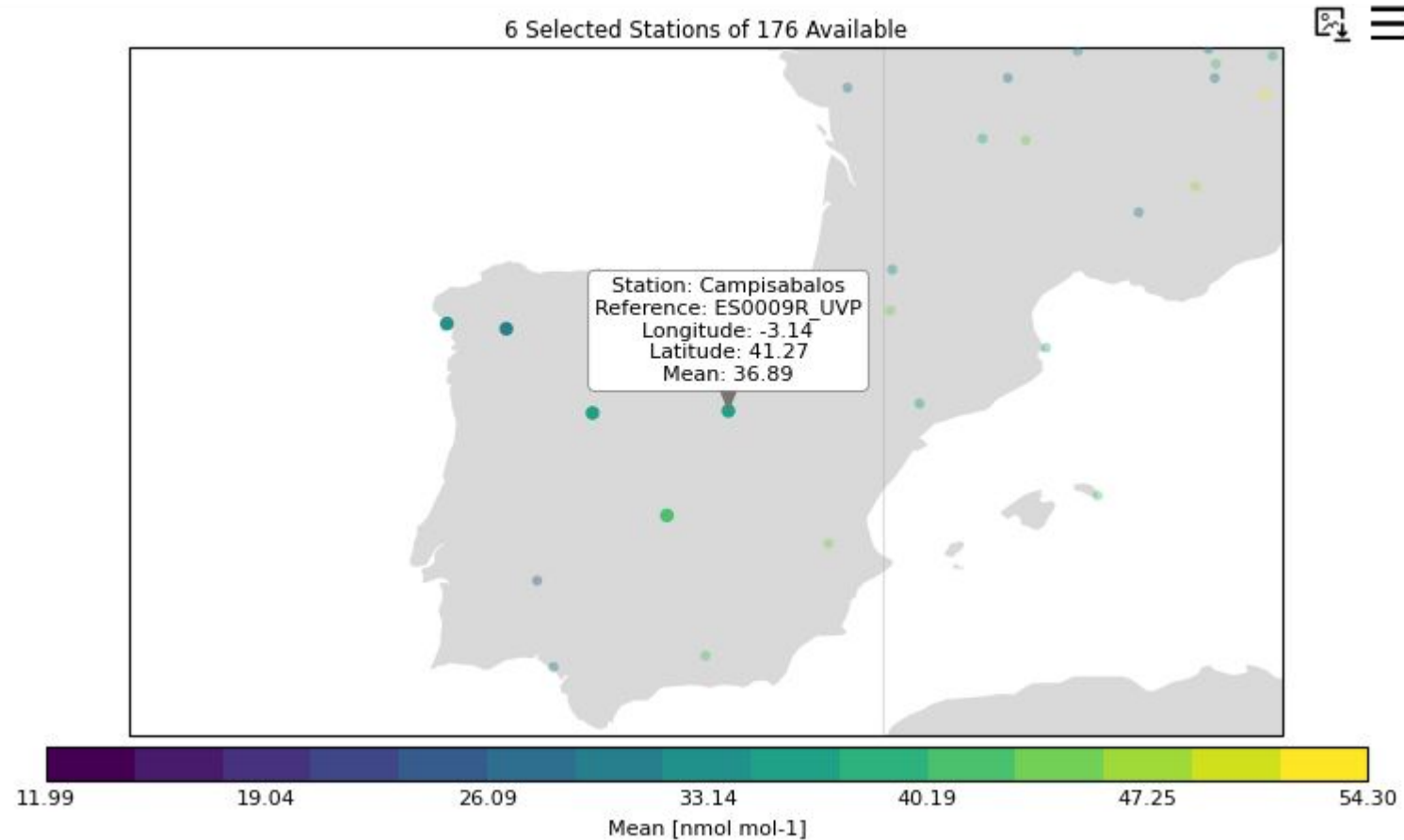
Choosing the statistics

The statistics in other plots can also be updated from the burger menus.



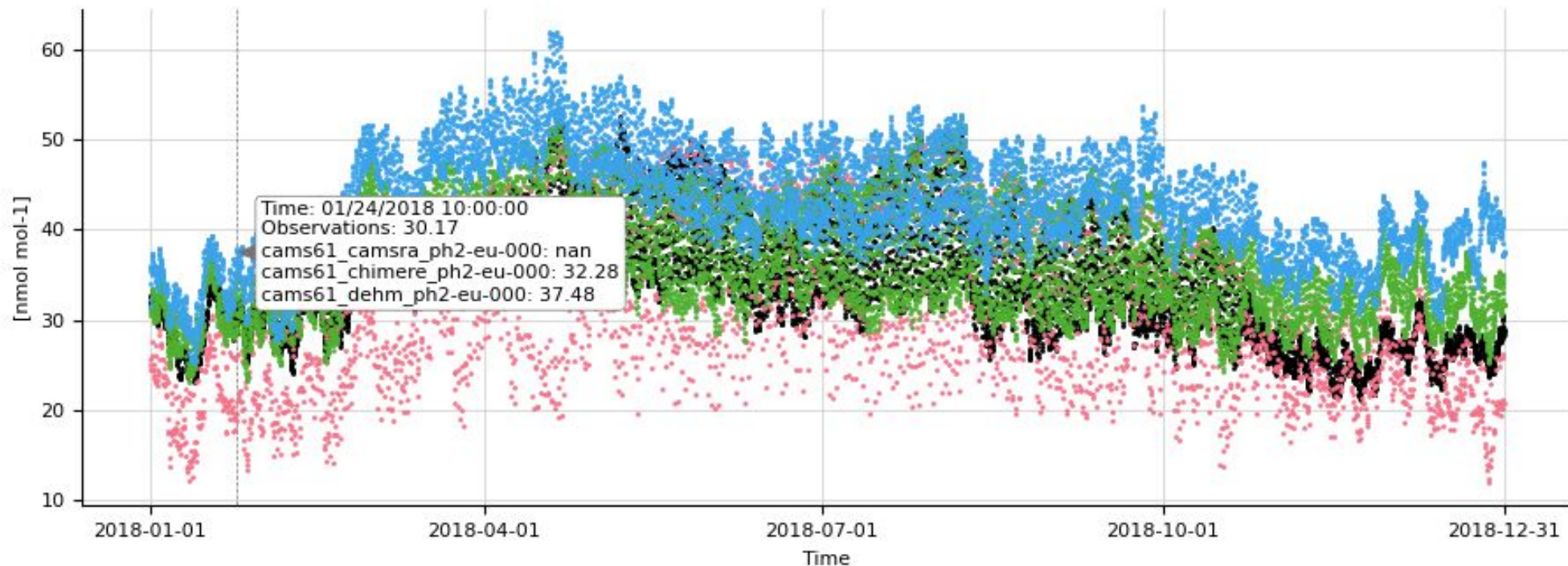
Information on hover

We can see the stations details and data by hovering on the map.



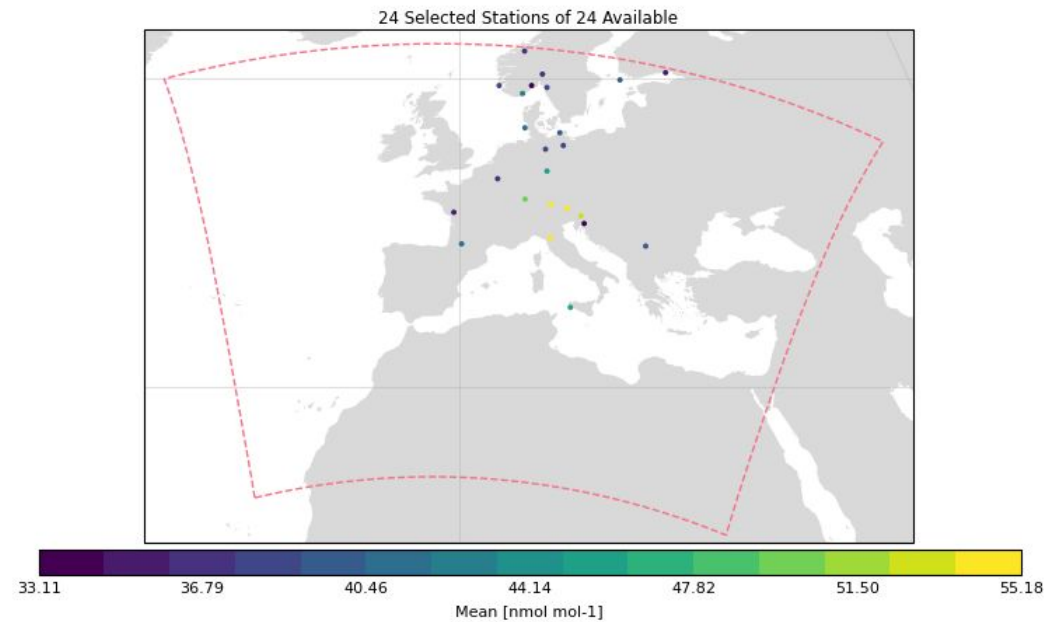
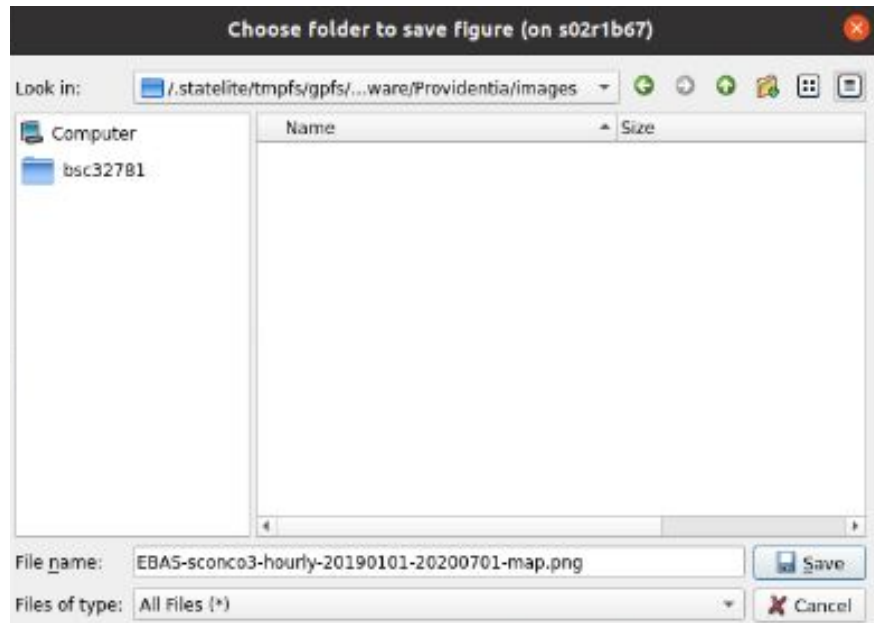
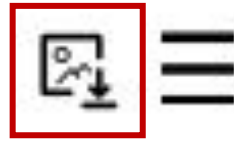
Information on hover

We can also check the values of each dataset by hovering on the other plot types.



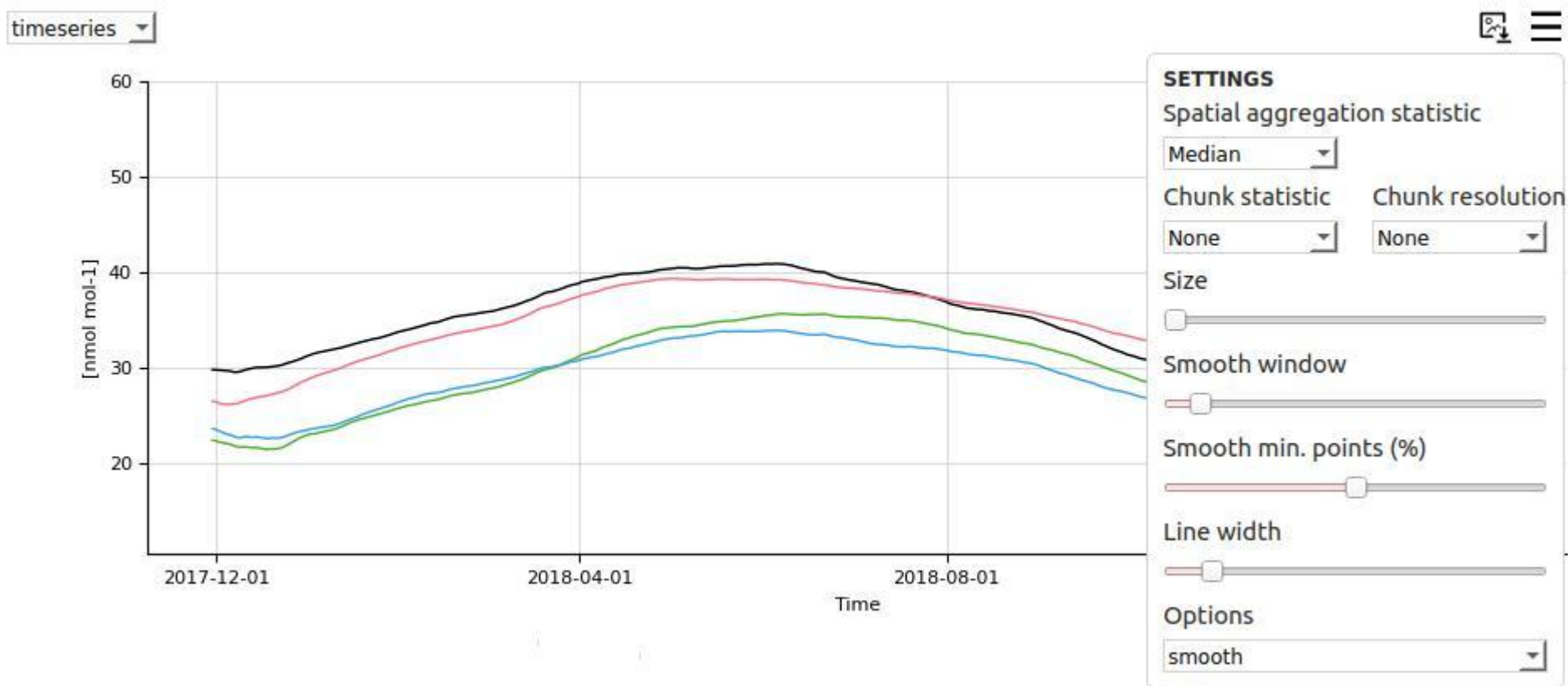
Saving plot figures

Saving plot figures is now possible by clicking on the image icons next to the burger menus



Smoothing

It is possible to add a smooth line in the timeseries plot by editing the smooth window, the data points can be then hidden by reducing their size to 0.



Dashboard specific configuration options

Variable	Purpose
active_dashboard_plots	Defines the plots that will be active in the dashboard once it is launched (e.g. timeseries, periodic-violin, scatter, distribution).

Advanced features



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Filtering



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Metadata filtering

The most common type of data filtering is by metadata, e.g. for **country**, **longitude**, **altitude** etc.

Metadata can either be **numerical** or **text** and the method for filtering for these slightly varies, as will be described.

One common question is what metadata available variables exist for the loaded data that can be filtered by. These can be seen empirically by looking through the sub-menus under the **META** button on the dashboard, with all available variables are organised into 5 categories: **station position**, **station classifications**, **station miscellaneous**, **globally gridded classifications** and **measurement process information**.

Metadata filtering: Numerical data

Dashboard

The dashboard interface includes several sections for configuring filters. At the top, a 'Filters' section has input fields for 'Bounds' (0.0 to 20.0) and buttons for '% REP', 'PERIOD', 'META' (circled in red), 'RESET', and 'FILTER' (circled in red). Below this, a section titled 'Select metadata type to filter stations by' lists options: 'STATION POSITION' (circled in red), 'STATION CLASSIFICATIONS', 'STATION MISCELLANEOUS', 'GLOBALLY GRIDDED CLASSIFICATIONS', and 'MEASUREMENT PROCESS INFORMATION'. The 'Filter stations by measurement position' section contains 'HOME' and 'PREVIOUS' buttons, followed by a table for setting numerical ranges and checkboxes.

	Min	Max	A
latitude	30	72	☒
longitude	511465454	870803833	☐
altitude	1.0	3578.0	☐
sampling_height	0.0	50.0	☐
measurement_altitude	3.0	3578.0	☐

At the bottom, another 'Filters' section is visible, identical to the top one, with the 'META' and 'FILTER' buttons circled in red.

Configuration file

latitude = 30, 72

Library

```
provi.filter('latitude', lower=30, upper=72)
```

Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#numerical-metadata>

Metadata filtering: Text metadata

Dashboard

Filters

Bounds

1

Select metadata type to filter stations by

2

Filter stations by unique country metadata

	K	R
Antarctica	☐	☐
Austria	☒	☐
Belgium	☐	☐
Bermuda	☐	☐
Bulgaria	☐	☐
Cape Verde	☐	☐
Chile	☐	☐
Cyprus	☐	☐
Czech Republic	☐	☐
Denmark	☐	☐
Estonia	☐	☐
Finland	☐	☐
France	☐	☒
Germany	☐	☒

3

Filters

Bounds

4

Configuration file

```
country = keep: Austria  
country = remove: Germany, France
```

Library

```
provi.filter("country", keep="Austria")  
provi.filter("country", remove=["Germany", "France"])
```

Multispecies filtering

Multispecies filtering refers to the ability to **filter loaded species data by the values of another species**.

For example, when performing investigations of dust in the atmosphere it is a common practice to filter the AOD by the Angstrom exponent, to isolate values associated with dust.

Dashboard

Data Selection

v3_lev1.5	aod	od550aero	QA	MODS
monthly	20100101	20110101	FLAGS	MULTI
				READ

Select Network/s and Specie/s to Filter by

ADD ROW

Network	Matrix	Species	Min	Max	Fill value	A
AERONET_v3_lev1.5	aod	140-870aero	>0.6	:	nan	✓

Data Selection

v3_lev1.5	aod	od550aero	QA	MODS
monthly	20100101	20110101	FLAGS	MULTI
				READ

Configuration file

```
network = AERONET_v3_lev1.5
species = od550aero
filter_species = AERONET_v3_lev1.5:ae440-870aero (>0.6, :, nan)
spatial_colocation = True
```

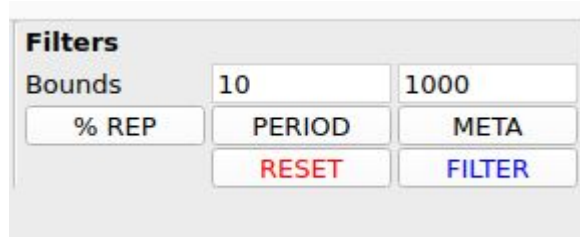
Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#multispecies-filtering>

Bounds

Often it is desired to remove values which exceed certain extreme bounds, as it is known that data should appear at such extremes. These bounds will be **by default active in Providentia**, with extreme bounds associated with a given species taken from definitions in GHOST.

Dashboard



Filters		
Bounds	10	1000
% REP	PERIOD	META
	RESET	FILTER

```
'extreme_lower_limit':0.0, 'extreme_upper_limit':400.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':1200.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':600.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':3000.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':30000.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':50000.0,
```

Configuration file

```
lower_bound = 10  
upper_bound = 1000
```

Link to [GHOST Standards](https://providentia.readthedocs.io/en/latest/Filtering.html#bounds)

Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#bounds>

Representativity

Observations often have missing data. If you compare them directly with complete model data, the results can be biased. Representativity filtering helps by only using time periods that have enough data.

You can either apply the representativity filter for an **entire time range** (**all_representativity_percent**) or by **periods**, which are: **day**, **month** and **year**.

As you change the temporal resolution of the data to be coarser, some of these fields will become unavailable. For example when using monthly data, you can not filter by daily representativity.

Reference: <https://providentia.readthedocs.io/en/latest/Filtering.html#available-filters>

Available representativity fields

By default they are 0:

Native representativity filters

hourly_native_representativity_percent = 0

daily_native_representativity_percent = 0

monthly_native_representativity_percent = 0

annual_native_representativity_percent = 0

Averaged representativity filters

daily_representativity_percent = 0

monthly_representativity_percent = 0

annual_representativity_percent = 0

all_representativity_percent = 0

Representativity

Dashboard

Filters

Bounds	0.0	20.0
% REP	PERIOD	META
	RESET	FILTER

Select Minimum Required % Data Representativity

Native

Monthly

Annual

Averaged

Annual

All

Filters

Bounds	0.0	20.0
% REP	PERIOD	META
	RESET	FILTER

Configuration file

```
monthly_native_representativity_percent = 20
```

Library

```
provi.filter('monthly_native_representativity_percent',  
limit=20)
```

Periods

It is often desired to select or remove certain periods over a time range, for example just keep daytime data or remove summertime data.

Providentia gives an easy way to filter in such a way using the **period** variable when using GHOST data.

The available period fields that that can be selected are: **Daytime, Nighttime, Weekday, Weekend, Spring, Summer, Autumn and Winter.**

Some fields depend on the current time resolution of the data. For example, you cannot select daytime values when monthly data is loaded.

Reference: <https://providentia.readthedocs.io/en/latest/Filtering.html#available-filters>

Periods

Dashboard

Filters

Bounds

Select Data Periods

	K	R
Daytime	☐	☒
Nighttime	☐	☐
Weekday	☐	☐
Weekend	☐	☐
Spring	☐	☐
Summer	☐	☐
Autumn	☐	☐
Winter	☒	☐

Configuration file

```
period = keep: Winter  
period = remove: Daytime
```

Library

```
provi.filter('period', keep='Winter', remove='Daytime')
```

QA



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GHOST QA

GHOST QA refers to quality assurance flags which come from checks designed in GHOST to inform on the quality of data, and can be used to filter measurements.

These can be set via the configuration file either by code or name as follows:

qa = 0, 1, 2

qa = Missing Measurement, Infinite Value, Negative Measurement

QA flags can also be added to, or subtracted from as follows (again, either code or name):

add_qa = Missing Measurement, Negative Measurement

subtract_qa = 0, 2

On the dashboard the GHOST QA flags can be set interactively via the **QA** button.

See [here](#) for specific GHOST QA flag definitions and codes.

QA flags

Select standardised quality assurance flags to filter by

Select AllClear AllSelect Default

Missing Measurement	☒	Above Preferential Upper Limit of Detection	☒
Infinite Value	☒	Insufficient Measurement Resolution - Documented	☐
Negative Measurement	☒	Insufficient Measurement Resolution - Reported	☐
Zero Measurement	☐	Insufficient Measurement Resolution - Preferential	☒
Not Maximum Data Quality Level	☐	Insufficient Measurement Resolution - Empirical	☒
Preliminary Data	☐	Persistent Recurring Values - 5/6	☐
Invalid Data Provider Flags - GHOST Decreased	☒	Persistent Recurring Values - 9/12	☐
Invalid Data Provider Flags - Network Decreased	☐	Persistent Recurring Values - 16/24	☐
No Valid Data to Average	☒	Monthly Fractional Unique Values <= 1%	☐
Methodology Not Mapped	☐	Monthly Fractional Unique Values <= 5%	☐
Assumed Primary Sampling	☐	Monthly Fractional Unique Values <= 10%	☐
Assumed Sample Preparation	☐	Monthly Fractional Unique Values <= 30%	☐
Assumed Measurement Methodology	☐	Monthly Fractional Unique Values <= 50%	☐
Unknown Primary Sampling Type	☐	Monthly Fractional Unique Values <= 70%	☐
Unknown Primary Sampling Instrument	☐	Monthly Fractional Unique Values <= 90%	☐
Unknown Sample Preparation Type	☐	Data Outlier - Exceeds Scientifically Decreased Lower/Upper Limit	☒
Unknown Sample Preparation Technique	☐	Data Outlier - Monthly Median Exceeds Scientifically Decreased Upper Limit	☒
Unknown Measurement Method	☐	Data Outlier - Network Decreased	☒
Unknown Measuring Instrument	☐	Data Outlier - Manually Decreased	☒
Erroneous Primary Sampling	☒	Possible Data Outlier - Monthly Adjusted Boxplot	☐
Erroneous Sample Preparation	☒	Probable Data Outlier - Monthly Adjusted Boxplot	☒
Erroneous Measurement Methodology	☒	Monthly Distribution Consistency - Zone 1	☐
Invalid QA Measurement Methodology	☐	Monthly Distribution Consistency - Zone 2	☐
Corrected Parameter	☐	Monthly Distribution Consistency - Zone 3	☐
Sample Gas Volume - Network Standard	☐	Monthly Distribution Consistency - Zone 4	☐
Sample Gas Volume - Unknown	☐	Monthly Distribution Consistency - Zone 5	☐
Unit Conversion - Network Standard Sample Gas Volume Assumption	☐	Monthly Distribution Consistency - Zone 6	☐
Unit Conversion - Educated Guess Sample Gas Volume Assumption	☐	Monthly Distribution Consistency - Zone 7	☐
Station Position Doubt - DEM Decreased	☐	Monthly Distribution Consistency - Zone 8	☐
Station Position Doubt - Manually Decreased	☐	Monthly Distribution Consistency - Zone 9	☐
Data Product	☐	Monthly Distribution Consistency - Zone 10	☐
Insufficient Data to Calculate Data Product	☐	Monthly Distribution Consistency - Unclassified	☐
Local Precipitation	☐	Systematic Inconsistent Monthly Distributions - 2/3 Months >= Zone 6	☐
Local Extreme Weather	☐	Systematic Inconsistent Monthly Distributions - 4/6 Months >= Zone 6	☒
Local Atmospheric Obscuration	☐	Systematic Inconsistent Monthly Distributions - 8/12 Months >= Zone 6	☒
Local Contamination	☐		
Local Exceptional Event	☐		
Non-Integer Local Timezone (relative to UTC)	☐		
Timezone Doubt	☐		
Below Documented Lower Limit of Detection	☐		
Below Reported Lower Limit of Detection	☐		
Below Preferential Lower Limit of Detection	☒		
Above Documented Upper Limit of Detection	☐		
Above Reported Upper Limit of Detection	☐		

Network QA

Network QA refers to standardised quality assurance flags which come from networks processed in GHOST which inform on the quality of data, and can be used to filter measurements.

These can be set via the configuration file either by code or name as follows:

flags = 0, 1, 2

flags = Valid Data, Preliminary Data, Missing Data

Network flags can also be added to, or subtracted from as follows (again, either code or name):

add_flags = Valid Data, Missing Data

subtract_flags = 0, 2

On the dashboard the network flags can be set interactively via the **FLAGS** button.

See [here](#) for specific Network QA flag definitions and codes.

Network flags

Select standardised data reporter provided flags to filter by

Select All Clear All Select Default

☐ Valid Data	☐ Calibration	☐ Freezing Rain - Moderate	☐ Diamond Dust - Moderate	☐ Volcanic Eruptions
☐ Preliminary Data	☐ Accuracy Check	☐ Freezing Rain - Heavy	☐ Diamond Dust - Heavy	☐ Wildfire
☐ Missing Data	☐ Blank Check	☐ Freezing Rain Showers - Unspecified Intensity	☐ Glass	☐ Chemical/Industrial Accident
☐ Invalid Data - Unspecified	☐ Detection Limits Check	☐ Freezing Rain Showers - Light	☐ Haze	☐ Cleanup After a Major Disaster
☐ Un-Flagged Data	☐ Precision Check	☐ Freezing Rain Showers - Moderate	☐ Thunderstorm	☐ Demolition
☐ Estimated Data - Unspecified	☐ Response Time Check	☐ Freezing Rain Showers - Heavy	☐ Funnel Clouds	☐ Fireworks
☐ Estimated Data - Measured Negative Value	☐ Span Check	☐ Snow - Unspecified Intensity	☐ Squalls	☐ Infrequent Large Gathering
☐ Estimated Data - No Value Detected	☐ Zero Check	☐ Snow - Light	☐ Tropical Cyclone (Cyclone/Hurricane/Typhoon)	☐ Tornado Act
☐ Estimated Data - Value Below Detection Limit	☐ Instrumental Inspection	☐ Snow - Moderate	☐ Duststorm	☐ Visibility Distance Unlimited
☐ Estimated Data - Value Above Detection Limit	☐ Instrumental Repair	☐ Snow - Heavy	☐ Sandstorm	☐ Ceiling Height Unlimited
☐ Estimated Data - Value Substituted From Secondary Monitor	☐ Quality Control Audit	☐ Snow Showers - Unspecified Intensity	☐ Dustified Whirls	☐ AERONET L2 ADD
☐ Estimated Data - Multiple Parameters Aggregated	☐ Data Formatting/Processing Issue	☐ Snow Showers - Light	☐ High Winds	☐ AERONET L2 WITHOUT ADD
☐ Estimated Irregular Data - Unspecified	☐ Connected Data Formatting/Processing Issue	☐ Snow Showers - Moderate	☐ No Atmospheric Observation	
☐ Data Does Not Meet Internal Network Quality Control Criteria	☐ Aggregation/Representation Issue - Unspecified	☐ Snow Showers - Heavy	☐ Atmospheric Obscuration - Unknown	
☐ High Variability of Data	☐ Data Window Completeness < 50%	☐ Hail - Unspecified Intensity	☐ Dust	
☐ Irregular Data Manually Screened and Accepted	☐ Data Window Completeness < 75%	☐ Hail - Light	☐ Blowing Dust	
☐ Negative Value	☐ Data Window Completeness < 85%	☐ Hail - Moderate	☐ Drifting Dust	
☐ No Value Detected	☐ Data Window Completeness < 95%	☐ Hail - Heavy	☐ Sand	
☐ Reasonable/Unreasonable Data	☐ Data Window Completeness < 98%	☐ Hail Showers - Unspecified Intensity	☐ Blowing Sand	
☐ Value Close to Detection Limit	☐ >= 75% of Measurements in Window Below Detection Limit	☐ Hail Showers - Light	☐ Drifting Sand	
☐ Value Below Acceptable Range	☐ >= 50% of Measurements in Window Below Detection Limit	☐ Hail Showers - Moderate	☐ Blowing Snow	
☐ Value Above Acceptable Range	☐ No Significant Weather	☐ Hail Showers - Heavy	☐ Drifting Snow	
☐ Value Below Detection Limit	☐ Precipitation - Unspecified Intensity	☐ Ice Pellets - Unspecified Intensity	☐ Fog	
☐ Value Above Detection Limit	☐ Precipitation - Light	☐ Ice Pellets - Light	☐ Freezing Fog	
☐ Measurement Issue - Unspecified	☐ Precipitation - Moderate	☐ Ice Pellets - Moderate	☐ Ground Fog	
☐ Chemical Issue	☐ Precipitation - Heavy	☐ Ice Pellets - Heavy	☐ Ice Fog	
☐ Erroneous Sampling Operation	☐ Drizzle - Unspecified Intensity	☐ Ice Pellets Showers - Unspecified Intensity	☐ Haze	
☐ Extreme Internal Instrument Meteorological Conditions	☐ Drizzle - Light	☐ Ice Pellets Showers - Light	☐ Mist	
☐ Extreme External Meteorological Conditions	☐ Drizzle - Moderate	☐ Ice Pellets Showers - Moderate	☐ Smog	
☐ Extreme Ambient Laboratory Meteorological Conditions	☐ Drizzle - Heavy	☐ Ice Pellets Showers - Heavy	☐ Sea Spray	
☐ Extreme Sample Transport Conditions	☐ Freezing Drizzle - Unspecified Intensity	☐ Snow Pellets - Unspecified Intensity	☐ Volcanic Ash	
☐ Invalid Flow Rate	☐ Freezing Drizzle - Light	☐ Snow Pellets - Light	☐ No Local Contamination	
☐ Human Error	☐ Freezing Drizzle - Moderate	☐ Snow Pellets - Moderate	☐ Local Contamination - Unspecified	
☐ Matrix Effect	☐ Freezing Drizzle - Heavy	☐ Snow Pellets - Heavy	☐ Agricultural Contamination	
☐ No Technician	☐ Rain - Unspecified Intensity	☐ Snow Pellets Showers - Unspecified Intensity	☐ Bird-Dropping Contamination	
☐ Operational Maintenance Check Issue	☐ Rain - Light	☐ Snow Pellets Showers - Light	☐ Construction Contamination	
☐ Physical Issue With Filter	☐ Rain - Moderate	☐ Snow Pellets Showers - Moderate	☐ Industrial Contamination	
☐ Power Failure	☐ Rain - Heavy	☐ Snow Pellets Showers - Heavy	☐ Insect Contamination	
☐ Sample Diluted for Analysis	☐ Rain Showers - Unspecified Intensity	☐ Snow Drains - Unspecified Intensity	☐ Internal Laboratory/Instrument Contamination	
☐ Unmeasured Key Meteorological Parameter	☐ Rain Showers - Light	☐ Snow Drains - Light	☐ Pollen/Leaf Contamination	
☐ Sample Not Analyzed	☐ Rain Showers - Moderate	☐ Snow Drains - Moderate	☐ Traffic Contamination	
☐ Operational Maintenance - Unspecified	☐ Rain Showers - Heavy	☐ Snow Drains - Heavy	☐ Exceptional Event - Unspecified	
	☐ Freezing Rain - Unspecified Intensity	☐ Diamond Dust - Unspecified Intensity	☐ Seismic Activity	
	☐ Freezing Rain - Light	☐ Diamond Dust - Light	☐ Biogenic Aerosol Situation	

Calibration



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Calibration factor

When a model has a known bias and you want to correct it, you can apply what we call the calibration factor by defining it in your configuration file.

- To add:

```
calibration_factor = +10
```

- To multiply:

```
calibration_factor = *10
```

- To subtract:

```
calibration_factor = -10
```

- To divide:

```
calibration_factor = /10
```

Calibration factor for multiple models/species

The calibration factor can also be defined independently for different models:

```
calibration_factor = a54s-regional-000 (*0.62), a4xf-regional-000 (*0.51)
```

It can also be defined independently for different species:

```
network = EEA_AQ_eReporting  
species = pm2p5, pm10  
calibration_factor = a54s-regional-000 (*0.62, *0.4),  
a4xf-regional-000 (*0.52, *0.9)
```

Colocation



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Spatial colocation

When loading more than one species (using **multispecies filtering** or in **multispecies plots** in the report and library modes) you may want to ensure that the available stations measure **data for all species** that are to be loaded. To do this, we need to activate spatial colocation.

After activating spatial colocation, any stations that do not have valid data for any of the loaded species are dropped.

Spatial colocation can be set in the configuration file by setting a boolean as follows (default: True):

```
spatial_colocation = False
```

Reference:

<https://providentia.readthedocs.io/en/latest/Colocation.html>

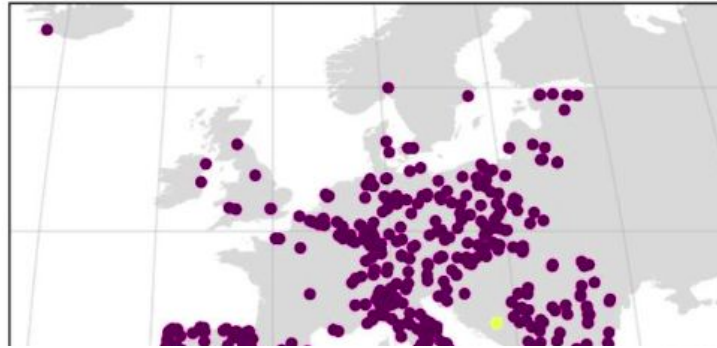
Surface O₃ (mean)
1731 stations

observations
CAMS2_40 (1731 stations)



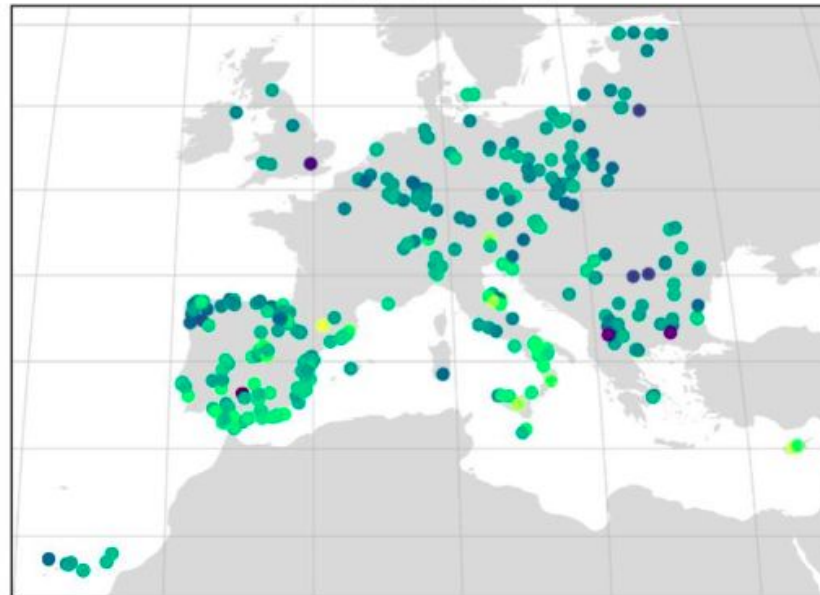
Surface CO (mean)
635 stations

observations
CAMS2_40 (635 stations)



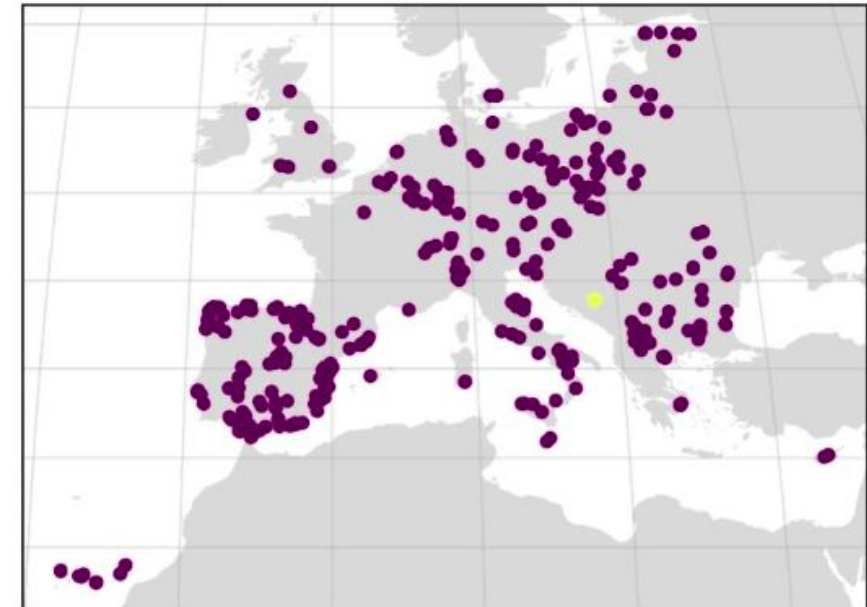
Surface O₃ (mean)
366 stations

observations
CAMS2_40 (366 stations)



Surface CO (mean)
366 stations

observations
CAMS2_40 (366 stations)



Without spatial
collocation

With spatial
collocation

Temporal colocation

Temporal colocation is used to **temporally pair observations and model data**, with any missing measurements in either the observational or model array, imposing missing measurements on the other.

When temporal colocation is active, you will have access to more plot types (scatter, taylor, fairmode-target, and fairmode-statsummary).

Temporal colocation can be set in the configuration file by setting a boolean as follows (default: True):

```
temporal_colocation = False
```

Reference: <https://providentia.readthedocs.io/en/latest/Colocation.html>

Without temporal collocation

Data Selection
EBAS gas sconco3 QA MODS
hourly 20180101 20190101 FLAGS MULTI
READ

Filters
Bounds 0.0 400.0
% REP PERIOD META
RESET FILTER

Statistics
Mode Temporal[S]
Aggregation Median

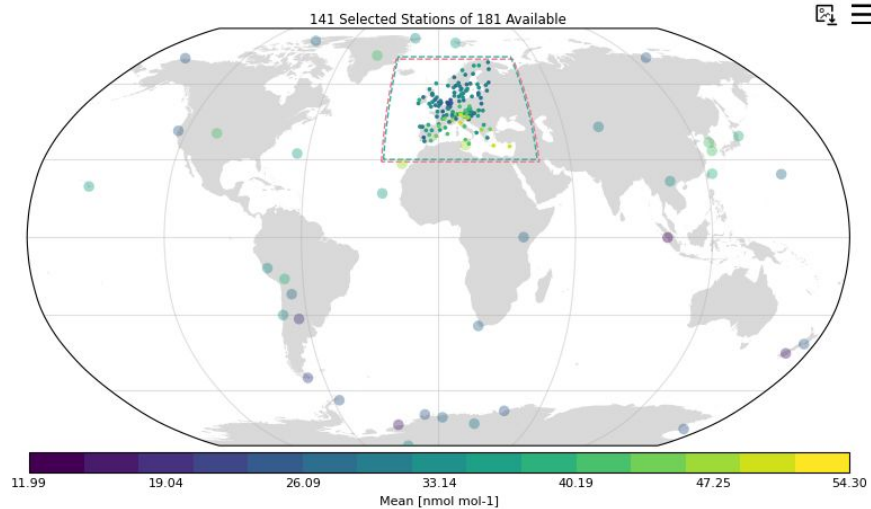
Colocation
☐ Temporal

Resampling
None

Site Selection
☒ All
☐ Intersect
☐ Extent

↓ 📄 🌐 🔍 ↶ ↷ 📏 🔄 📌

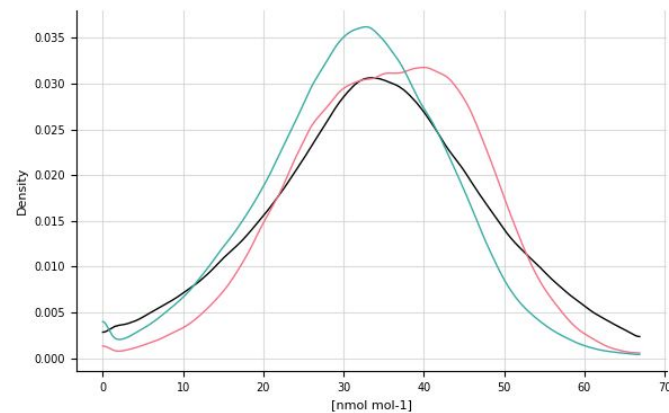
(x, y) = (8.1, 0.0303)



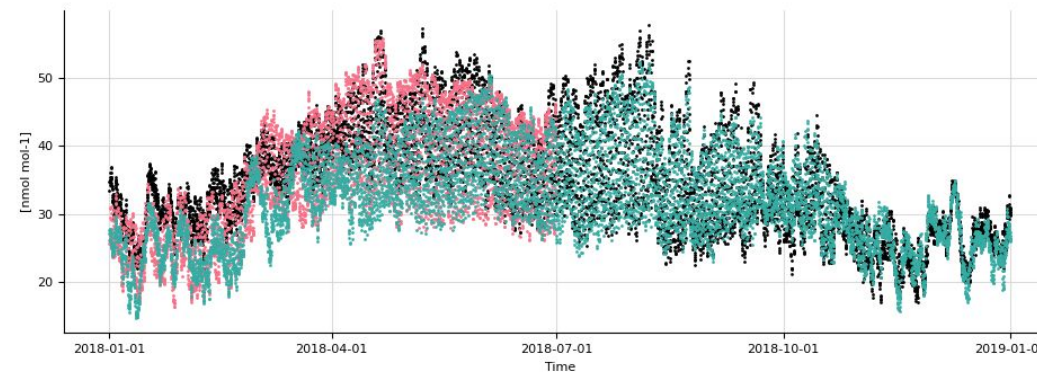
statsumma

	Mean	StdDev	p5	Median	p95
observations	32.85	11.89	13.24	32.80	54.20
cams61_match_ph2-eu-000	34.93	10.53	17.43	34.96	52.54
cams61_monarch_ph3-eu-000	30.71	9.98	14.36	30.85	47.83

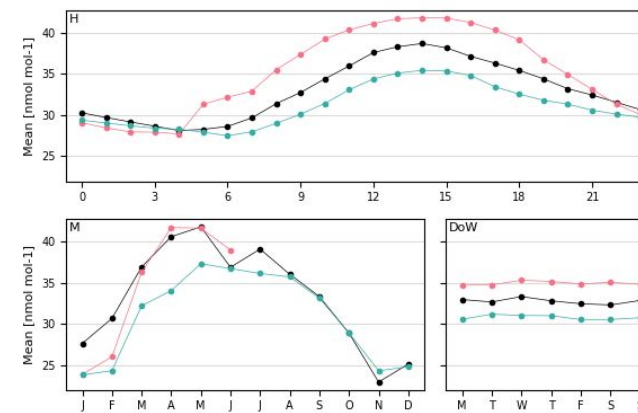
distribution



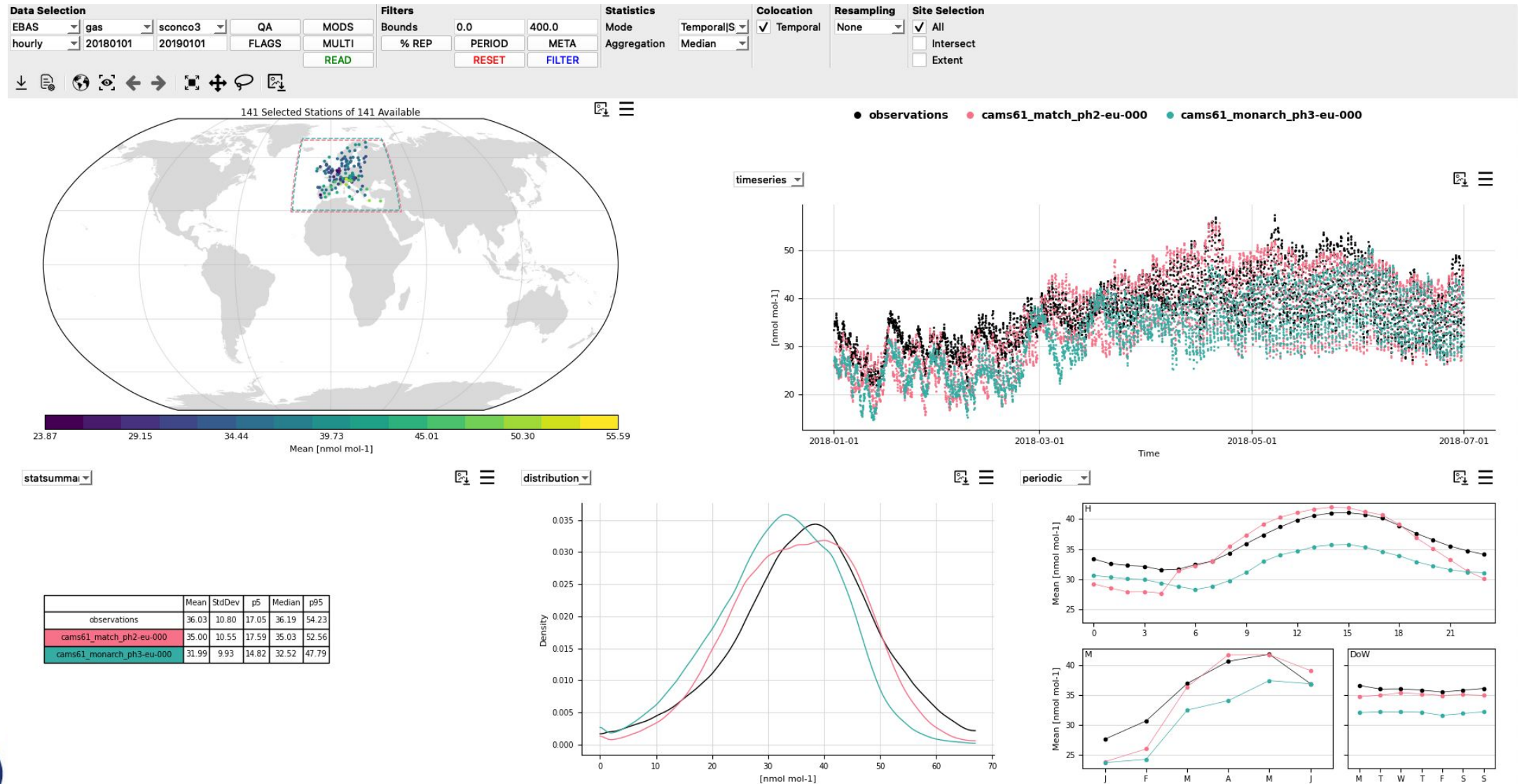
timeseries



periodic



With temporal collocation



Resampling



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Resampling

The resampling section allows the plotted data to be **resampled to a coarser temporal resolution** (by the mean), e.g. hourly to daily, by clicking on a resolution from the drop-down menu. When a resolution is selected, it updates the plots and statistics across the dashboard.

From the configuration file, we need to assign a temporal resolution to the key **resampling_resolution**:

```
resampling_resolution = daily
```

Providentia

Data Selection

EBAS

sconco3

QA

EXPS

hourly

20150101

20200101

FLAGS

MULTI

READ

Filters

Bounds

% REP

PERIOD

META

RESET

FILTER

Statistics

Mode

Aggregation

Colocation

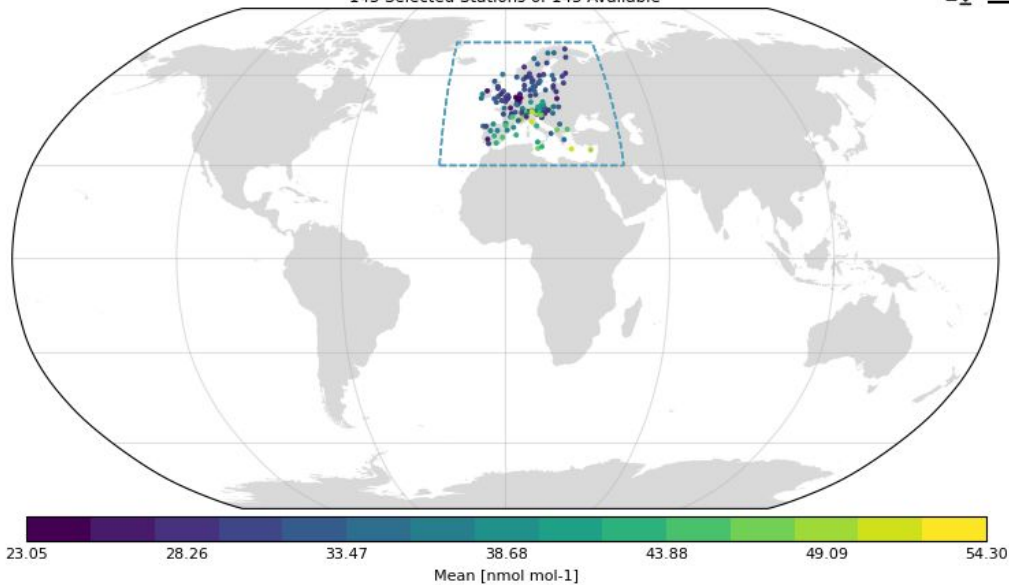
☒ Temporal

Resampling

Site Selection

☒ All
☐ Intersect
☐ Extent


143 Selected Stations of 143 Available

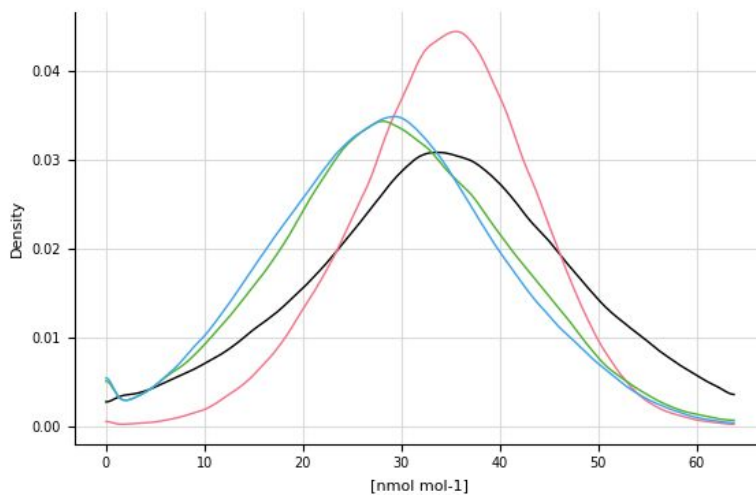


statsumma



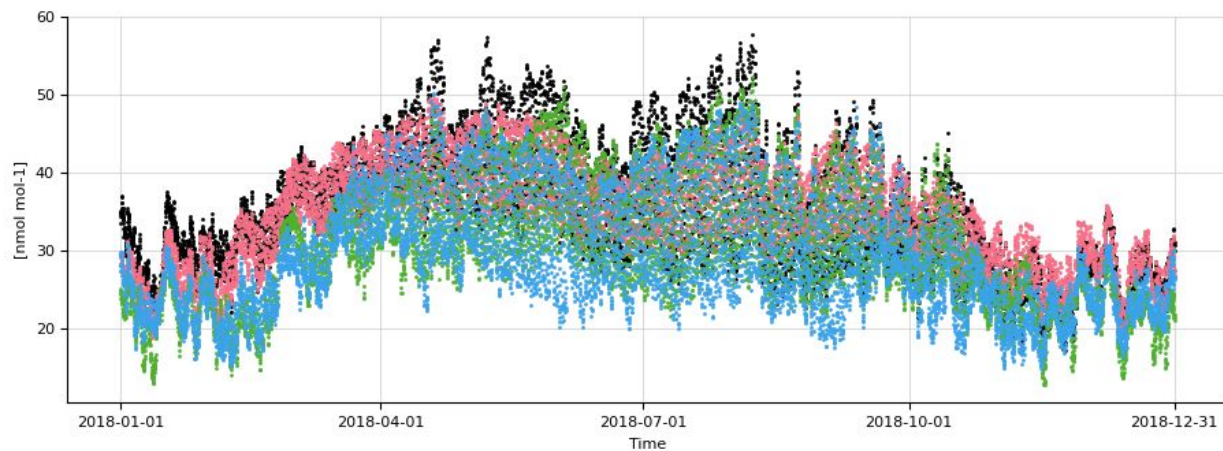
distribution

	Mean	StdDev	p5	Median	p95
observations	32.91	11.84	13.30	32.80	54.46
EMEP	33.96	8.47	18.77	34.14	47.72
MONARCH	27.88	10.57	12.19	27.44	47.20
SILAM	27.41	10.55	11.02	27.29	47.37

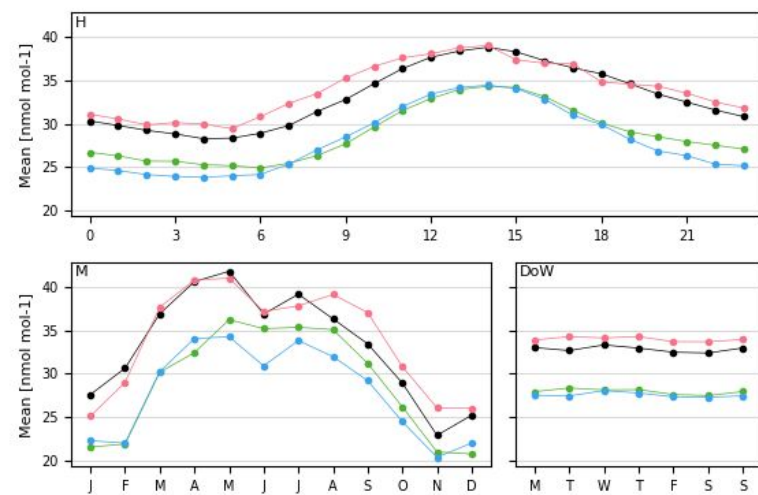


● observations ● MONARCH ● SILAM
● EMEP

timeseries



periodic



Providentia



Data Selection

EBAS

sconco3

EXPS

hourly

20150101

FLAGS

Filters

Bounds

400.0

% REP

RESET

Statistics

Mode

Aggregation

Colocation

☒ Temporal

Resampling

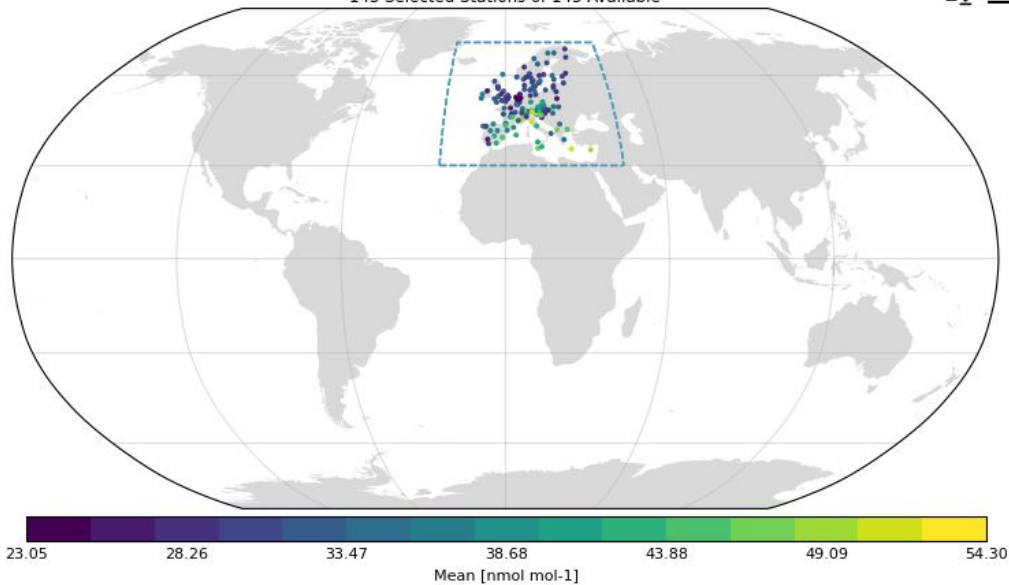
Site Selection

☒ All
☐ Intersect
☐ Extent

(x, y) = (0.256, 0.92)



143 Selected Stations of 143 Available

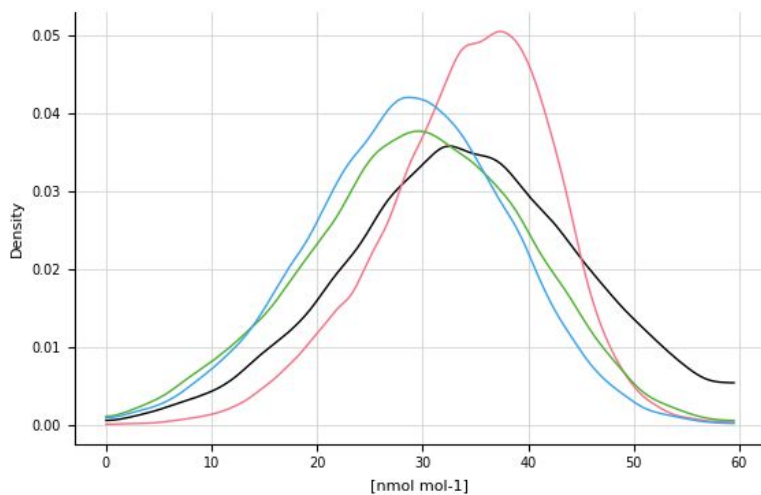


statsumma

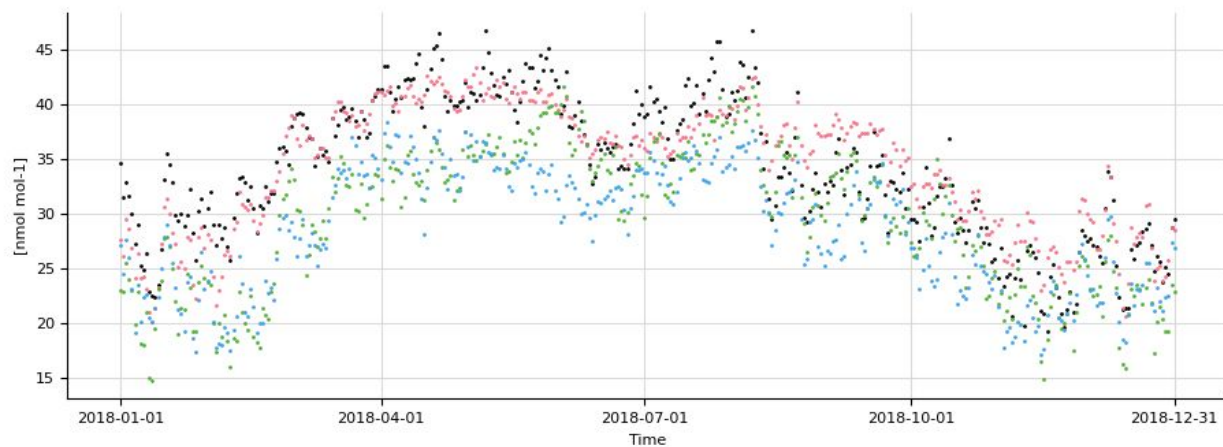


distribution

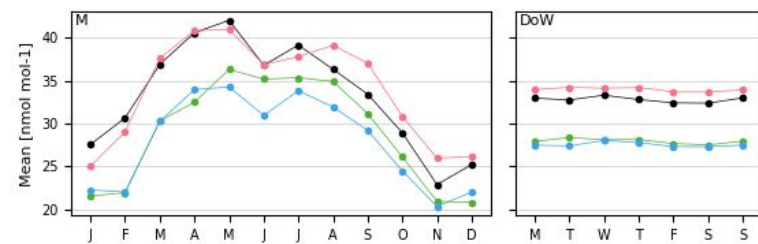
	Mean	StdDev	p5	Median	p95
observations	32.91	9.41	17.20	33.11	48.63
EMEP	33.88	7.28	19.98	34.89	44.72
MONARCH	27.86	8.72	13.73	28.71	42.28
SILAM	27.42	8.15	14.17	27.71	40.97



timeseries



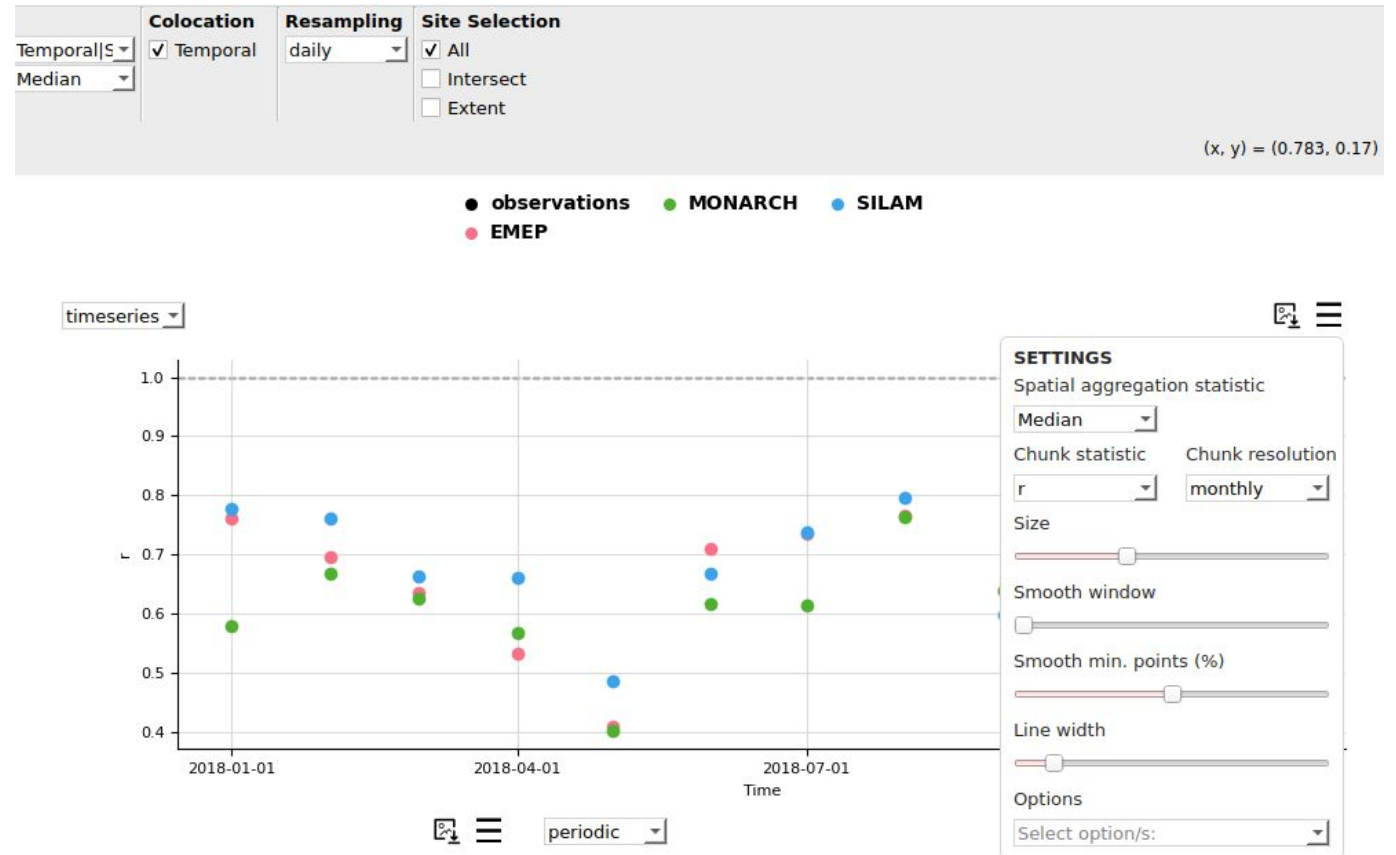
periodic



Timeseries chunking

For the timeseries there is an additional selection for the temporal resolution
(**chunk resolution**) which works on top of the resampling if active.

For each chunk a statistic can be calculated
(**chunk statistic**), allowing to see how that stat evolves over time.



Statistics



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Statistical modes

Most of the plotted data in Providentia depends on certain logical choices made when calculating statistics.

Statistics can be calculated in a variety of ways, and Providentia allows a number of ways to alter the calculation, via 3 statistical modes. These are as follows:

- **Temporal|Spatial (default)**
- **Spatial|Temporal**
- **Flattened**

Statistical modes

The name of each statistical mode relates to how the dimensions of the selected data are reduced to calculate the statistical metrics, e.g. mean, median etc, going from 2D to 0D.

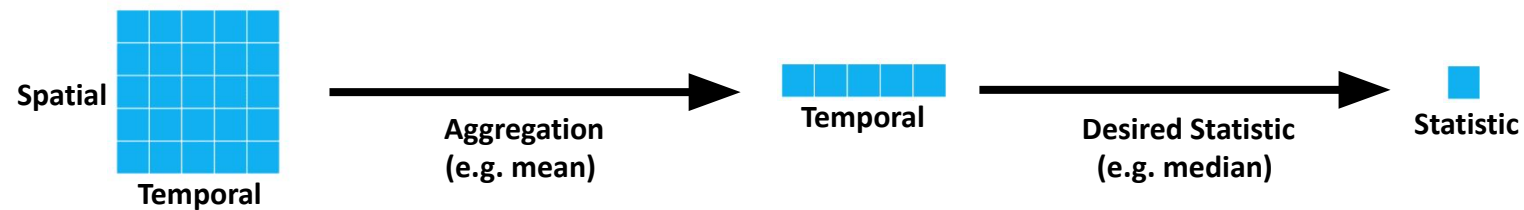
When selecting data across multiple stations, it has 2 dimensions: **Spatial** and **Temporal**.

For **Temporal|Spatial**, aggregation is performed first temporally per station (e.g. median), going to 1D, before then calculating the desired statistic across stations (e.g. p50), going to 0D.

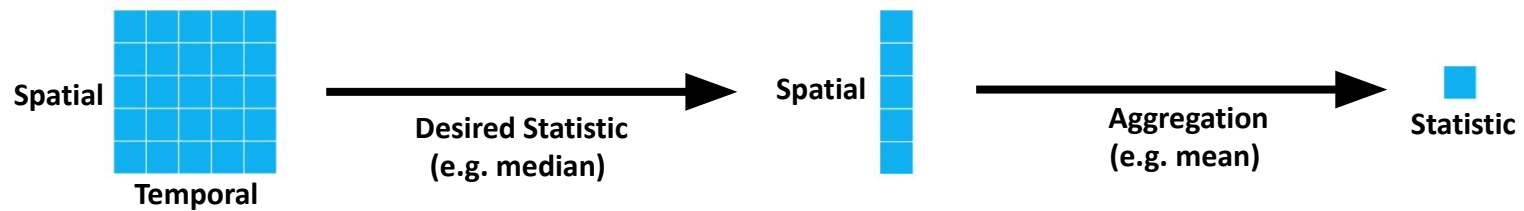
For some plot types the full dimensional reduction is not possible, e.g. map. Therefore, they are locked in certain reduction configurations.

Statistical modes

Spatial | Temporal



Temporal | Spatial



Flattened



2D

1D

0D

Dimensional reduction per plot type

timeseries



Temporal

map



Spatial

1D -
Locked

scatter



Temporal & Spatial

Temporal | Spatial

Flattened



Temporal

Spatial | Temporal

distribution



Temporal & Spatial

Temporal | Spatial

Flattened



Temporal

Spatial | Temporal

1D -
Semi-Locked

boxplot



Temporal & Spatial

Temporal | Spatial

Flattened



Temporal

Spatial | Temporal

periodic-violin



Temporal & Spatial

Temporal | Spatial

Flattened



Temporal

Spatial | Temporal

periodic



Statistic
(per periodic
timestep)

statsummary



Statistics

table



Statistic

heatmap



Statistic

0D -
Unlocked

Statistical modes

On the dashboard, in the statistics menu at the top, the mode and aggregation can be selected via the dropdown menus.

In the configuration files these can be set like so:

statistic_mode = Temporal | Spatial

statistic_aggregation = Median

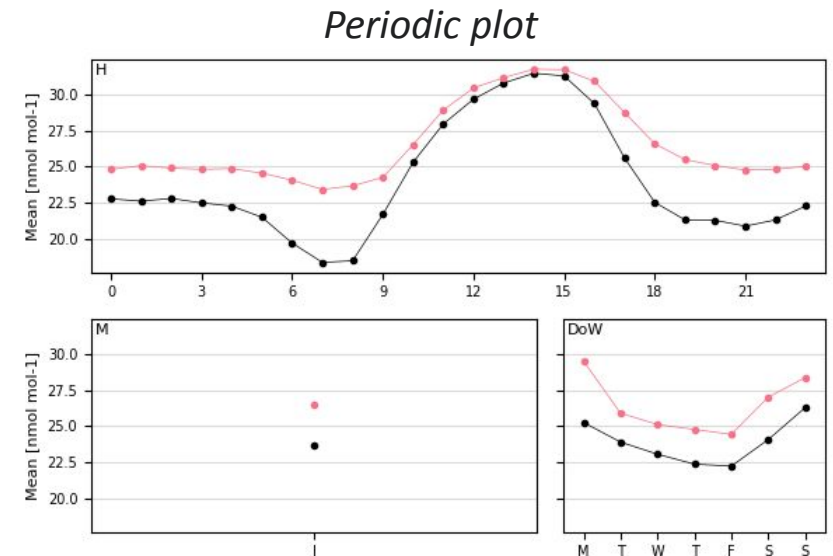
Note: For the Flattened mode, there is no aggregation statistic.

Periodic statistical modes

The periodic plot gives statistical information for grouped data in individual timesteps. Thus it can be seen how each individual timesteps compare for observations vs model/s.

However, statistics can also be calculated which assess the entirety of each periodic cycle, i.e. diurnal, weekly, annual. These statistics are available via the **statsummary**, **table** and **heatmap** plot types.

Note: Only statsummary is available on the dashboard



Periodic statistical modes

There exist 2 modes for calculating these periodic statistics:

- **Independent (default)**
- **Cycle**

Independent works by calculating the desired statistic (e.g. r) per periodic timestep (i.e. as seen in periodic plot), before aggregating across the timesteps (e.g. median).

Cycle works by aggregating the grouped data per timestep (e.g. mean), before then calculating the desired statistic across the timesteps (e.g. MB).

Periodic statistics

On the dashboard, in the plot options of the **statsummary** plot, the periodic statistic mode and aggregation, and periodic statistics can be selected for specific diurnal, weekly and monthly cycles via the dropdown menus.

In the configuration file these can be set like so:

periodic_statistic_mode = Independent

periodic_statistic_aggregation = Median

For non-dashboard modes, the periodic statistic can be set via adding the specific cycle to the statistic, e.g. **Mean-diurnal**, in the **statsummary** section of **settings/plot_characteristics.yaml**.

Forecast



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Forecast options

Forecast models predict the state of the atmosphere in the near future, most typically on the scale of days. Forecast data comes in a different format to most other kinds of model data, with data per day for the subsequent N days.

This data can be visualised in Providentia in different ways, set by the **forecast** variable in the configuration file with the following valid fields:

- **day, dayN**
- **combined, daily**
- **dailyN**

It is possible to compare different models where one has forecast data and the other does not. In these cases the non-forecast data is treated as day 1 of a forecast for comparison purposes.

If no forecast option is set for a model with forecast data, data from the first forecast day will simply be taken to represent the model (i.e. treating the data as a non-forecast type).

Forecast data on the dashboard

When using the dashboard, models that have have interpolated forecast data are automatically detected.

When clicking on the **MODS** button on the menu bar, if the model has forecast data, then when you select the model, an extra drop-down menu will appear to the right. This will allow you to select the forecast option, i.e. combined, daily or day .If no option is set, data from the first forecast day will be taken to represent the model (i.e. treating the data as a non-forecast type).

If a forecast option is selected, another drop-down menu will then appear to the right of that, allowing you to select specific forecast days. If no specific forecast days are selected but a forecast option is, then all forecast days will be utilised.

Select Model/s

Select All Clear All

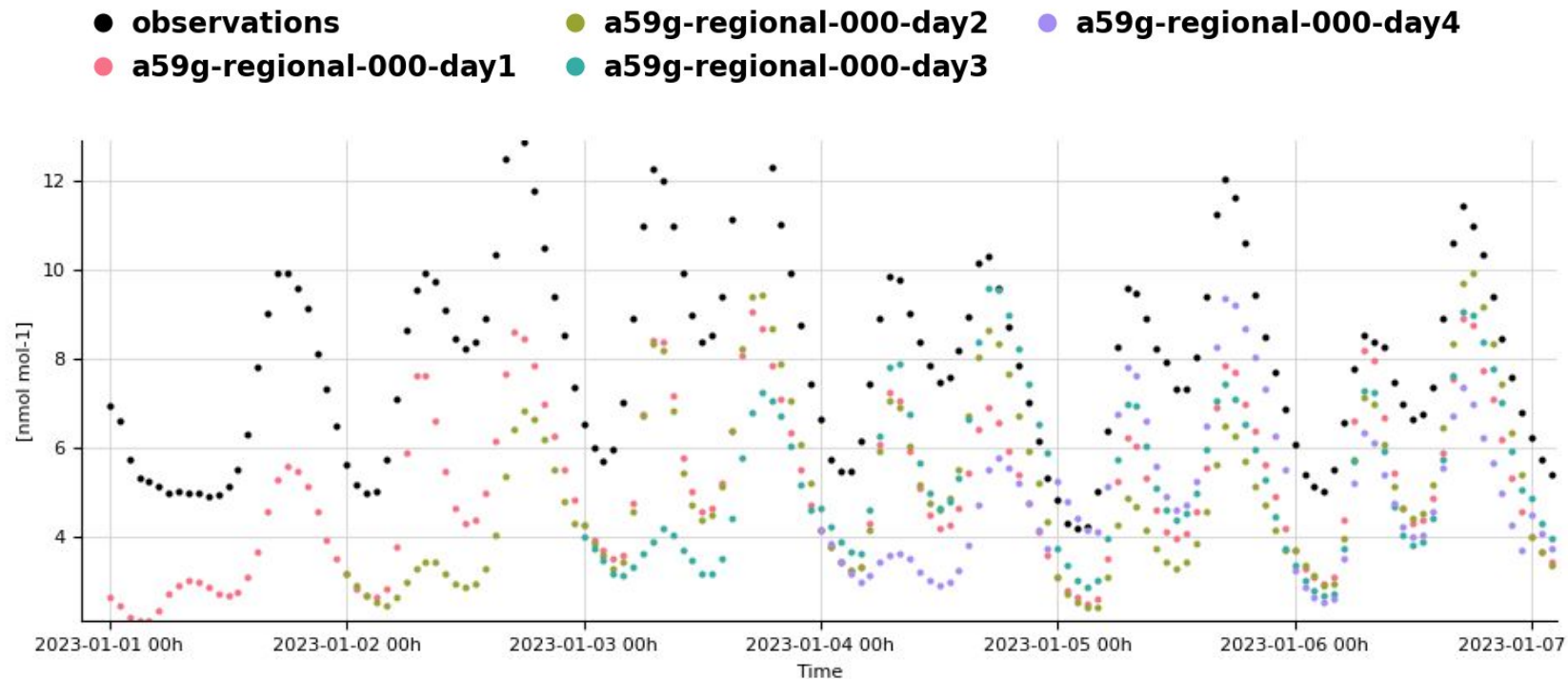
a59g-regional-000	☒	daily	day 3, day 4
a59j-regional-006	☐		

☒ day 1
☒ day 2
☒ day 3
☒ day 4

*Interactive selection of
model forecast data*

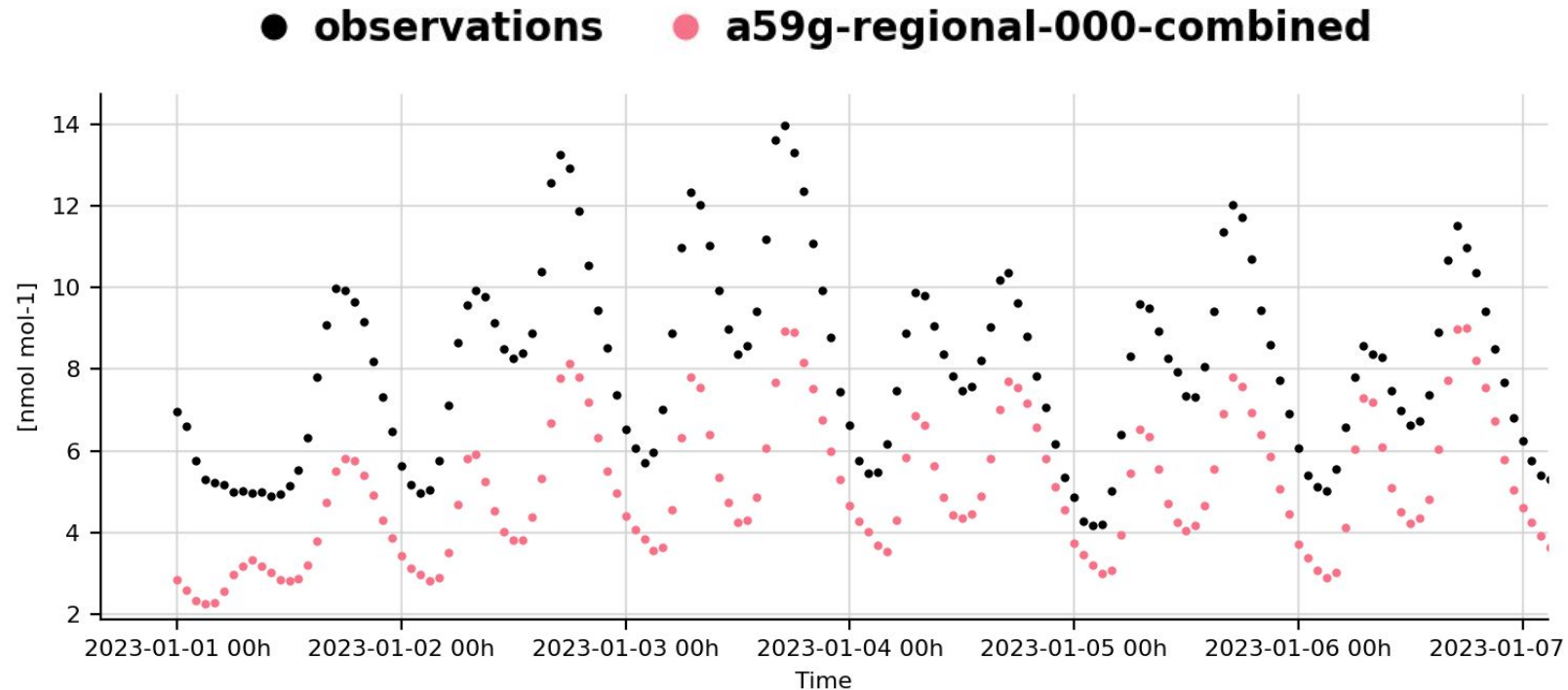
Day forecast

The **day** option is used when wanting to compare different forecast days. The effect this has in Providentia is essentially to treat each forecast day as an independent model. Thus statistics for each forecast day can be directly compared.



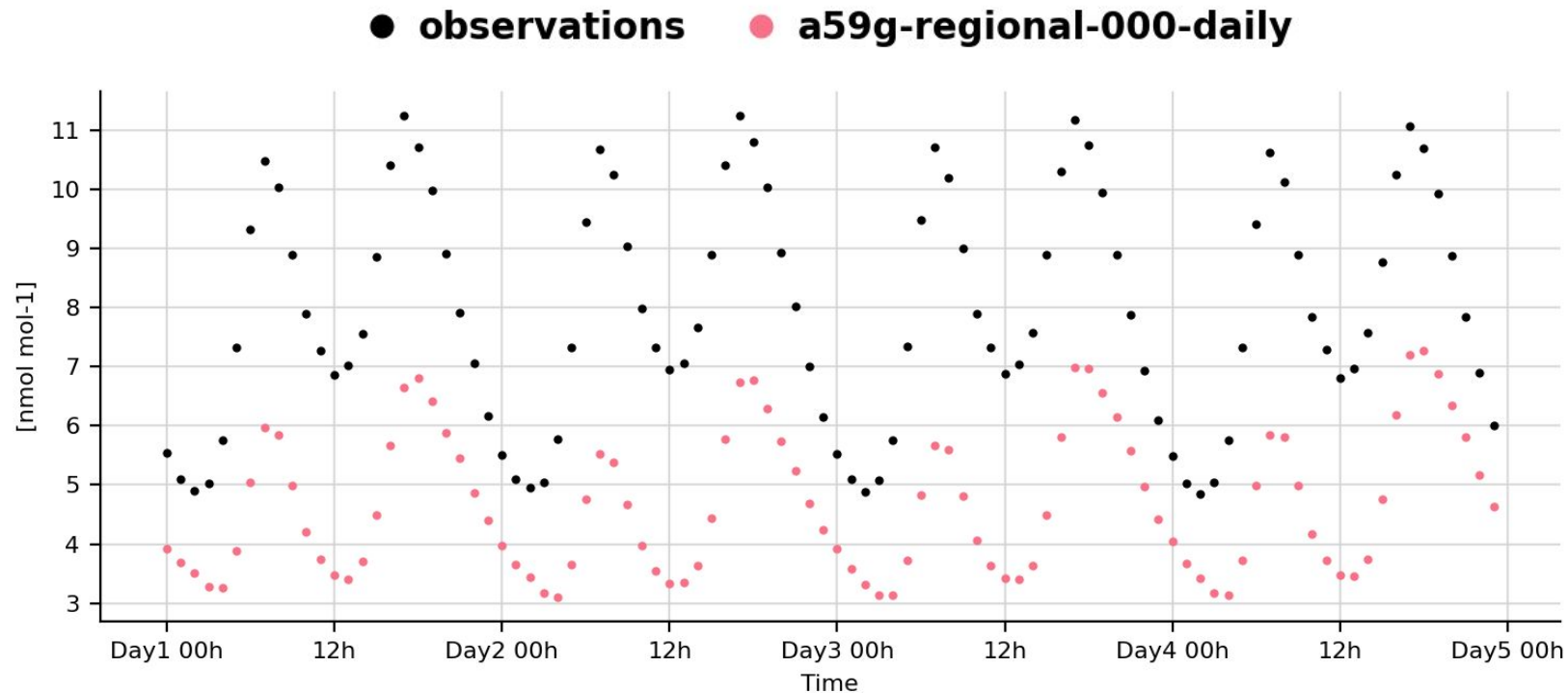
Combined forecast

The **combined** option is used when wanting to integrate data across all forecast days, providing statistics which represent the average state across all forecast days. The timeseries plot shows the average timeseries across all forecast days, for the entire time range.



Daily forecast

The **daily** option is also used when wanting to integrate data across all forecast days, however the timeseries plot is transformed to display the averaged data per forecast day, i.e. the plotted timeseries data for forecast day 1 in this case would be the average day 1 forecast across the entire time range.



Dashboard practical



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Dashboard practical

1. Perform a visualisation for **hourly ozone** from **EEA**, in **January 2023**, using **GHOST version 1.6**, for the **a59g-regional-000** model, for **Spanish stations** focused on the **bias** between observations and model, making a **map r bias, timeseries bias, periodic r bias, distribution bias, and statsummary bias**.
2. Perform a visualisation for **hourly ozone** from **EEA**, in **January 2023**, using **GHOST version 1.6**, for the **a59j-regional-006** and **a59g-regional-000** models, which keeps only stations with **>= 40% total representativity**, for strictly **daytime** periods, at stations **1km from the coast**.
3. Perform a **forecast day** visualisation for **hourly nitrogen dioxide** from **EEA**, in **January 2023**, using **GHOST version 1.6**, for the **a59g-regional-000** model, only keeping stations along the **coast of the United Kingdom and Ireland**, increase the selected map point size, **resample** the data to 3 hourly, and **save the data** to a netCDF file.



BREAK

10 minutes



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Report



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Choosing the plots

Users need to define the **report_type** in the configuration file, which will be linked to the plot types that appear in **settings/report_plots.yaml**.

configurations/configuration_name.conf

```
[PRV_sconco3_CHIMERE]
network = EBAS
species = sconco3
resolution = hourly
start_date = 20180101
end_date = 20180601
model = cams61_chimere_ph2-eu-000 (CHIMERE)
report_type = docs
report_summary = True
report_stations = True
report_title = Providentia Report
report_filename = CHIMERE
```



settings/report_plots.yaml

```
"docs": ["map-Mean", "timeseries", "periodic-Mean",
"periodic-violin", "boxplot", "distribution", "scatter",
"heatmap-RMSE", "table-RMSE", "statsummary",
"taylor-r", "fairmode-target",
"fairmode-statsummary", "contingencytable",
"metadata"]
```

Summary or station plots?

We can show data for all stations (`report_summary=True`) or per station (`report_station=True`). It is also possible to show certain plots as summary and others only per station by using a dictionary to define the report type in **settings/report_plots.yaml**.

configurations/configuration_name.conf

```
[PRV_sconco3_CHIMERE]
network = EBAS
species = sconco3
resolution = hourly
start_date = 20180101
end_date = 20180601
model = cams61_chimere_ph2-eu-000 (CHIMERE)
report_type = docs
report_summary = True
report_stations = True
report_title = Providentia Report
report_filename = CHIMERE
```



settings/report_plots.yaml

```
"docs": ["map-Mean", "timeseries", "periodic-Mean",
"periodic-violin", "boxplot", "distribution", "scatter",
"heatmap-RMSE", "table-RMSE", "statsummary", "taylor-r",
"fairmode-target", "fairmode-statsummary", "contingencytable",
"metadata"]
```

```
"docs":
{"summary": ["map-Mean", "timeseries", "periodic-Mean",
"periodic-violin", "boxplot", "distribution",
"scatter", "heatmap-RMSE", "table-RMSE",
"statsummary", "taylor-r", "fairmode-target",
"fairmode-statsummary", "contingencytable",
"metadata"],
"station": ["contingencytable"]}
}
```

Report specific configuration options

Parameter	Description	Default
report_type	Type of report to generate that defines which plots the report will contain, from the options given in report_plots.yaml.	standard
report_summary	Boolean variable to set if you wish to make specific plots for each station in subsection.	True
report_stations	Boolean variable to set if you wish to make summary plots across station subsection.	False
report_title	The header in the first page of the report (as in the PDF).	Providentia Report
report_filename	The filename of the report or the path to create the report (as in the PDF).	PROVIDENTIA_Report
harmonise_stations	Boolean variable to set if you wish to harmonise axes limits across stations for stations report.	True
harmonise_summary	Boolean variable to set if you wish to harmonise axes limits across subsections for summary report.	True

Plot types



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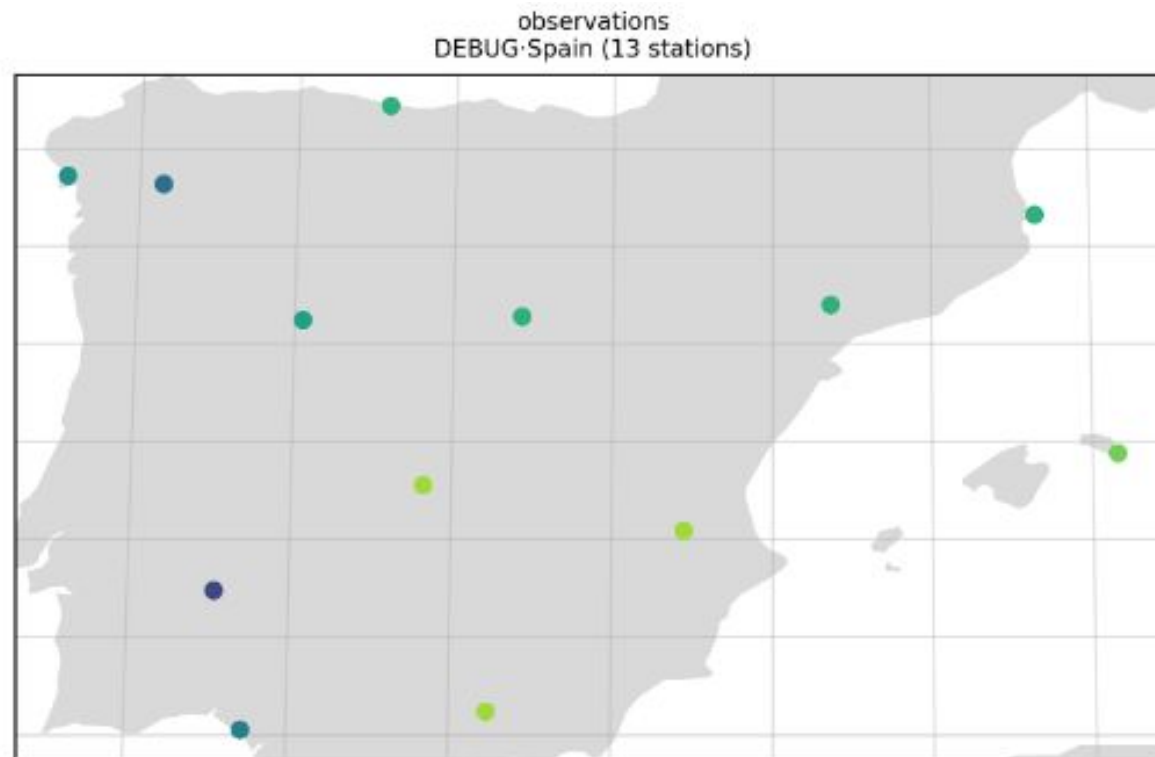
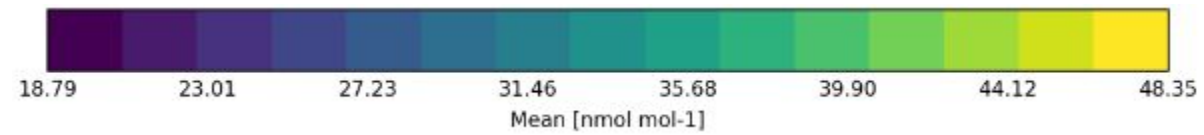
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Plot types

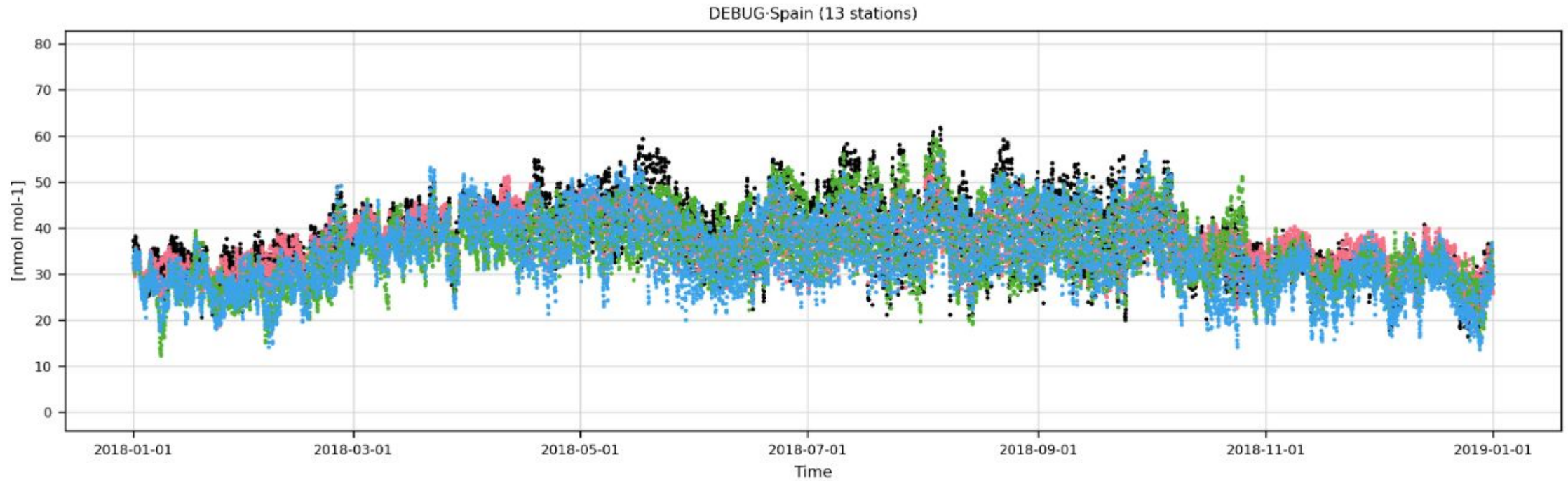
- Map (**map**)
- Metadata summary (**metadata**)
- Timeseries (**timeseries**)
- Periodic plot (**periodic**)
- Periodic violin plot (**periodic-violin**)
- Box plot (**boxplot**)
- Distribution plot (**distribution**)
- Scatter plot (**scatter**)
- Heat map (**heatmap**) - Not available in dashboard
- Table that gives one statistic per subsection per model (**table**) - Not available in dashboard
- Table that gives multiple statistics per model (**statsummary**)
- Taylor Diagram (**taylor**)
- FAIRMODE target plot (**fairmode-target**)
- FAIRMODE statistics summary plot (**fairmode-statsummary**)
- Contingency table (**contingencytable**)

Reference: <https://providentia.readthedocs.io/en/latest/Plot-types-and-options.html>

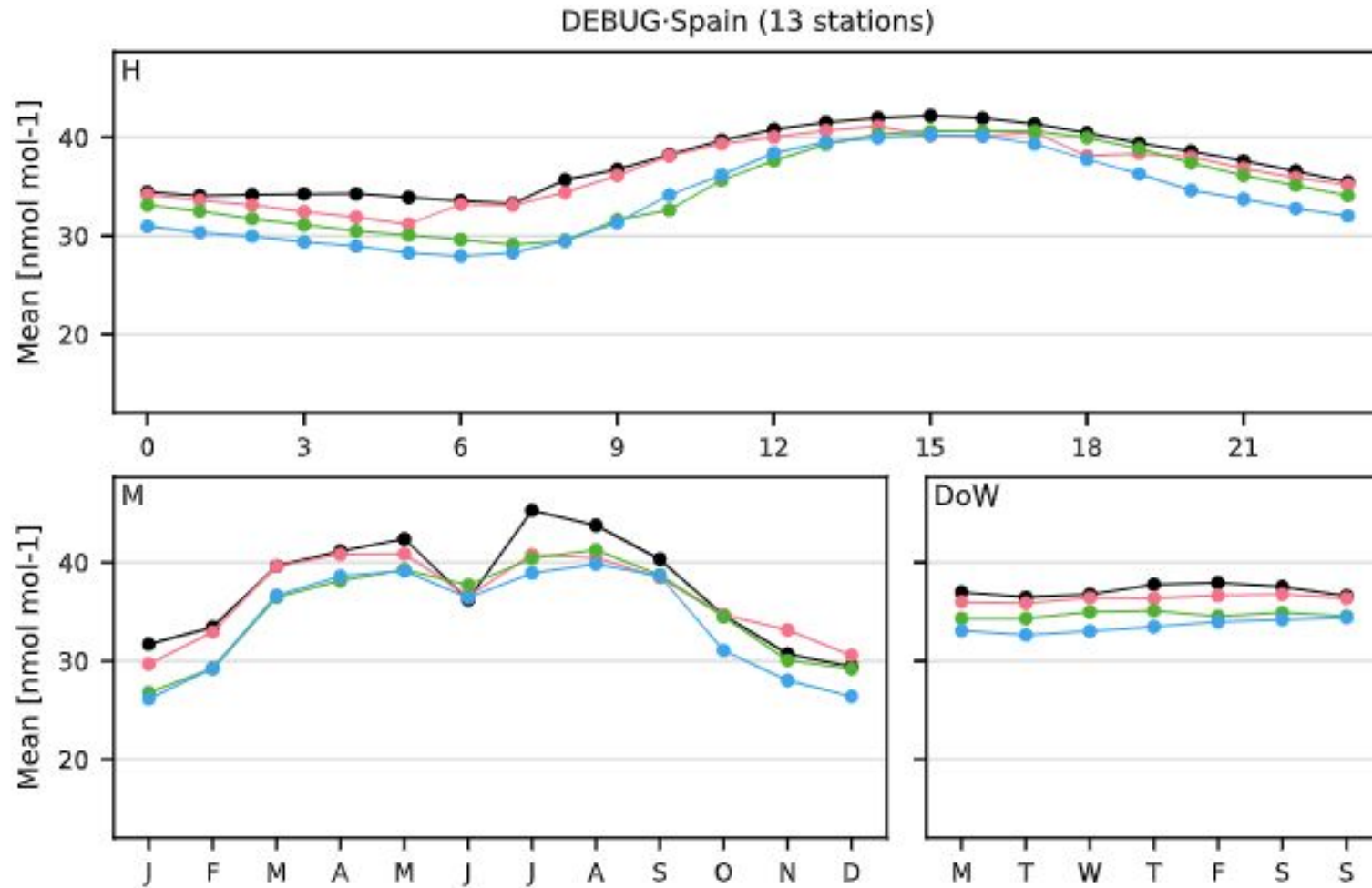
map-[stat]



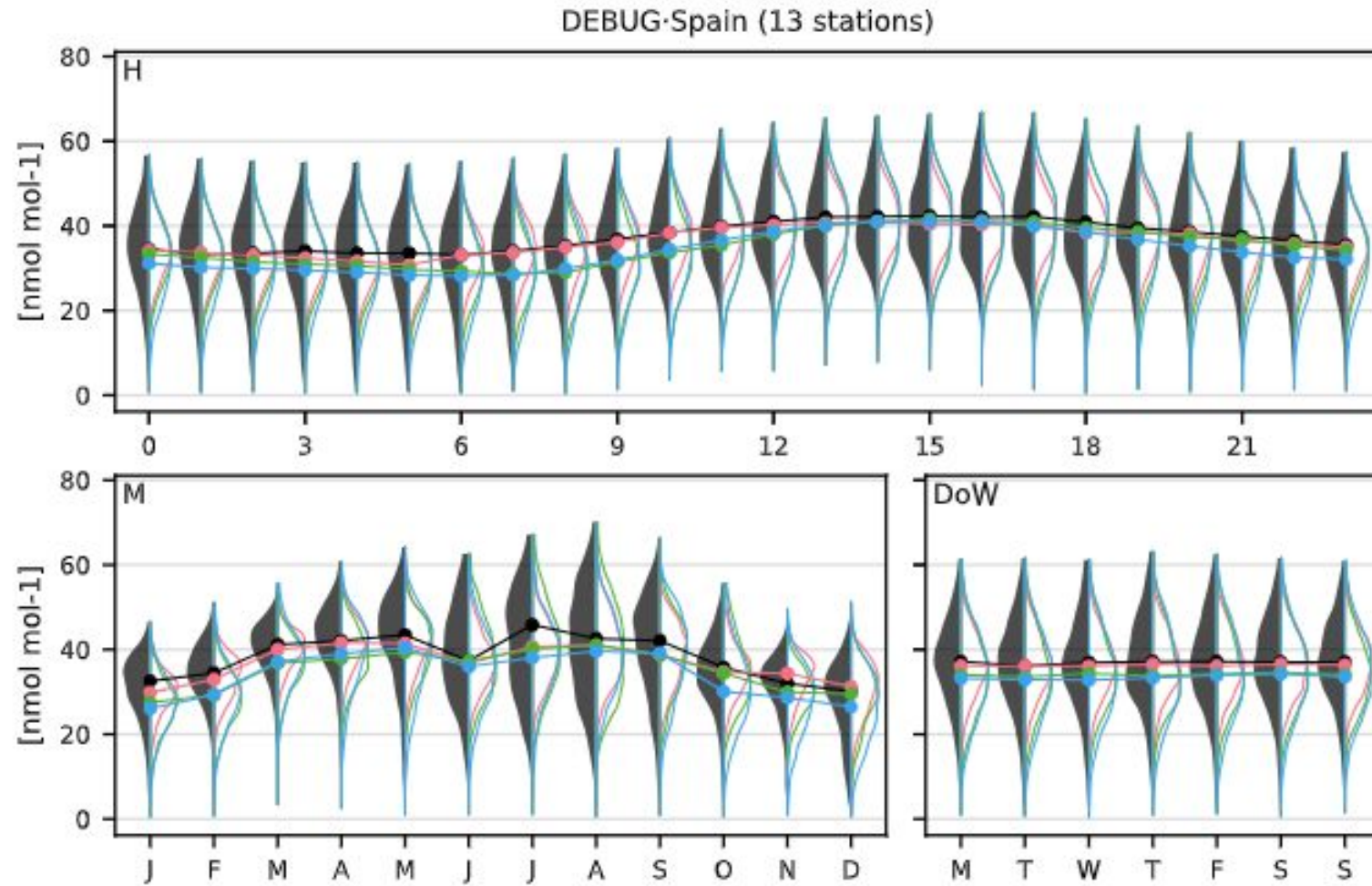
timeseries / timeseries-[stat]-[resolution]



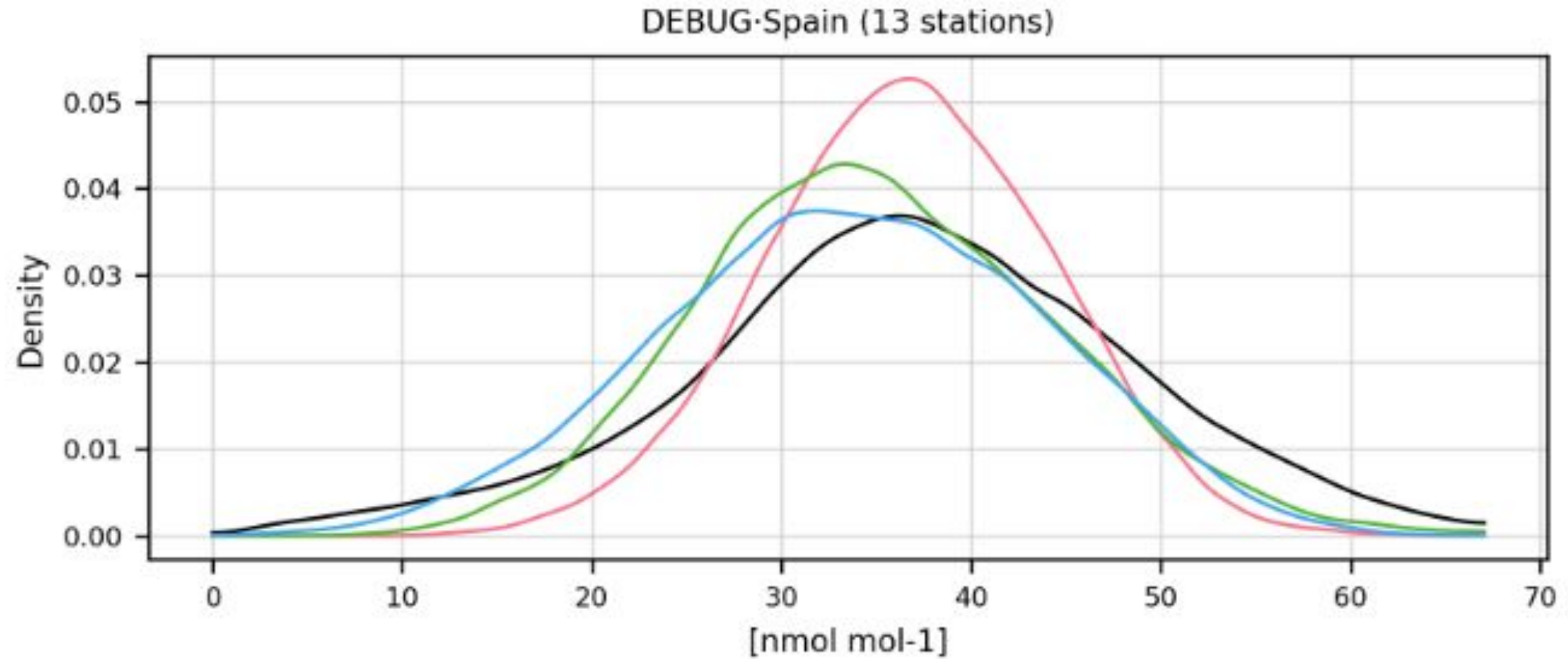
periodic-[stat]



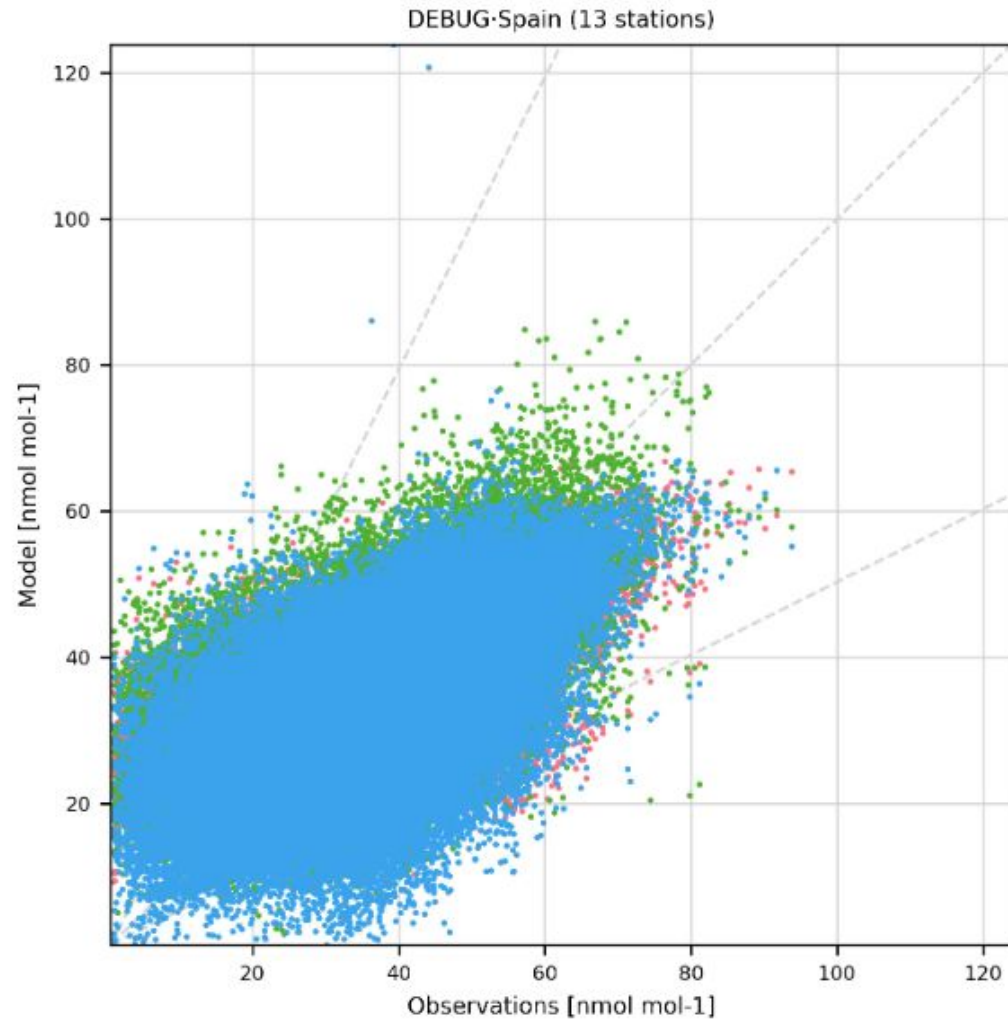
periodic-violin



distribution



scatter



heatmap-[stat]

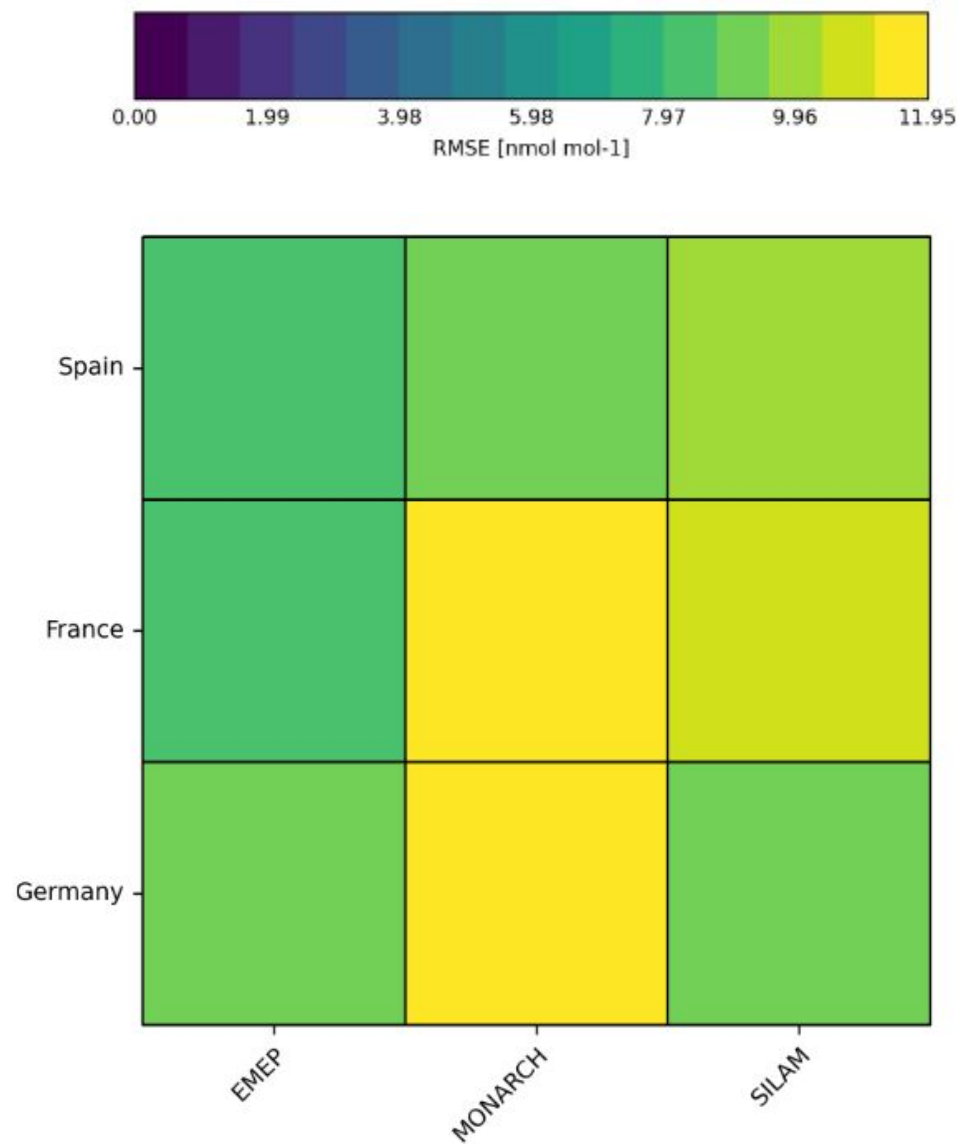
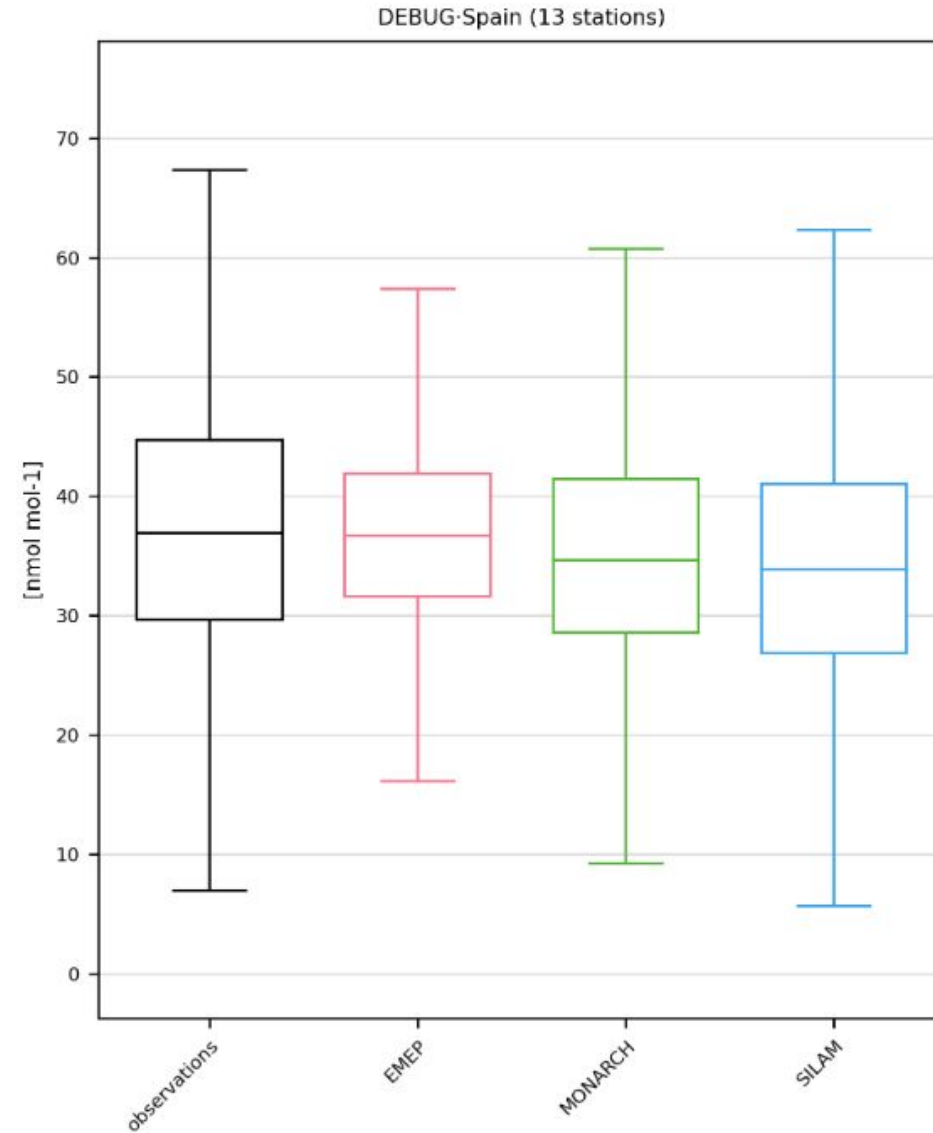


table-[stat]

	EMEP	MONARCH	SILAM
Spain	8.32	9.28	9.77
France	8.55	11.20	10.82
Germany	9.46	11.95	9.25

boxplot



statsummary

		Mean	StdDev	p5	Median	p95
Spain	observations	37.14	10.24	22.13	37.14	55.03
	EMEP	36.30	6.98	24.61	36.43	48.10
	MONARCH	34.56	8.88	21.43	33.95	51.45
	SILAM	33.64	9.58	18.37	33.25	49.69
France	observations	34.26	11.73	13.65	34.00	57.00
	EMEP	35.80	7.98	20.89	35.86	47.50
	MONARCH	29.35	10.81	11.48	28.21	46.11
	SILAM	29.33	10.80	11.37	28.76	47.61
Germany	observations	33.67	14.89	11.20	34.90	58.30
	EMEP	34.52	10.08	17.48	35.58	48.91
	MONARCH	26.74	12.56	6.73	27.10	47.03
	SILAM	28.81	11.95	8.92	29.04	48.23



metadata

DEBUG·Spain (13 stations)

13 Stations

Median Measurement Altitude: 506.00 m

Median To Coast: -55.00 km

Median Population Density: 0.0 xx/km-2

Country: Spain

Area Classification: rural:76.9%,

urban-suburban:23.1%

Station Classification: background:76.9%,

nan:23.1%

MODIS Land Use: 4 uniques

GHSL Classification: low density rural:23.1%,

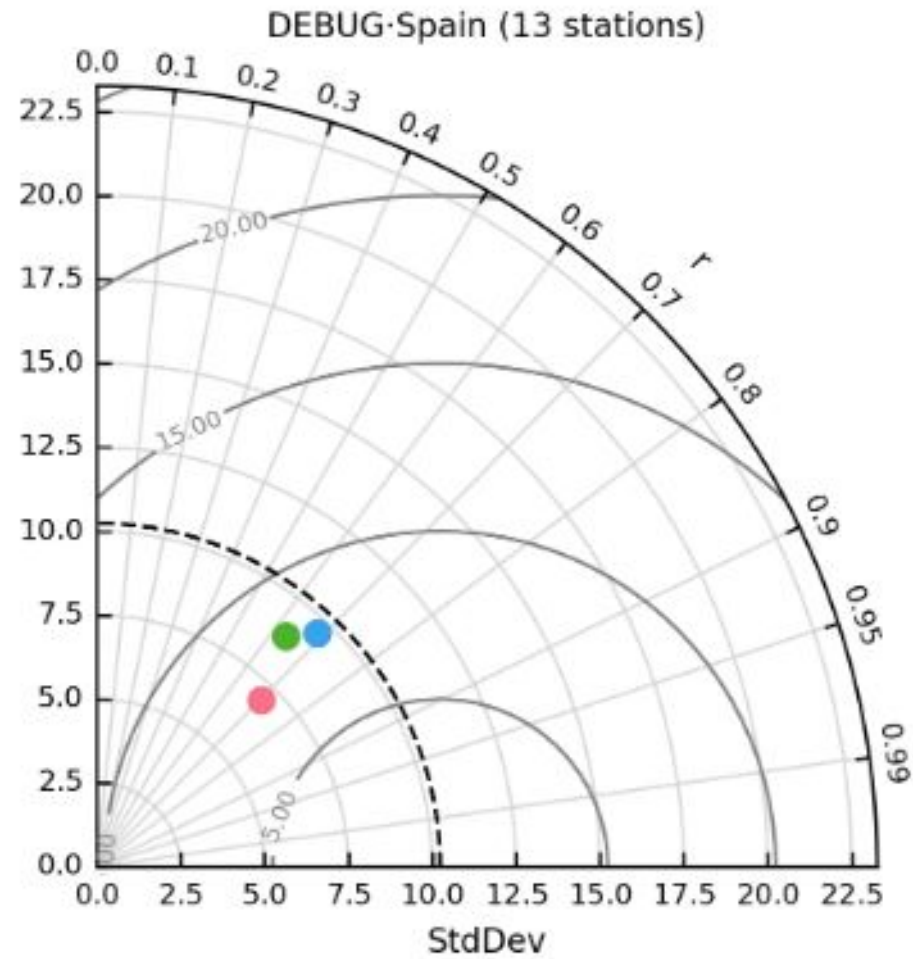
very low density rural:76.9%

Method: ultraviolet photometry

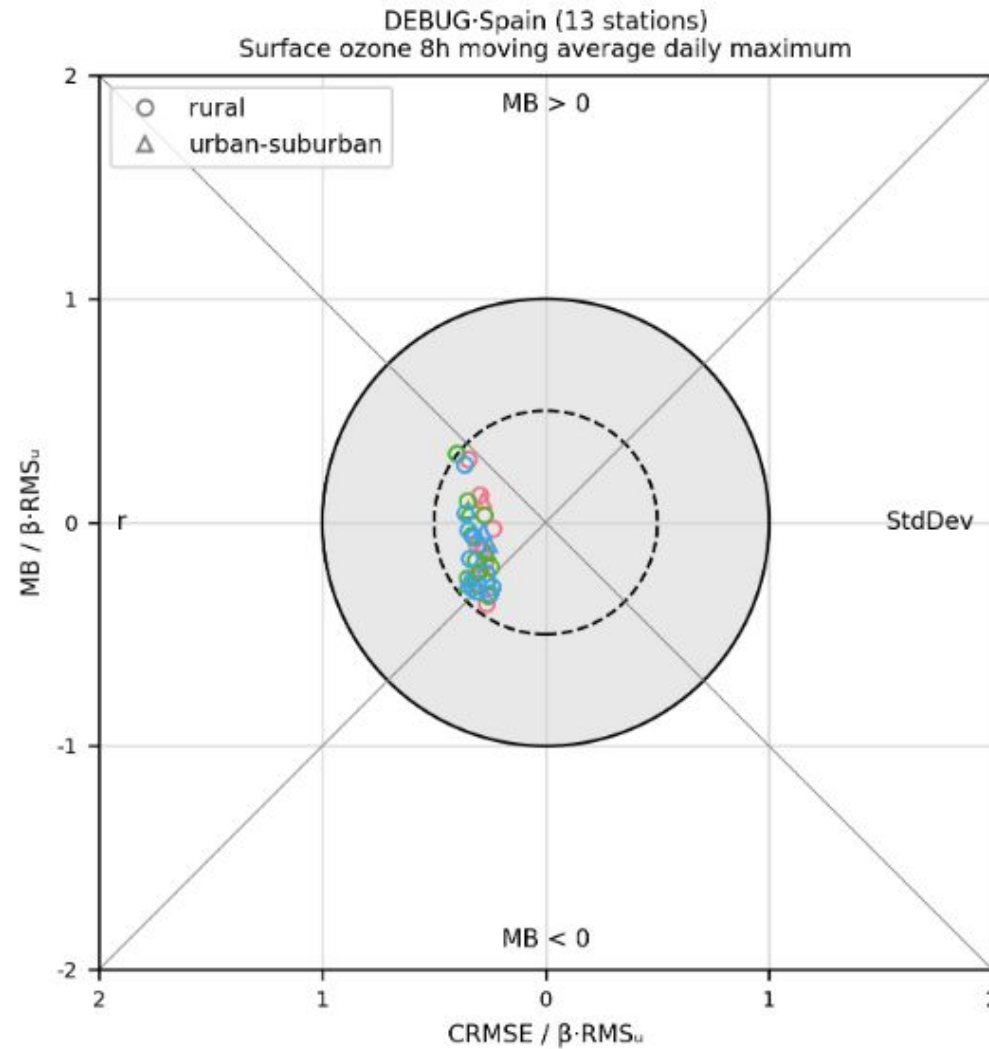
Instrument: nan



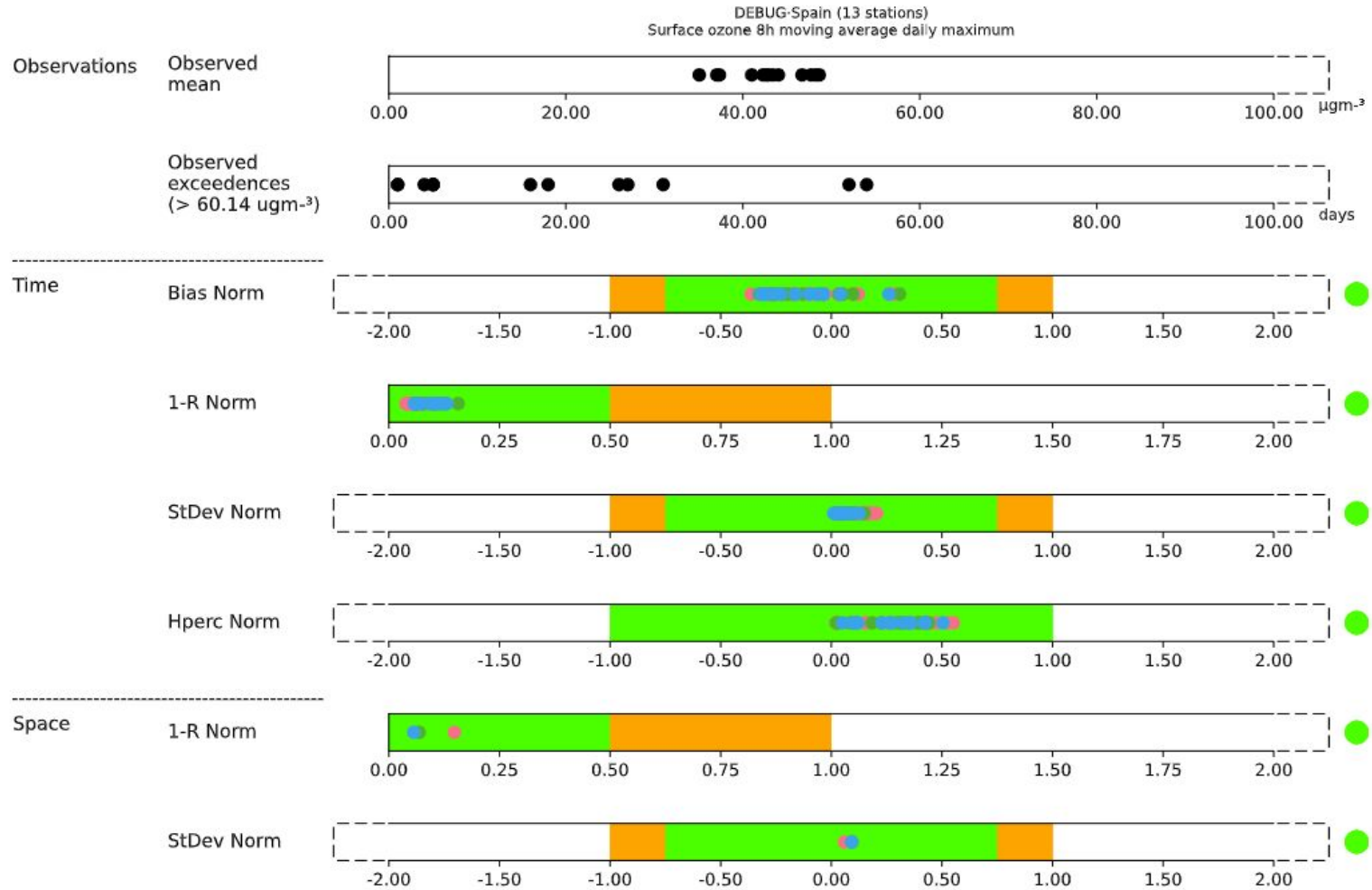
taylor-[stat]



fairmode-target



fairmode-statsummary



contingencytable

ES0001R_UVP, San Pablo De Los Montes (-4.35, 39.55)

	EMEP					
observations	Good	Fair	Moderate	Poor	Very poor	Extremely poor
Good	32	249	0	0	0	0
Fair	208	5934	24	0	0	0
Moderate	6	1858	278	0	0	0
Poor	0	22	34	0	0	0
Very poor	0	0	0	0	0	0
Extremely poor	0	0	0	0	0	0



Statistics in plots



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-[stat]

It must be added to create **maps, periodic plots, heatmaps, tables and Taylor diagrams.**

Basic statistics

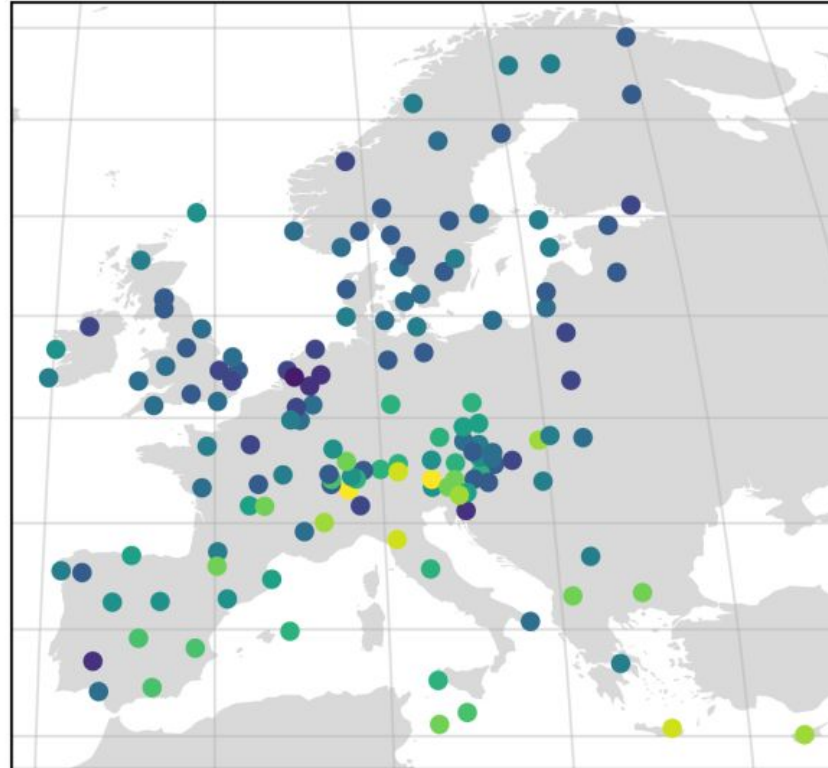
Statistic	Meaning
Mean	Mean
StdDev	Standard deviation
Var	Variance
Min	Minimum
Max	Maximum
Data%	Data availability
Exceedances	Number of exceedances
p1, p5, p10, p25, p50, p75, p90, p95, p99	Percentiles
NStations	Number of stations
MDA8	Daily maximum 8 hour average

Model bias statistics

Statistic	Meaning
MB	Mean bias
NMB	Normalised mean bias
ME	Mean error
NME	Normalised mean error
MNB	Mean normalised bias
MNE	Mean normalised error
MFB	Mean fractional bias
MFE	Mean fractional error
RMSE	Root mean square error
NRMSE	Normalised root mean square error
COE	Coefficient of efficiency
FAC2	Fraction of model values within a factor of two of observed values
IOA	Index of agreement
R	Pearson correlation coefficient
R ²	Coefficient of determination
UPA	Unpaired peak accuracy

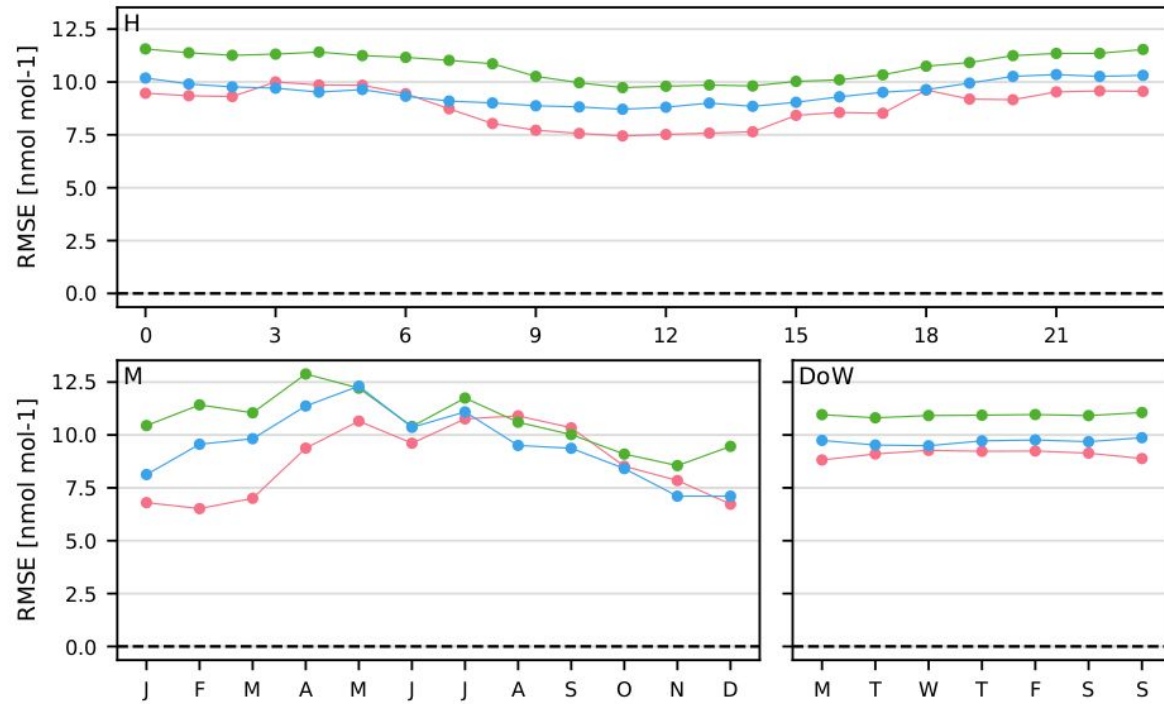
-[stat]

map-Mean

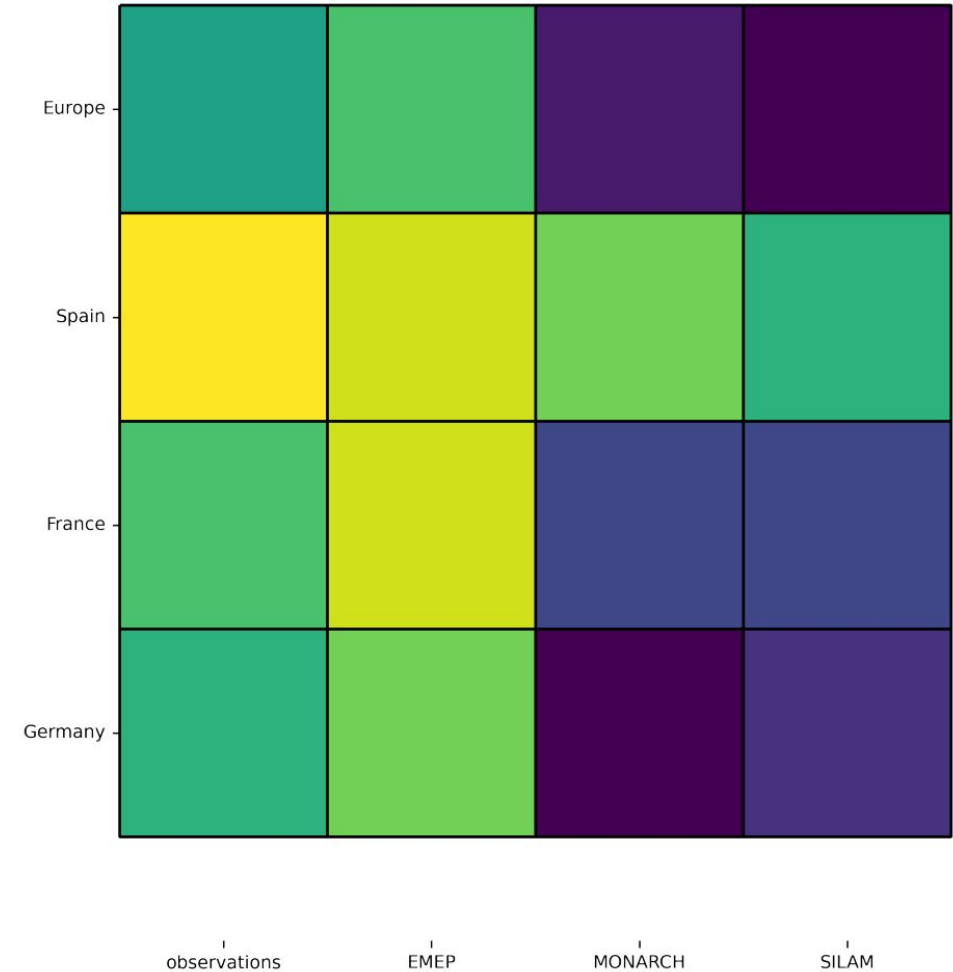


-[stat]

periodic-RMSE



heatmap-Mean

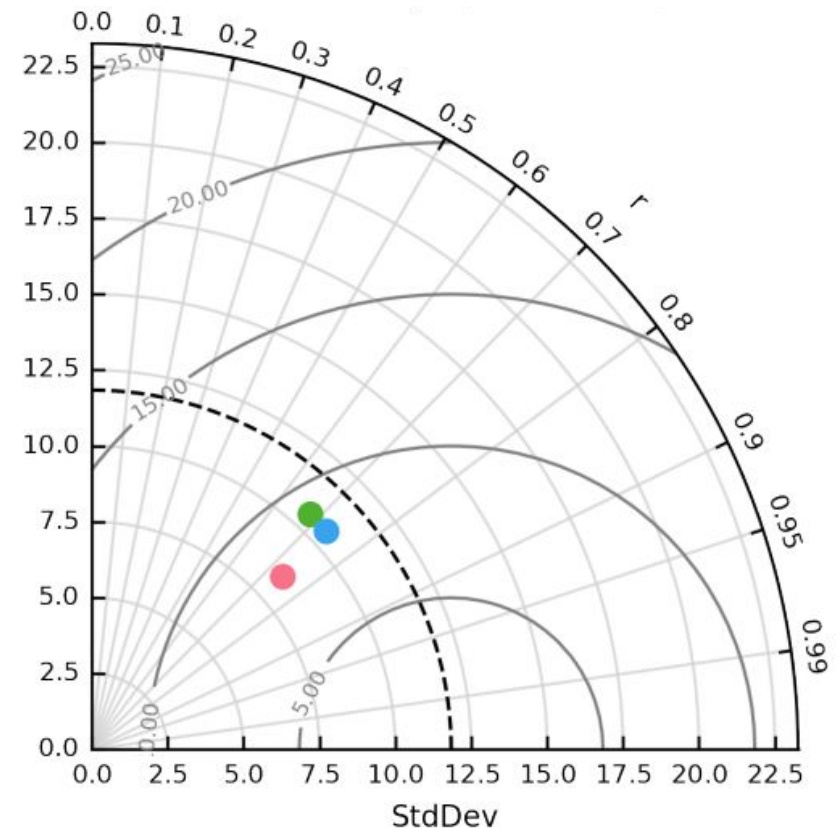


-[stat]

table-Data%

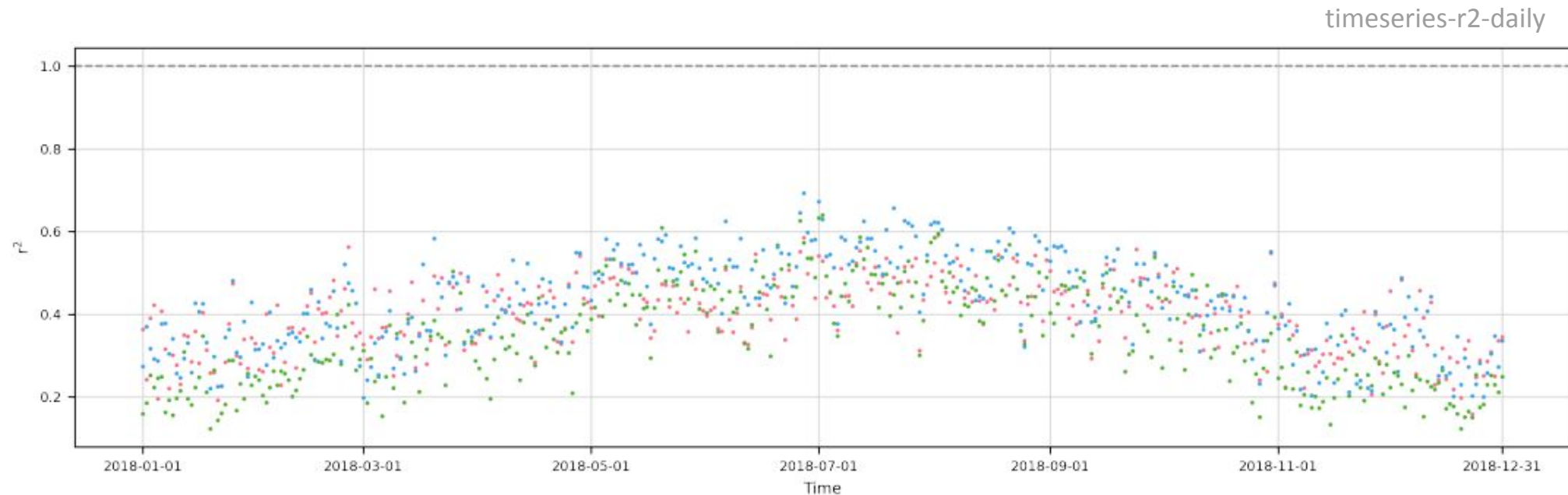
	observations	EMEP	MONARCH	SILAM
Europe	19.16	19.16	19.16	19.16
Spain	19.56	19.56	19.56	19.56
France	19.29	19.29	19.29	19.29
Germany	17.41	17.41	17.41	17.41

taylor-r



-[stat]-[resolution]

The timeseries can also be used to show how statistics vary in time. In order to do this, we need to add **-[stat]** and the temporal resolution after the plot type name (e.g. *timeseries-Mean-daily*, *timeseries-r2-monthly*, *timeseries-r-annual*).



Plot options



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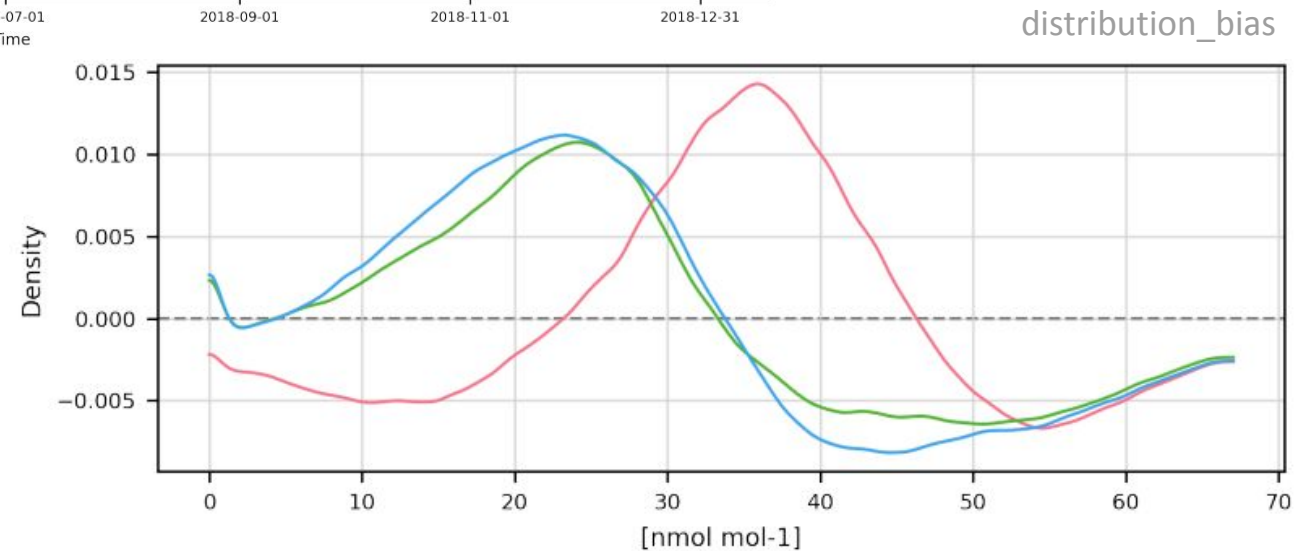
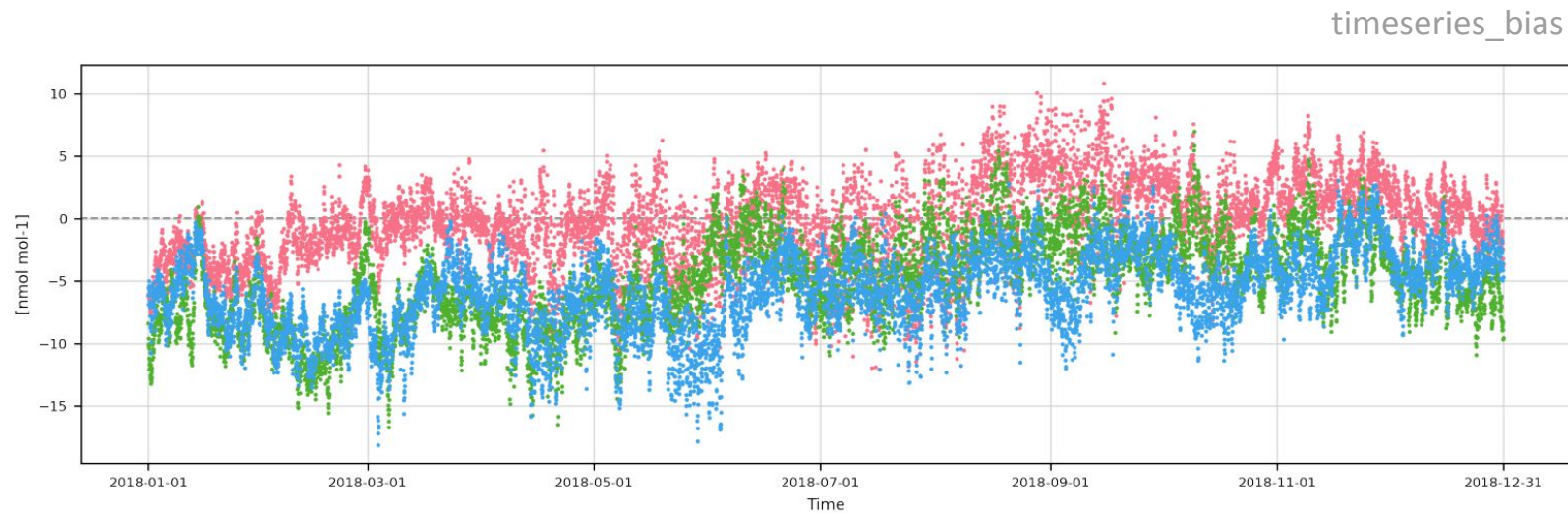
Plot types

- Make bias (**_bias**)
- Show annotate (**_annotate**)
- Show label plots (**_individual**)
- Show observation plots (**_obs**)
- Make multispecies (**_multispecies**)
- Show logarithmic in x axis (**_logx**)
- Show logarithmic in y axis (**_logy**)
- Add regression line (**_regression**)
- Add smooth line (**_smooth**)
- Hide points (**_hidedata**)
- Show domain (**_domain**)
- Add threshold line (**_threshold**)
- Normalise y axis (**_normalise**)
- Show Gerrity scores (**_gerrity**)

Reference: <https://providentia.readthedocs.io/en/latest/Plot-types-and-options.html>

_bias

- **Dashboard:** timeseries, periodic, distribution, statsummary
- **Report:** map, timeseries, periodic, distribution, heatmap, table, statsummary
- **Library:** timeseries, periodic, distribution, heatmap, table, statsummary

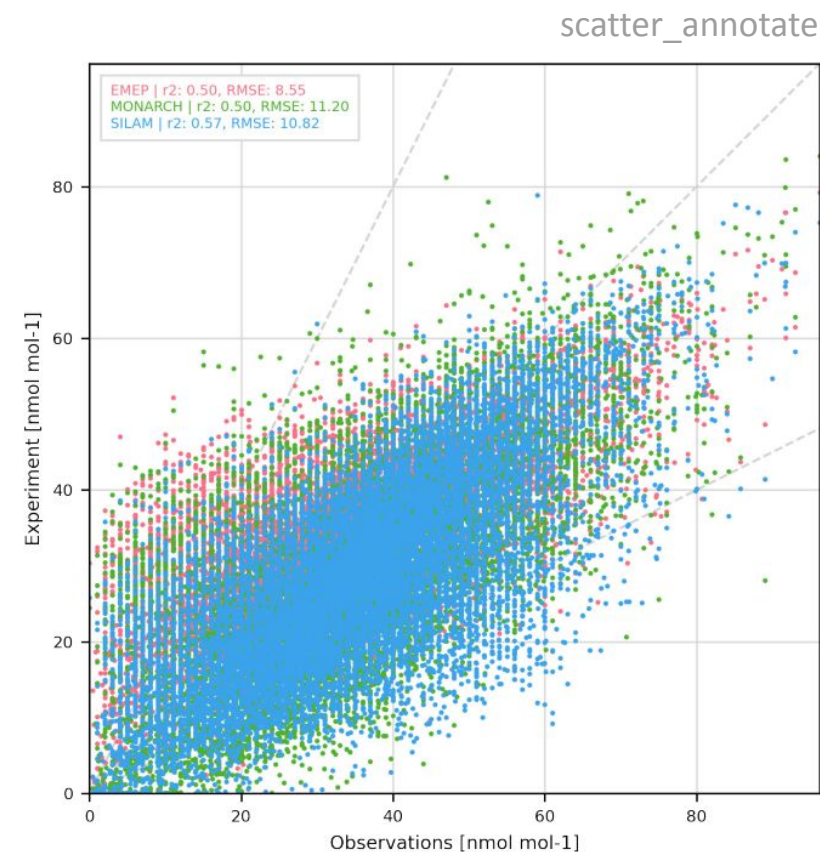


_annotate

- Dashboard / Report / Library: map, timeseries, periodic, periodic-violin, distribution, scatter, boxplot, taylor, fairmode-target

The statistics to annotate are defined per plot type in **settings/plot_characteristics.yaml**.

```
"scatter":  
  {  
    "grid": { "axis": "both", "color": "lightgrey", "alpha": 0.8 },  
    "annotate_stats": ["r2", "RMSE"],  
    "annotate_offset": { "loc": "upper left" },  
    "annotate_bbox":  
      {  
        "facecolor": "white",  
        "edgecolor": "gainsboro",  
        "alpha": 1,  
        "zorder": 100,  
      },  
    ...  
  }
```

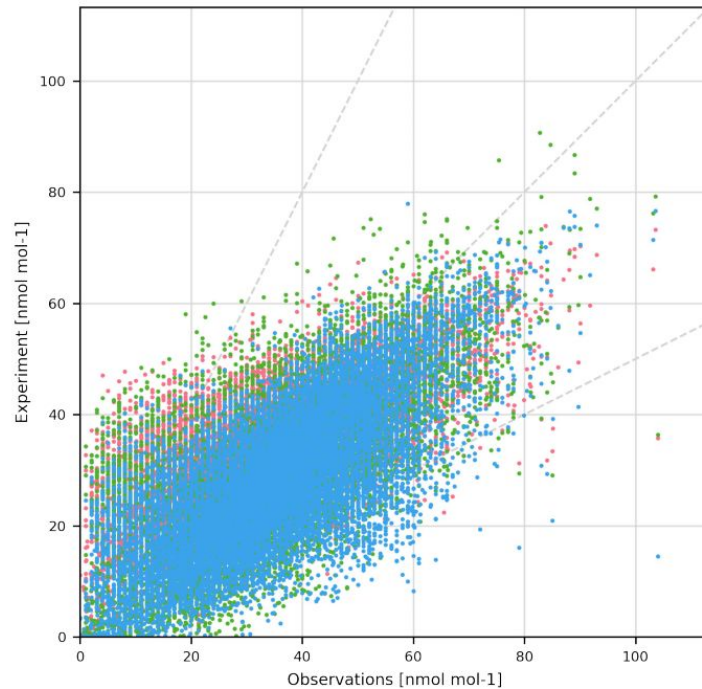


_individual

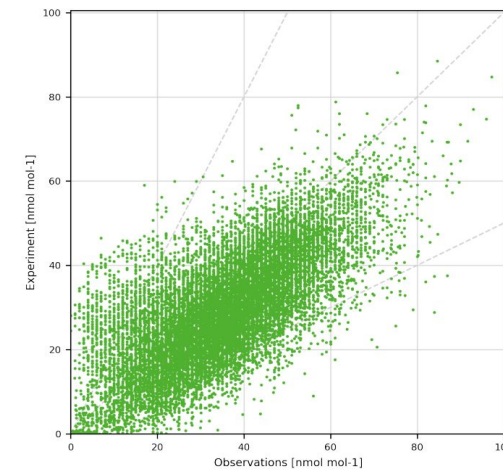
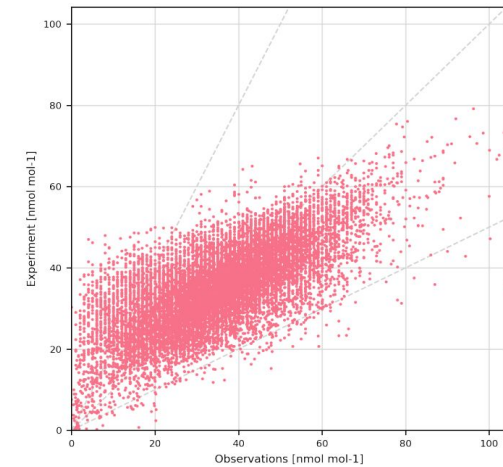
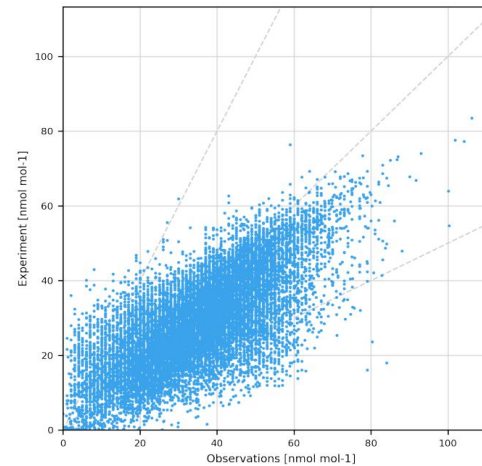
- **Dashboard / Library:** Not applicable
- **Report:** timeseries, periodic, periodic-violin, distribution, scatter, boxplot, taylor, fairmode-target, fairmode-statsummary

Sometimes you might want to show the data per model to see it with more detail.

scatter



scatter_individual

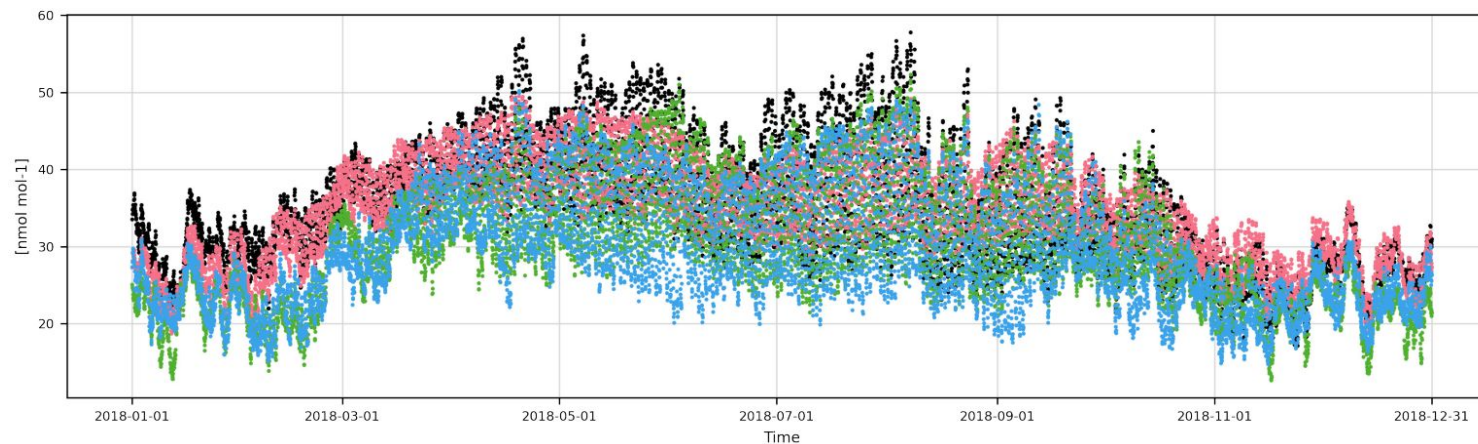


_obs

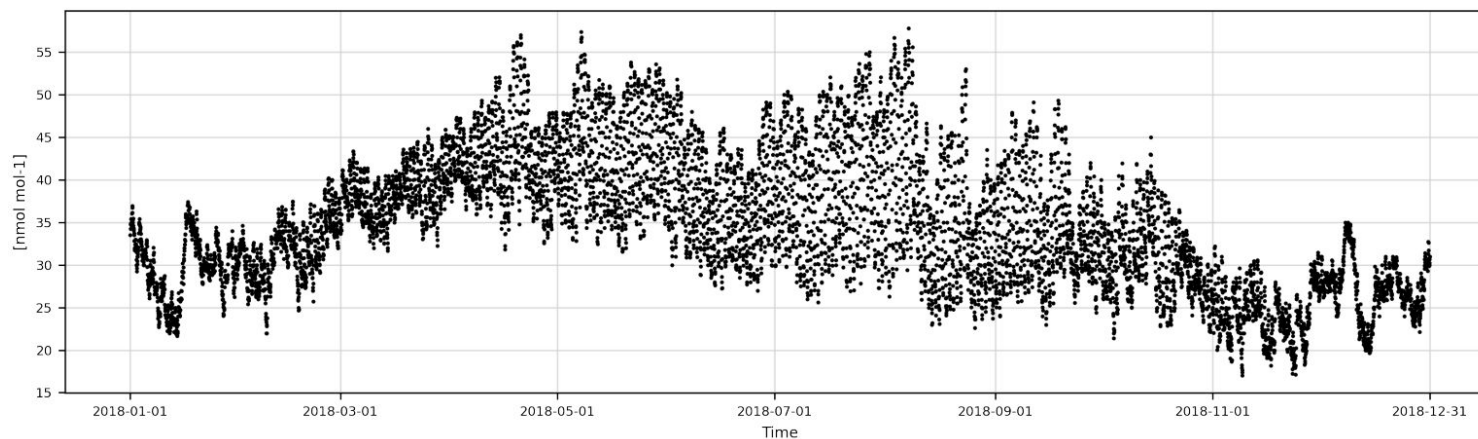
- Dashboard / Library: Not applicable
- Report: [map](#), [timeseries](#), [periodic](#), [periodic-violin](#), [distribution](#), [boxplot](#)

To isolate the observations data

timeseries



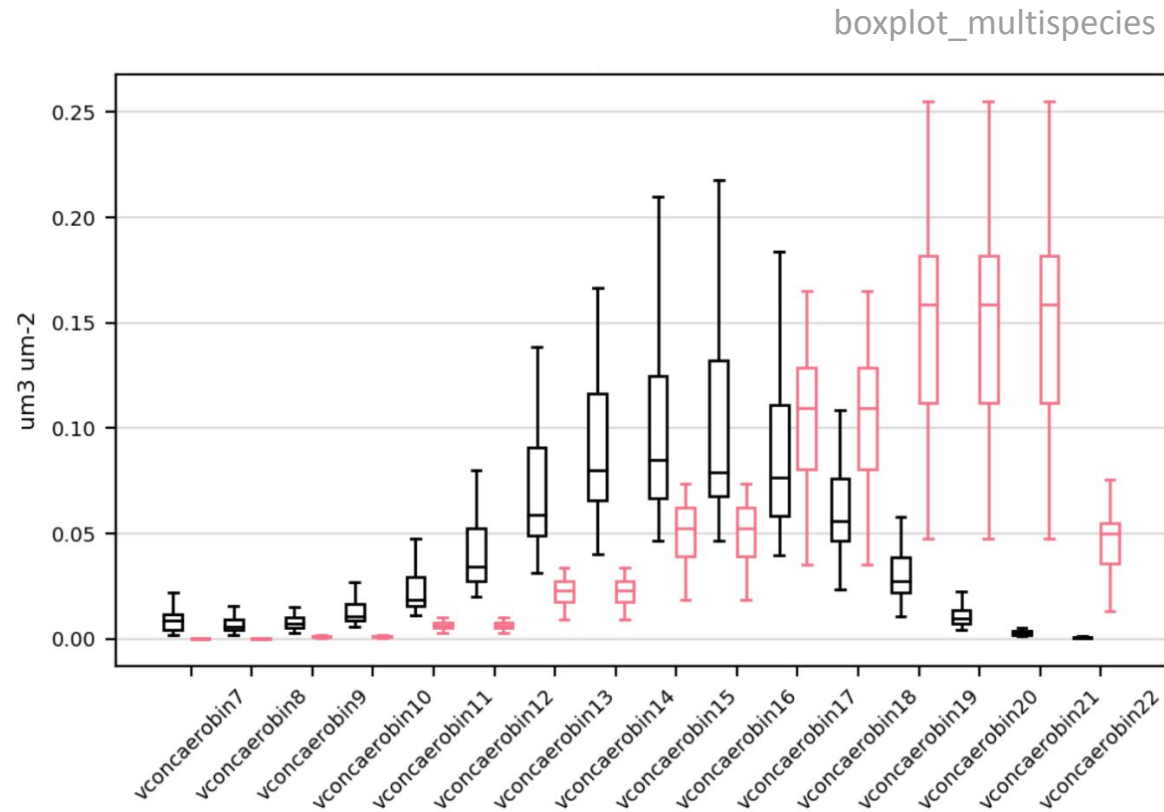
timeseries_obs



_multispecies

- **Dashboard:** Not applicable
- **Report / Library:** **boxplot**, **heatmap**, **table**, **statsummary**

The report mode is the only one able to plot multiple species per plot.

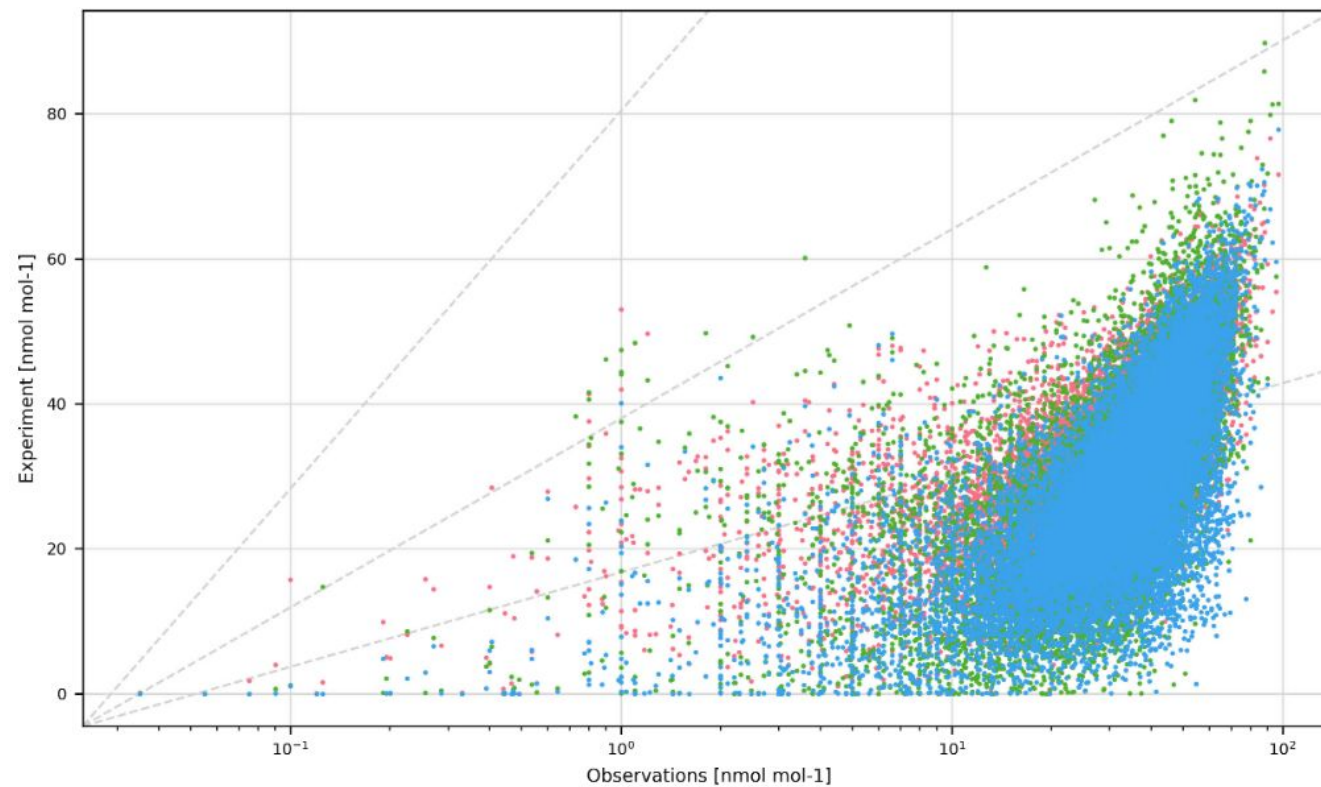


statsummary_multispecies

		p5	Mean	StdDev	p50	p95
sconco3	Observations	50.0	71.03	14.59	69.42	96.0
	Forecast	40.44	60.34	13.72	60.12	83.31
	Analysis	47.14	67.25	13.7	65.81	89.43
sconco2	Observations	5.64	8.5	2.21	8.0	13.08
	Forecast	1.66	4.59	2.34	4.31	9.09
	Analysis	2.92	5.65	2.09	5.34	9.69
sconcco	Observations	0.2	0.24	0.02	0.24	0.28
	Forecast	0.13	0.15	0.01	0.14	0.17
	Analysis	0.16	0.18	0.01	0.18	0.2
sconco2	Observations	2.0	2.1	0.16	2.0	2.49
	Forecast	0.45	0.81	0.24	0.82	1.2
	Analysis	0.95	1.45	0.29	1.52	1.82
pm10	Observations	12.01	15.91	2.43	16.19	19.46
	Forecast	5.99	8.44	1.6	8.17	10.96
	Analysis	10.43	13.91	2.27	14.06	17.3
pm2p5	Observations	5.7	8.83	1.84	9.02	11.55
	Forecast	4.18	6.17	1.33	6.03	8.52
	Analysis	5.12	8.15	1.72	8.29	10.84

_logx

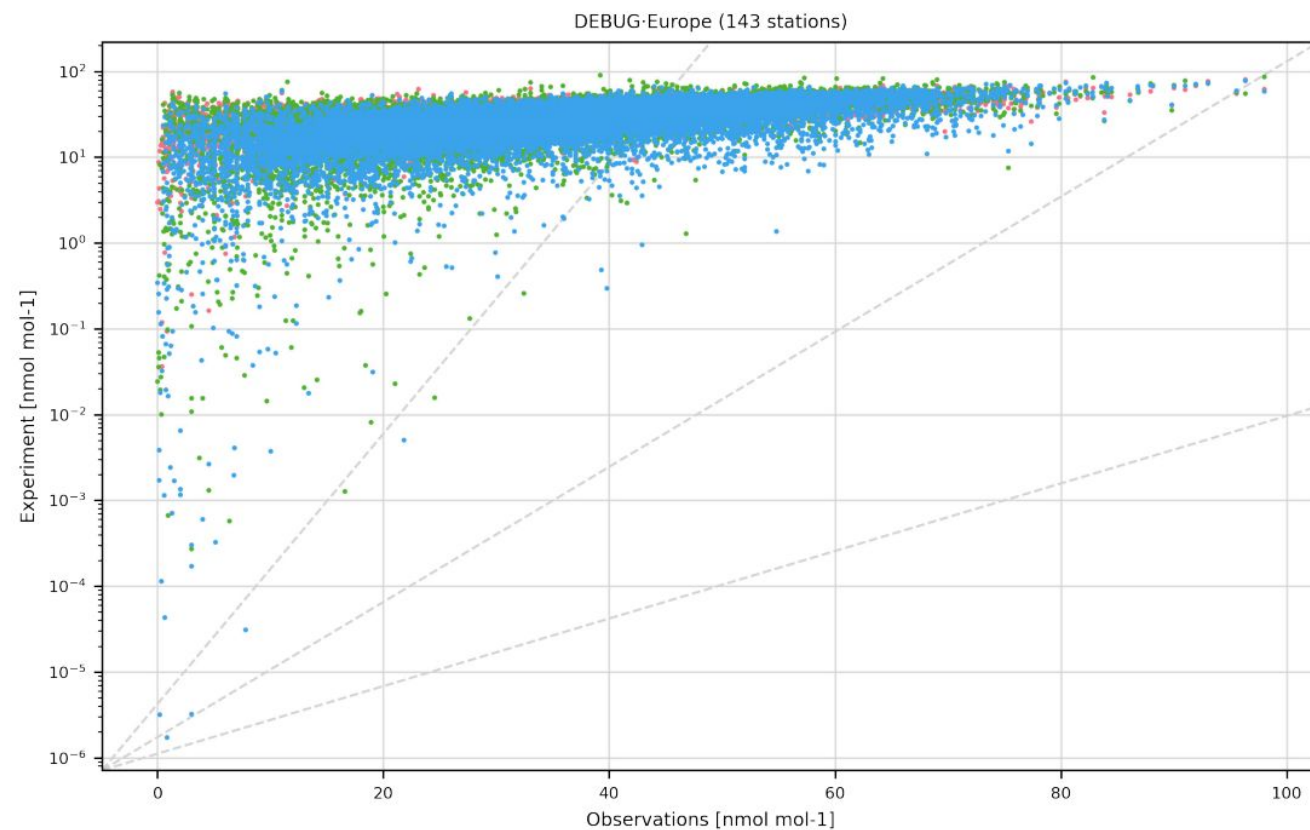
scatter_logx



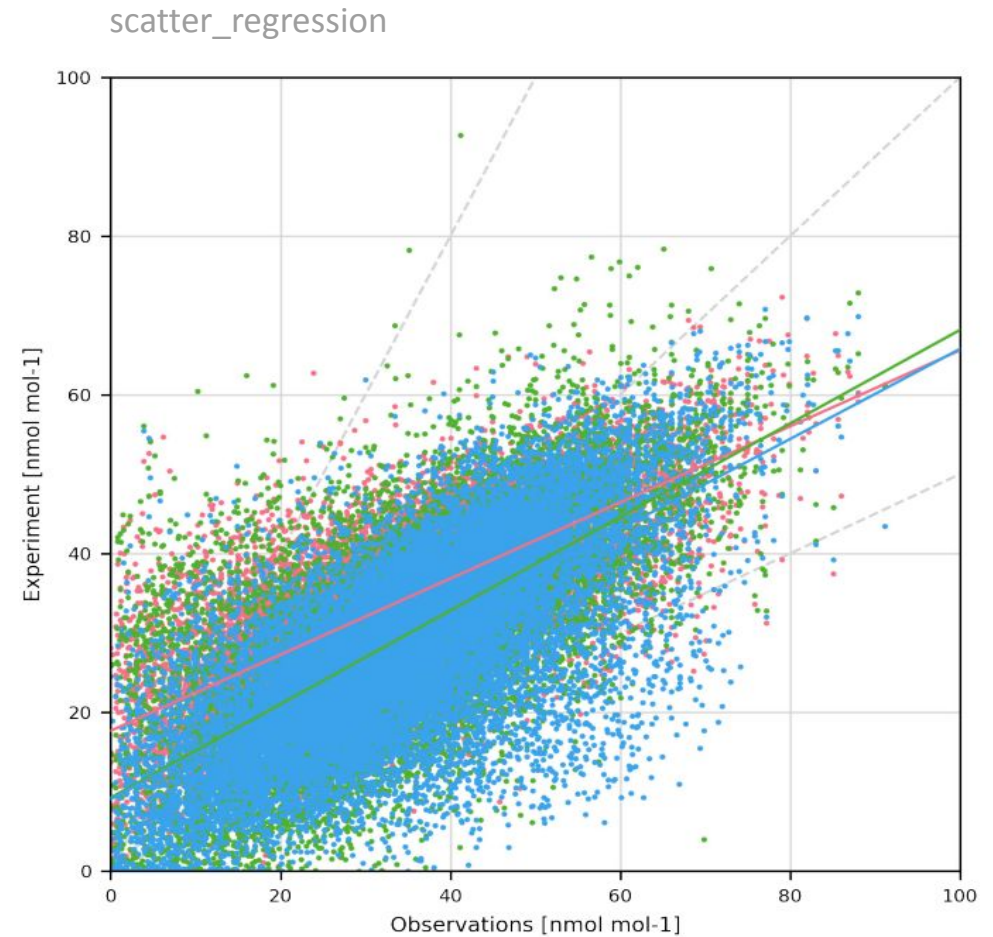
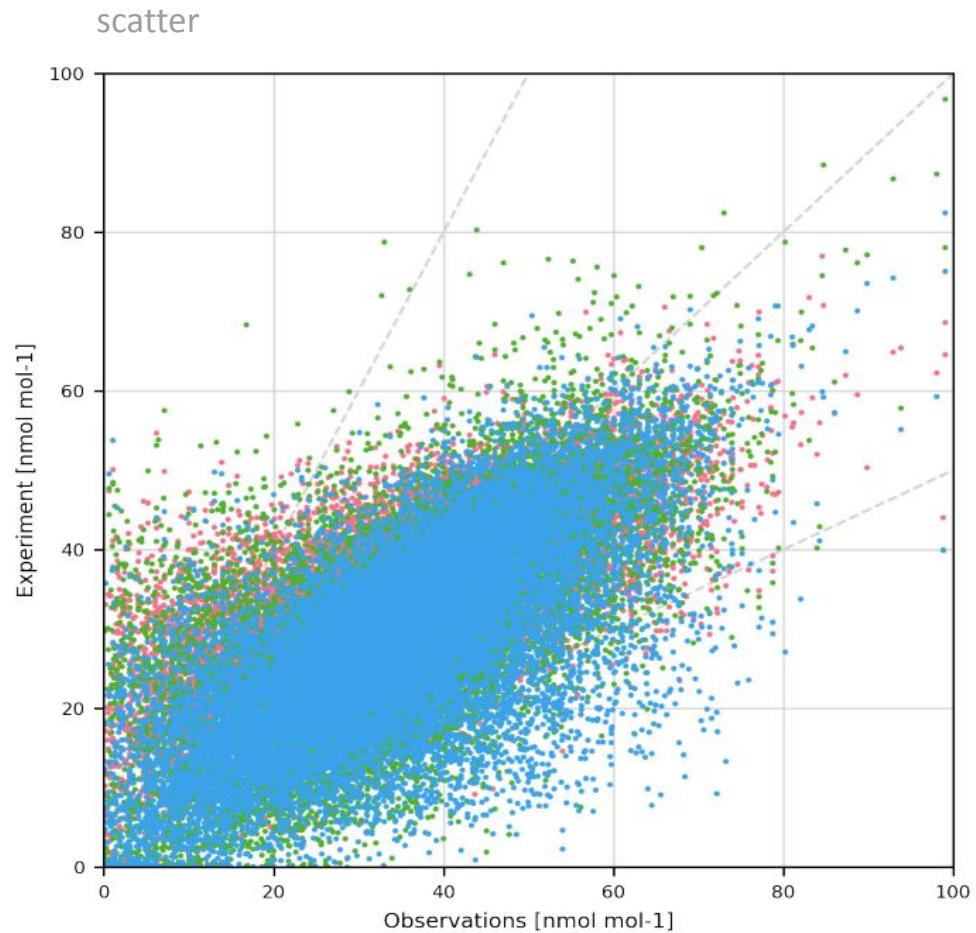
_logy

- Dashboard / Report / Library: timeseries, periodic, periodic-violin, distribution, scatter, boxplot

scatter_logy



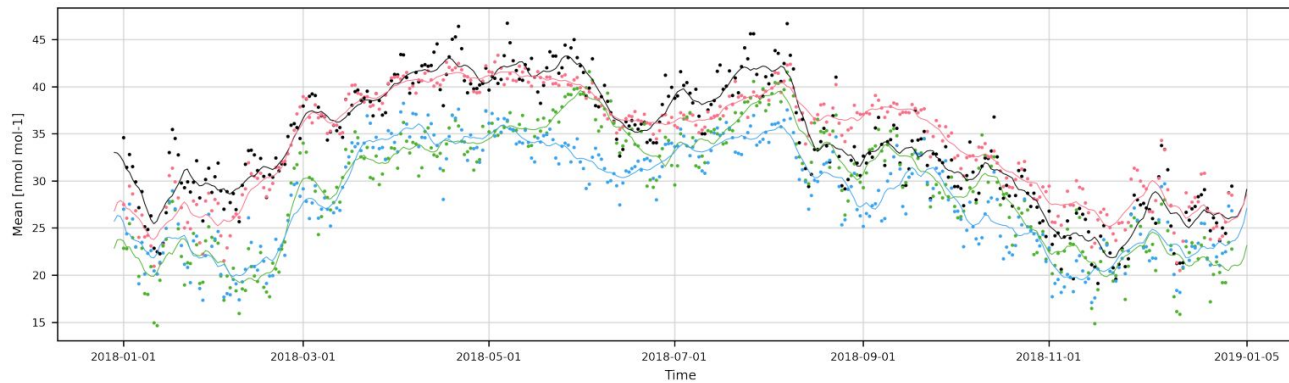
_regression



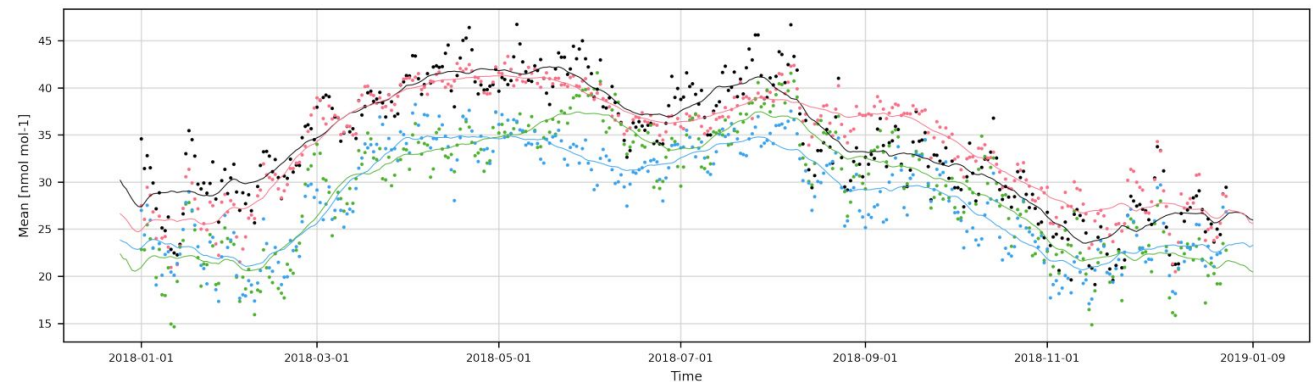
_smooth

The smooth line in the timeseries plot can be adjusted by setting the window and minimum points percentage in **settings/plot_characteristics.yaml**.

timeseries-Mean-daily_smooth (window=10)



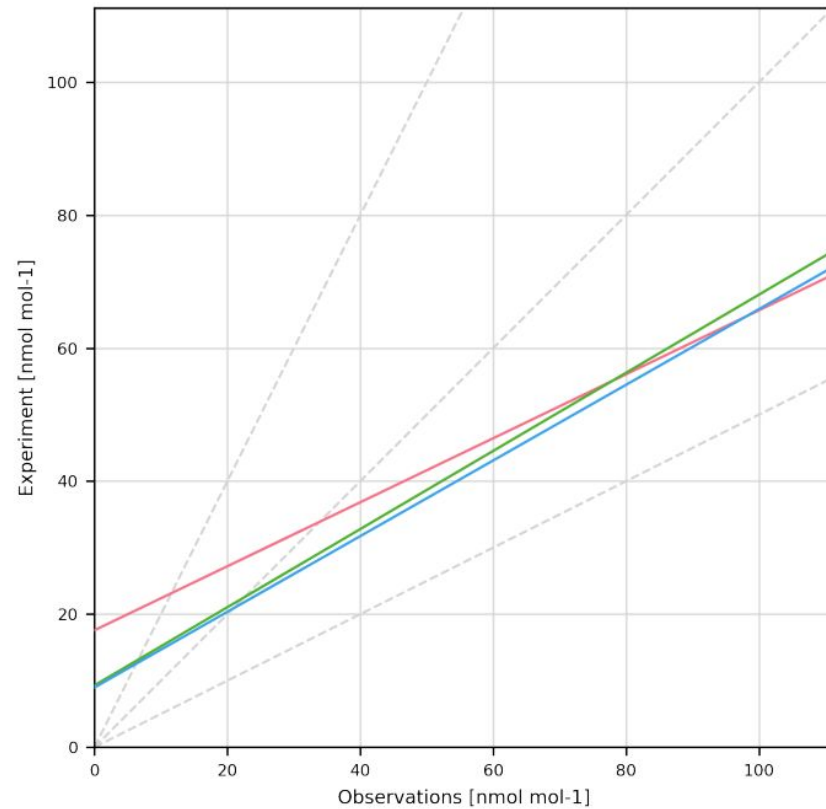
timeseries-Mean-daily_smooth (window=30)



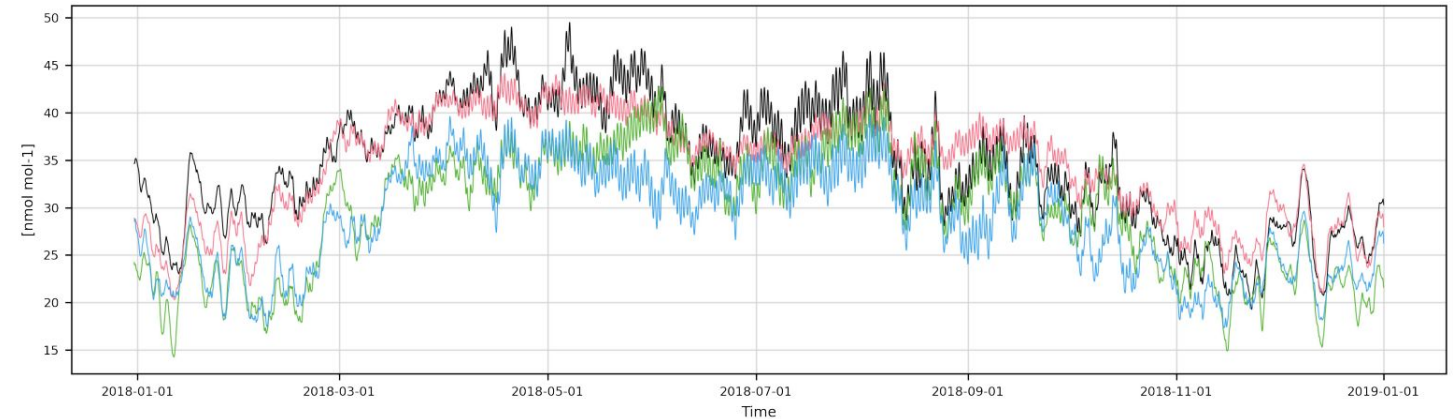
_hidedata

You can hide the points to only show the smooth or regression lines by using this option.

scatter_regression_hidedata

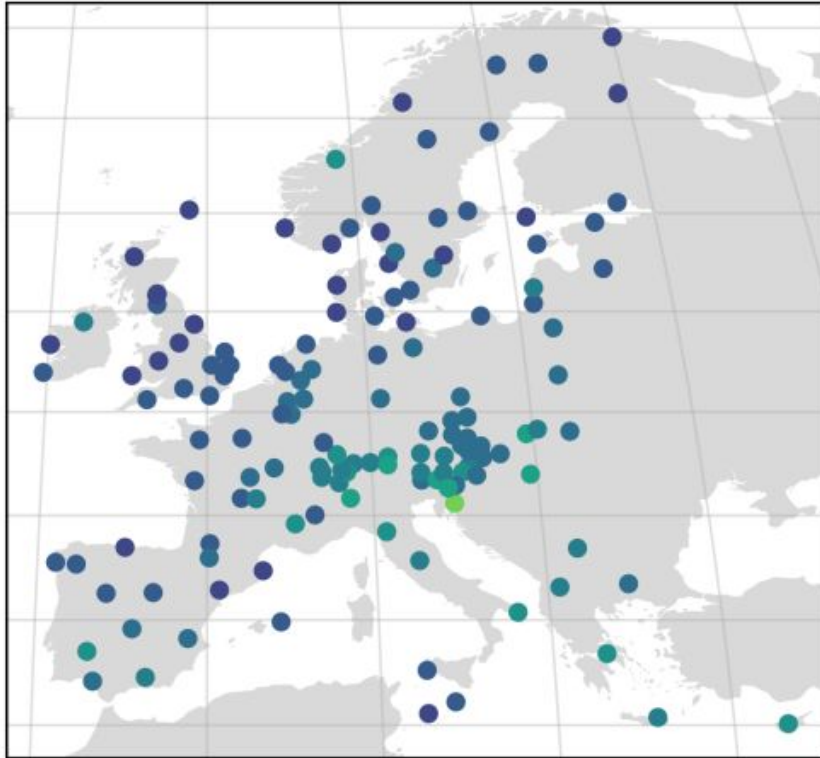


timeseries_smooth_hidedata

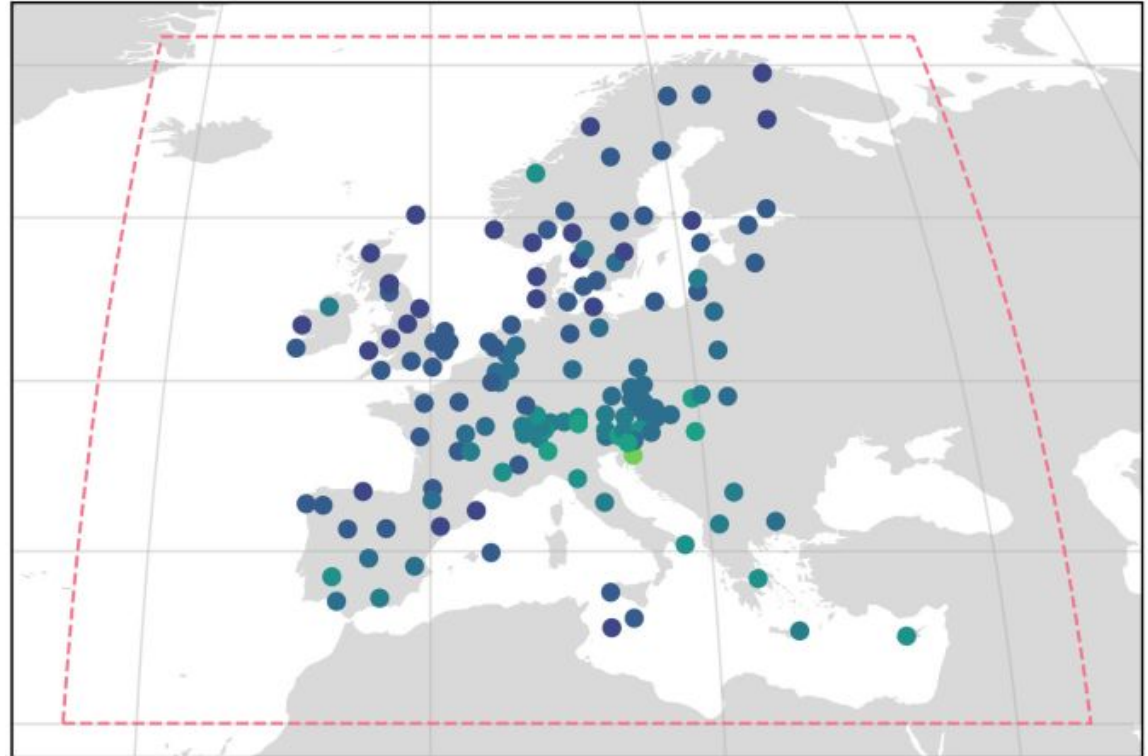


_domain

map-RMSE

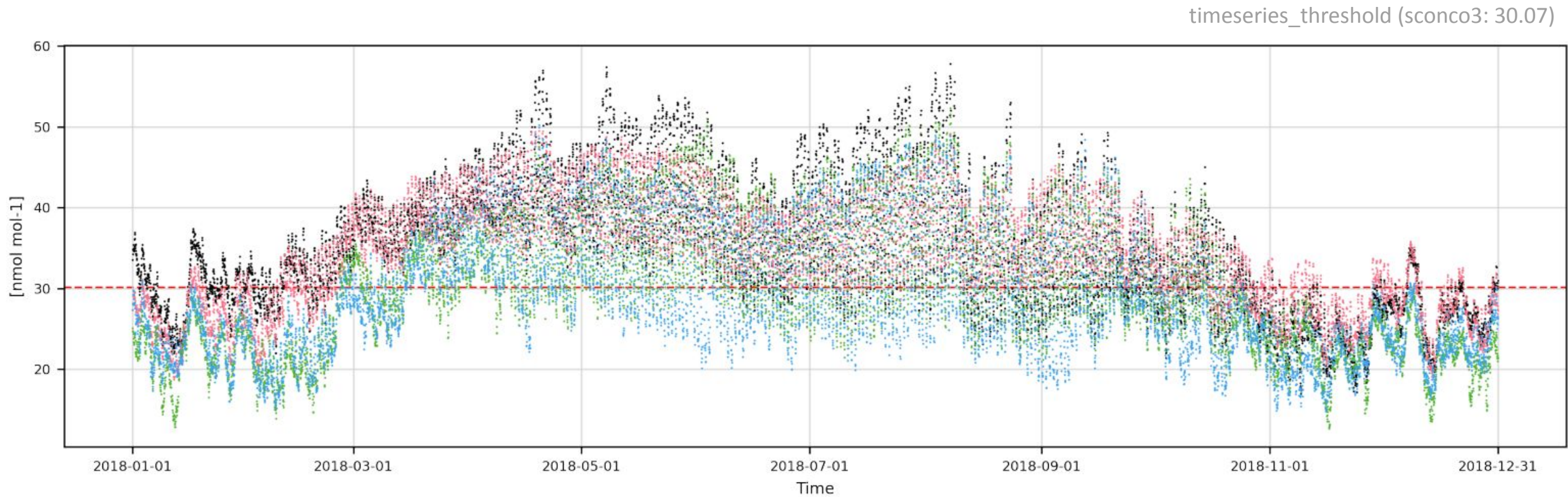


map-RMSE_domain



_threshold

- Dashboard / Report / Library: timeseries, periodic, periodic-violin, distribution, scatter, boxplot



_threshold

The **Exceedances** statistic gives the number of instances above a threshold. The threshold values can now be set in the file **settings/exceedances.yaml** per component, or network-component pair, as so:

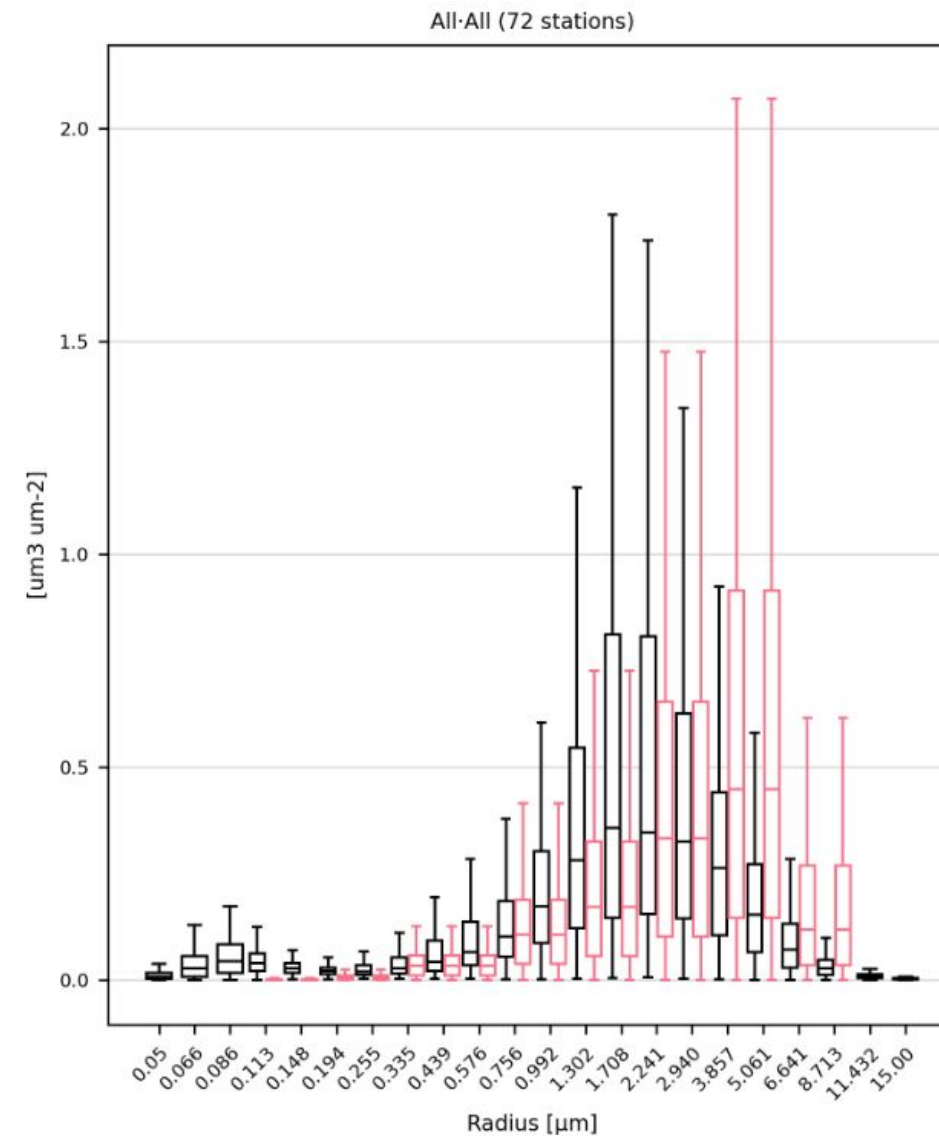
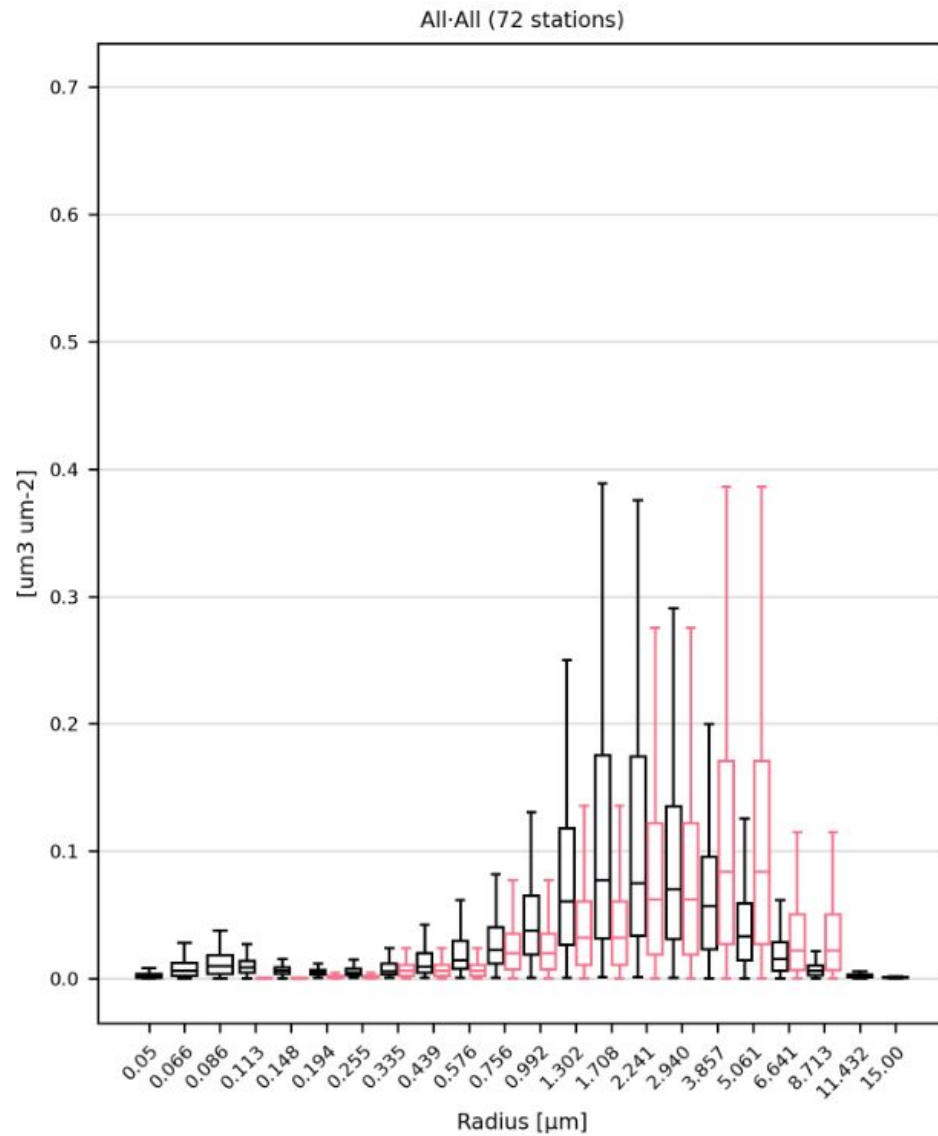
settings/exceedances.yaml

```
sconco3: 30.07  
sconco2: 5.23  
sconco2: 15.02  
sconco: 3435.18  
pm2p5: 5.0  
pm10: 15.0
```

In the case a threshold is set for a specific component, and per network-component, then the threshold for network-component is taken preferentially.

_normalise

- Dashboard: Not applicable
- Report/Library: **boxplot**



Station reference	Station name	Latitude	Longitude	Gerrity Score
AT0002R_UVP	Illmitz	47.77	16.77	0.2
AT0005R_UVP	Vorhegg	46.68	12.97	0.06
AT0030R_UVP	Pillersdorf Bei Retz	48.72	15.94	0.17
AT0032R_UVP	Sulzberg	47.53	9.93	0.14
AT0034G_UVP	Sonnblick	47.05	12.96	0.18
AT0040R_UVP	Masenberg	47.35	15.88	0.14
AT0041R_UVP	Haunsberg	47.97	13.02	0.17
AT0042R_UVP	Heidenreichstein	48.88	15.05	0.14
AT0043R_UVP	Forsthof	48.11	15.92	0.16
AT0045R_UVP	Dunkelsteinerwald	48.37	15.55	0.17
AT0046R_UVP	Gänserndorf	48.33	16.73	0.2
AT0047R_UVP	Stixneusiedl	48.05	16.68	0.19
AT0048R_UVP	Zoebelboden	47.84	14.44	0.12
AT0049R_UVP	Grebenzen Bei St. Lamprecht	47.04	14.33	0.13
AT0050R_UVP	Graz Lustbuehel	47.07	15.49	0.13
DE0054R_UVP	Zugspitze-Schneefernerhaus	47.42	10.98	0.06

Report practical



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Report practical

Make 3 reports showing hourly data in January 2023 of the EEA network (GHOST version 1.6). Make a report that visualises:

1. The distribution of **rural** ozone using the a59j-regional-006 model in **Spain**.
2. The differences in ozone concentrations given by the a59g-regional-000 **forecast** in the **Mediterranean** and **Scandinavia**. Estimated coordinates:
 - Mediterranean: longitudes are -5 to 38, latitudes 32 to 43.
 - Scandinavia: longitudes are -1 to 32, latitudes from 55 to 72.
3. The change of ozone in rural areas from the a59j-regional-006 model during the month when nitrogen dioxide is **below 2 ppbv** at **altitudes under 1000m**.

Reference: <https://providentia.readthedocs.io/en/latest/Configuration.html>

Useful metadata fields

area_classification

GSFC_coastline_proximity

country

latitude

longitude

altitude

station_reference

map_extent

station_name

Exercise 1

The distribution of **rural** ozone using the a59j-regional-006 model in **Spain**.

```
[REPORT1]
ghost_version = 1.6
network = EEA
species = sconco3
resolution = hourly
start_date = 20230101
end_date = 20230201
model = a59j-regional-006
area_classification = keep: rural
country = keep: Spain
report_type = training
report_filename = Report1
```

Exercise 2

The differences in ozone concentrations given by the a59g-regional-000 **forecast** in the **Mediterranean** and **Scandinavia**. Estimated coordinates:

- Mediterranean: longitudes are -5 to 38, latitudes 32 to 43.
- Scandinavia: longitudes are -1 to 32, latitudes from 55 to 72.

```
[REPORT2]
ghost_version = 1.6
network = EEA
species = sconco3
resolution = hourly
start_date = 20230101
end_date = 20230201
model = a59g-regional-000
forecast = day
report_type = training2
report_filename = Report2
```

```
[[Mediterranean]]
longitude = -5, 38
latitude = 32, 43
map_extent = -6, 39, 32, 44
```

```
[[Scandinavia]]
longitude = -1, 32
latitude = 55, 71
```

Exercise 3

The change of ozone in rural areas from the a59j-regional-006 model during the month when nitrogen dioxide is **below 2 ppbv** at **altitudes under 1000m**.

```
[REPORT3]
ghost_version = 1.6
network = EEA
species = sconco3
resolution = hourly
start_date = 20230101
end_date = 20230201
filter_species = EEA:sconcn02 (:, <2.0,
nan)
model = a59j-regional-006
report_type = training3
altitude = 0, 1000
report_filename = Report3
area_classification = keep: rural
```

Library



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Library

The library functionality allows Providentia to be used as a Python module, **with access to its backend functions**, allowing different parts of Providentia to be used in scripts, as required. In that way, we can import it:

```
import providentia as prv
```

From a Python script or notebook outside Providentia it is important to add the path to your directory to the system paths before importing Providentia:

```
import sys  
sys.path.append("path/to/your/Providentia/directory")
```

Examples: <https://github.com/BSC-ES/providentia/tree/master/tutorials>

Opening Jupyter notebook session



One typical use case is the interactive evaluation in a Jupyter notebook. Providentia has an in-built manner of opening a Jupyter notebook. If we add `--notebook` or `--nb` as a launch option in the command line, a Jupyter session will start:

```
./bin/providentia --notebook
```

Running locally, Jupyter **will open automatically**.

BSC HPC users:

If you are on a HPC machine this this **will launch an associated slurm job**, and a file will be created (notebook.out) containing the command to create an SSH tunnel on your local machine, which should be pasted into the terminal, and the path to open Jupyter in your browser.

Loading a configuration file

Once imported the first step is calling the class Providentia with the configuration you plan to use.

```
provi = prv.Providentia("interactive_template.conf")
```

This class' methods can be used:

- To access the modes: download, interpolate, report and dashboard.
- To load, filter, save and plot the data.

You can specify the section or subsection you want to work with by. Otherwise, if you have multiple sections, the first one will be selected unless it is explicitly defined.

```
provi = prv.Providentia("interactive_template.conf",  
section="SECTIONNAME", subsection="SECTIONNAME.SUBSECTIONNAME")
```

Overwriting configuration parameters

If wanting to overwrite any variables defined in the .conf file, each variable can be simply passed as an argument when loading the data, e.g.:

```
provi.load("interactive_template.conf", network="EANET")
```

These variables can be any variable that can be set in a .conf file.

Reference: <https://providentia.readthedocs.io/en/latest/Configuration.html>

Viewing active configuration

It is possible to view the active configuration variables in two different ways:

- Using the method *config*:

```
provi.config()
```

This method can also be used to view another configuration file, by passing the filename:

```
provi.config("important.conf")
```

- By printing the active class instance:

```
print(provi)
```

Accessing the modes

The *download*, *interpolate*, *report* and *dashboard* functionalities can be integrated in workflows with little effort.

```
provi.download()
```

```
provi.interpolate()
```

```
provi.report()
```

```
provi.dashboard()
```

Loading the data

When the data is available in your machine, you can load it by doing:

```
provi.load()
```

Once the data has been loaded, it can be easily accessed by these methods:

- Retrieving a specific variable, by providing a variable name to the *variable* method.

```
var_data = provi.variable("network|component_variablename")
```

- Retrieving all metadata / data variables by using the *data* method. The returned data can be in different formats: nc (netCDF), np (numpy) or xr (xarray) by passing the format argument:

```
data = provi.data(format="xr")
```

Applying filters

To apply a filter not set in the configuration file, the *filter* method is used. The fields to filter by can be any data /metadata variable standardised in GHOST, and can be textual or numeric.

- If the field is numeric, use *lower* and/or *upper*, e.g.:

```
provi.filter(field, lower=28, upper=31)
```

- If the field is textual, use *keep* and/or *remove*, e.g.:

```
provi.filter("country", keep="Spain")
```

```
provi.filter("country", remove=["Spain", "France"])
```

- If the field is a representativity field, use *limit*:

```
provi.filter(rep_field, limit=20)
```

Selecting stations

To select station for one or more stations, you can use the function `filter_station`:

```
provi.filter_station(['AT0002R_UVP','SK0007R_UVP'])
```

Reset filters

If at any time any active data / metadata filters are wished to be reset, this can be done by using the *reset* method.

To reset all filters to the Providentia defaults, including those set from the original read configuration file:

```
provi.reset()
```

To keep the filters defined in the configuration file, but reset the ones added later, pass the *initialise* argument as True:

```
provi.reset(initialise=True)
```

Plotting

You can use the plotting functions of Providentia with the *plot* method. Any of the available plot types in Providentia can be made by passing the desired type.

```
provi.plot(plot_type)
```

Plots can also be saved to file, by passing *save* and a filename with the wanted extension (e.g. png, etc.):

```
provi.plot(plot_type, save="myplot.png")
```

The plotting object can also be returned for additional tinkering, by setting *return_plot* to True:

```
plot_obj = provi.plot(plot_type, return_plot=True)
```

Reference: <https://providentia.readthedocs.io/en/latest/Plotting.html>

Calculating statistics

All defined basic and model bias statistics defined in Providentia can be calculated from the loaded data, using the *statistic* method:

```
stat_calc = provi.statistic(stat, labela="OBS")
```

where *stat* is the statistic wished to be calculated, and *labela* is “observations”/model name or the alias set in the conf file. If wanting to calculate a bias statistic, you need to set *labelb*:

```
stat_calc = provi.statistic(stat, labela="OBS", labelb="EMEP")
```

For bias statistics for which a subtraction is involved, it is always done as *data_b* - *data_a*.

If wanting to calculate statistics per station, *per_station* can be passed as True:

```
stat_calc = provi.statistic(stat, labela="OBS", labelb="EMEP", per_station=True)
```

Saving the data

Save the current configuration:

```
provi.save(format='conf')
```

Save the data as netCDF:

```
provi.save(format='nc')
```

Save the data as numpy:

```
provi.save(format='np')
```

If you want to save the data to a specific location, use the argument `fname`. By default, the data is saved in the folder **providentia/saved_data**:

```
provi.save(fname='data/output/config.conf', format='conf')
```

Library practical



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Library practical

Launch the notebook mode, create a notebook and load the training configuration file. Use **hourly** data and **GHOST version 1.6**.

1. Create a **map** showing the **median** of the a59j-regional-006 model for EEA **ozone** stations in **Spain**.
2. Create a map showing the **RMSE** of the **rural nitrogen dioxide** stations using the a59j-regional-006 model in **Germany**.
3. Read the data to find out the available **station references** and select **one particular station** to create a **timeseries** with **statistical annotations** of the ozone concentrations.
4. Reset the filter and show a **multispecies statsummary bias** and **heatmap** showing the **r** statistic in the ozone concentrations given by the **two models**.
5. Save the heatmap as an **image** and the data as a **netCDF** file.

Customisation



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Plot customisation

To update the style of plots in the dashboard, reports and library, edit the plot characteristics in **settings/plot_characteristics.yaml**.

settings/plot_characteristics.yaml

All

Dashboard

Report

Library

```
"timeseries":  
{  
  "grid": { "axis": "both", "color": "lightgrey", "alpha": 0.8 },  
  "threshold_line":  
    { "linestyle": "--", "linewidth": 1.0, "color": "red", "zorder": 0 },  
  ...  
  "dashboard":  
    { "remove_spines": ["top", "right"], ... },  
  "report":  
    { "orientation": "landscape", ... },  
  "library":  
    { "figsize": [17, 6], ... },  
}
```

Reference: <https://providentia.readthedocs.io/en/latest/Plot-customisation.html>

Colorbar customisation - Bounds

There are two ways to change the colorbar bounds:

- If you want to set the same bounds for all statistics, you can edit the parameters *vmin_absolute*, *vmax_absolute*, *vmin_bias* and *vmax_bias* under *map* in the plot characteristics files.
- If you want to set the bounds for each statistic (recommended), you should edit the same parameters in **settings/basic_stats.yaml** and **settings/model_bias_stats.yaml**.

Colorbar customisation - Color

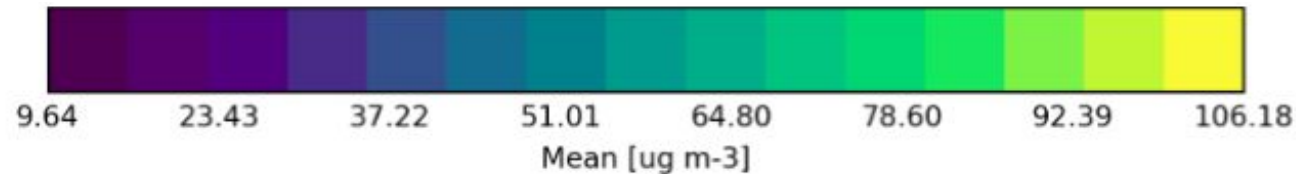
To change the colors, you need to edit the cmap:

- If you want to set the same colors for all statistics, edit the parameters *cmap_absolute* and *cmap_bias* under *map* in the plot characteristics files.
- If you want to set the colors for each statistic (recommended), you should edit them in **settings/basic_stats.yaml** and **settings/model_bias_stats.yaml**.

Colorbar customisation - Discretisation

You might also want to change the number of breaks if you have a discrete colorbar. For this you will need to change the number of ticks (*n_ticks*) and number of discrete colors (*n_discrete*) under *map* in the plot characteristics files.

In this example, we have set *n_ticks* to be 8 and *n_discrete* to be 15.



If you prefer to have a continuous colorbar and want to remove the breaks, you need to set *discrete* to be false.

Colorbar customisation - Per statistic

Users can define the color and bounds of the colorbar per species using a dictionary.

settings/basic_stats.yaml

```
"Mean": {"function": "calculate_mean",
        "order": 0,
        "label": "Mean",
        "arguments": {},
        "units": "[measurement_units]",
        "minimum_bias": [0.0],
        "vmin_absolute": {"sconco3": 0, "sconco2": 0},
        "vmax_absolute": {"sconco3": 20, "sconco2": 5},
        "vmin_bias": {},
        "vmax_bias": {},
        "n_ticks": {},
        "n_discrete": {},
        "cmap_absolute": {"sconco3": "viridis", "sconco2": "viridis"},
        "cmap_bias": "RdYlBu_r"}
```



Legend customisation

If you want to change the color of the legend, you can edit the *legend_color_palette* under *general* in the plot characteristics files.

The default palette is *husl*:



Legend and colorbar customisation - Color

Now the legend and colorbar colormaps can be customized by adding a new schema in **settings/color_palettes.yaml**.

```
"cams": ["#ED1B23", "#2D2F92", "#7CFC00", "#009B55"]
```

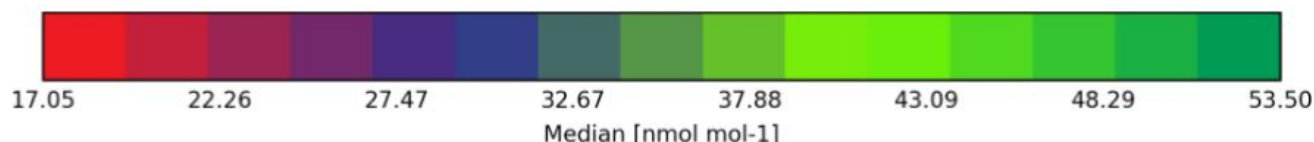


To update the legend colors, you will need to set the custom name in *legend_color_palette* in the plot characteristics files.

● observations ● EMEP ● MONARCH ● SILAM

For the colorbar, you should define *cmap_absolute* or *cmap_bias*, depending on the statistic type.

The colors will interpolate depending on *n_discrete*. Below *n_discrete* is 15.



Labels customisation

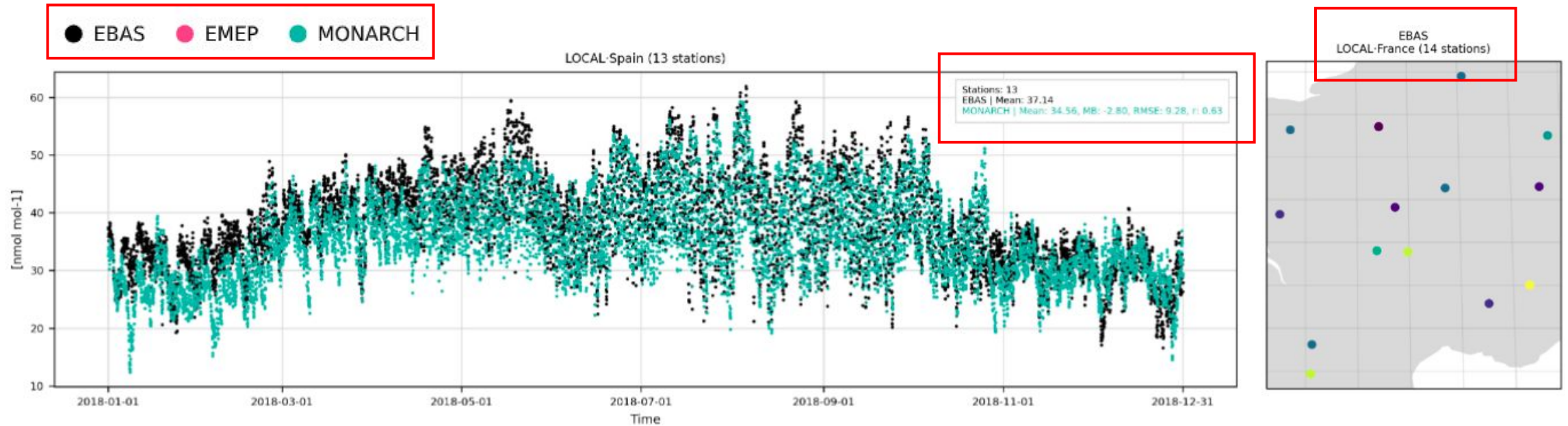
By defining **observations_data_label** in our configuration files, we can customise the name of the observations in the legend, plots and maps.

```
observations_data_label = EBAS
```

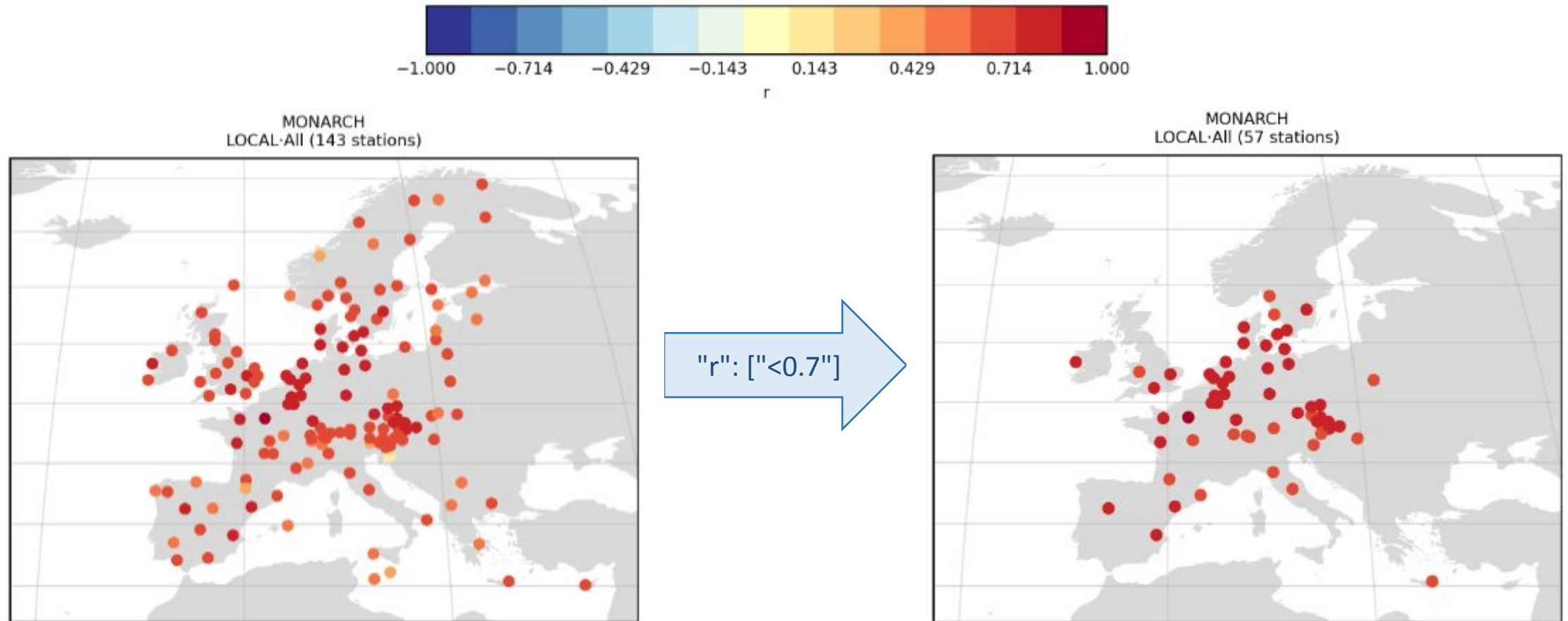
If you want to edit the names of the models, you can only do it launching Providentia with a configuration file, where you will define the alternative names as in:

```
model = cams61_emep_ph2-eu-000, cams61_monarch_ph2-eu-000 (EMEP, MONARCH)
```

Labels customisation



Remove stations by statistics



Remove stations by statistics

If you want to automatically remove stations that have certain statistical values, you will need to add your criteria in the file **settings/remove_extreme_stations.yaml**. An example of this exists for CAMS:

configurations/configuration_name.conf

```
[CAMS2_40]
network = ineris/eionet-cams2_40-ira
species = sconco3,sconco2,sconcco,sconco2,pm10,pm2p5
resolution = hourly
start_date = 20230101
end_date = 20230215
model = a59g-regional-000, a59j-regional-006 (Forecast,
Analysis)
temporal_colocation = True
spatial_colocation = False
report_type = operational
report_summary = True
report_stations = False
report_title = CAMS2_40 Forecast and Analysis Report
report_filename = operational_report
statistic_mode = Temporal|Spatial
statistic_aggregation = Median
periodic_statistic_mode = Independent
periodic_statistic_aggregation = Mean
remove_extreme_stations = CAMS
```



settings/remove_extreme_stations.yaml

```
"CAMS": {"r": ["<0.3"],
"NMB": ["<-100.0", ">100.0"],
"NRMSE": [">100.0"]}
```

Remove stations by statistics

Any absolute statistic can be set to be a bias statistic by adding `_bias` e.g.:

```
"p95_bias": ["<10", ">20"]
```

The statistics can be general, across all components, or they can be specific per component, for example:

settings/remove_extreme_stations.yaml

```
"CAM5": {"r": {"sconco3": ["<0.3"],  
               "sconco2": ["<0.55"]},  
  "NMB": {"sconco3": ["<-100.0", ">100.0"],  
          "sconco2": ["<-20.0", ">20.0"]},  
  "NRMSE": {"sconco3": [">100.0"],  
            "sconco2": [">200.0"]}}
```

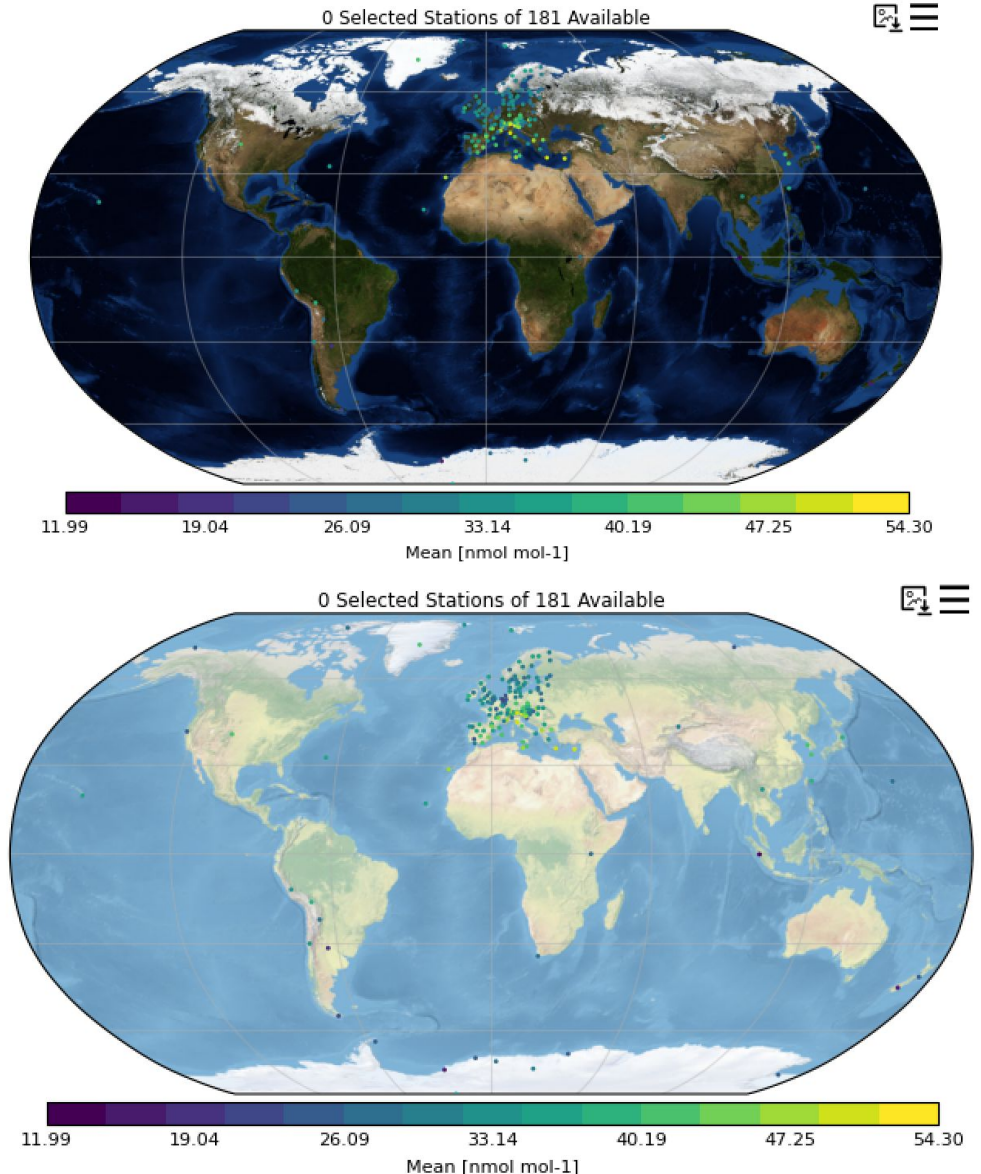
Map background

Users have the ability to plot a background image on the map axis by changing the ***background*** variable under the ***map*** section in the **settings/plot characteristics.yaml**.

There are 3 available standard options:

1. ***providentia*** (default)
2. ***blue_marble*** (NASA's blue marble product)
3. ***shaded_relief*** (imagery showing changes in evolution)

However, any image can be set. Check the Plot Customisation wiki page for all the details.



Report cover page

EVALUATION REPORT

Network: FRAS
Species: O3
Temporal resolution: Hourly
Date range: 01/01/2017 - 01/01/2020
Experiments: EMEP, MONARCH, SILAM
Subsections: DEBUG-Europe, DEBUG-Spain

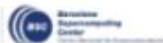
Earth Sciences
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EVALUATION REPORT

Network: EBAS
Species: O3
Temporal resolution: Hourly
Date range: 01/01/2017 - 01/01/2020
Experiments: EMEP, MONARCH, SILAM
Subsections: DEBUG-Europe, DEBUG-Spain

Earth Sciences
Department



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Report cover page

The cover page can be customised by editing the parameters under **header** in **settings/plot_characteristics.yaml** file. The most interesting ones are these:

- **dark_mode** to set the background to be dark (blue tone) or light (white).
- **variables** to specify which variables you want to show. The options are network, species, resolution, dates, model, temporal_colocation, spatial_colocation, filter_species, calibration and subsections. By specifying the value under those keys, you can overwrite the default variable values and write anything you want.
- **logo** to display any logo on your report cover. To use it, you must specify the path to your PNG file for the corresponding background (dark or light mode).



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Thank you for your attention!

More information at:

<https://github.com/bsc-es/providentia>

Join the #providentia Slack channel!

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